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MIL-T-24398(SHIPS)
29 January 1971
(See 6.5)

MILITARY SPECIFICATION

TURBINE, STEAM AND REDUCTION GEAR,

AUXILIARY, GENERATOR-DRIVE (NAVAL

SHIPBOARD USE)

1. SCOPE

1.1 Scope. This specification covers auxiliary steam turbine and reduction gear (where used) for generator sets for Naval ships and submarines and includes couplings, and additorial equipment, fittings and accessories. Propulsion generator turbines and gears are not included.

1.2 Classification. Generator-drive auxiliary-steam turbines shall be of the following types and classes, as specified (see 6.2.2(b)):

Type I - Condensing turbine for package type generator sets.
Type II - Condensing turbines for conventional (rionpackage) type generator sets.

Class A - Turbines and gears for submarines,
Clasa B - Turbines and gears operating on saturated (1 percent wet) steam, for combatant surface ships.
Class C - Turbines and gears operating on superheated steam, for combatant surface ships.
Class D - Turbines and gears, operating on superheated steam for non-combatant surface ships.

2. APPLICABLE DOCUMENTS

2,1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

:PEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical and Bearing, Roller, Self-Aligning.
FJ:'-B-187 - Bearings, Roller, Tapered.
QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
QQ-S-624 - Steel Bar, Alloy, Hot Rolled and Cold Finished (General Purpose).
QQ-S-630 - Steel, Bar, Carbon, Hot Rolled (Merchant Quality).
QQ-S-631 - Steel, Bar, Carbon, Hot Rolled (Special Quality).
QQ-S-634 - Steel Bar, Carbon, Cold Finished, (Standard Quality).
QQ-S-635 - Steel Plate, Carbon.
QQ-S-691 - Steel Plate, Carbon-Silicon, Carbon Molybdenum and Manganese-Molybdenum Alloys Hot Rolled, (Marine Boiler Quality).
QQ-S-699 - Steel Sheets, Carbon, Hot-Rolled, Heavy Gage Structural Quality.
QQ-S-763 - Steel Bars, Wire, Shapes and Forgings, Corrosion-Resisting.
QQ-S-764 - Steel Bar, Corrosion Resisting, Free Machining.
QQ-S-766 - Steel Plates, Sheet and Strip--Corrosion Resisting.
QQ-T-390 - Tin Alloy Ingots and Casting and Lead Alloy Ingots and Castings (Anti-Friction Metal) For Bearing Application.
TT-L-201 - Linseed Oil, Heat Polymerized.

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MILITARY

JAN-W-562 - Wire, Nickel-Alloy, Spring (Heat-Resistant and Age-Hardenable).
UL-S-860 - Steel Forgings for Steam Turbine Rotors,
MIL-S-861 - Steel Bars, Corrosion Resisting, Naval Steam Turbine Parts Use.
MIL-S-866 - Steel: Bars and Billets (For Carburizing).
MIL-S-869 - Steel Bar, Billet and Forging-Alloy Nitriding Application.
MIL-S-870 - Steel Castings, Molybdenum Alloy.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-S-1222 - Studs, Continuous Thread (Bolt Studs); Nuts, Plain, Hexagon; and Steel Bars, Round-High Temperature Service.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-G 2860 - Glasses, Sight, Flat, Clear, Borosilicate.
MIL-G-3087 - Generator Sets, Steam Turbine, (Direct and Alternating Current) (Naval Shipboard Use).
MIL-F-3541 - Fi tings, Lubrication.
MIL-N-7786 - Nickel-Chromium Alloy, Sheet and Strip, Age Hardenable Annealed.
MIL-Q-9858 - Quality Program Requirements.
MIL-S-15083 - Steel castings.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts For Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15345 - Castings, Nonferrous, Centrifugal.
MIL-S-15464 - Steel, Alloy, Chromium-Molybdenum; Castings.
MIL-L-15719 - Lubricating Grease (High-Temperature, Electric Motor, Ball and Roller Bearings).
MIL-C-15730 - coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hydraulic oil, and Fresh Water.
MIL-T-16049 - Tachometers: Electric; Self-Generating; Mechanical, Fixed Mounting and Hand Held; and Vibrating Reed.
MIL-S-16113 - Steel Plate, High Tensile (HT) Hull and Structural.
MIL-S-16216 - Steel Plate, Alloy, Structural, High Yield Strength (HY-80 and HY-100).
MIL-G-16265 - Gaskets, Metallic-Asbestos, Spiral Wound (For Navy Flanged Joi ts In Piping Systems).
MIL-S-16974 - Steel Bars, Billets, Blooms and Slabs; Carbon and Alloy (For Reforging or Other Operations Before Heat Treatment).
MIL-S-16993 - Steel Castings (12-Percent Chromium).
MIL-M-17059 - Motors, 60-Cycle Alternating-Current, Fractional H.P. (Ship-board Use).
MIL-M-17060 - Motors, 60 cycle Alternating Current, Integral Horsepower (Shipboard use).
MIL-R-17131 - Rods, Welding, Surfacing.
MIL-I-17244 - Indicators, Temperature, Direct-Reading Bimetallic (3 and 5 Inch Dial).
MIL-L-17331 - Lubricating Oil, Steam Turbine (Noncorrosive).
MIL-S-17849 - Strainers, Sediment, Pipeline, Duplex (With and Without Magnet).
MIL-G-17859 - Gear Assembly, Propulsion (Naval Shipboard Use).
MIL-B-17931 - Bearings, Ball, Annular, For Quiet Operation.
MIL-S-18410 - Steel Bars, Billets and Forgings-Chromium-Molybdenum Alloy.
MIL-I-18997 - Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-P-19131 - Pumps, Rotary, Power Driven, Miscellaneous.
MIL-S-19434 - Steel Gear and Pinion Forgings, Carbon and Alloy, Heat Treated, Naval Shipboard Propulsion Unit and Auxiliary Turbine.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-G-19650 - Generator Sets, Steam Turbine, Direct Drive, 2000 KW, 450 Volt, 60 Cycle A.C.
MIL-V-20065 - Valves, Angle, Pressure Relief, Naval Shipboard, For Steam service.
MIL-T-20157 - Tube and Pipe, Carbon Steel, Seamless.
MIL-S-20166 - Steel Structural Shapes, Weldable Medium Carbon and High Tensile; Hull and Structural.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.
MIL-G-21410 Governing Systems, Speed and Load-Sensing (For Electric Generator Sets).
MIL-S-21427 - Strainer, Assemblies, Main Steam, High Pressure (Sizes 4 Inches and Above).
MIL-F-21467 - Fittings, Flareless, Fluid Connection (Shipboard Use).
MIL-S-22698 - Steel Plate, Carbon, Structural, For Ships.
MIL-C-23233 - Couplings for Propulsion Units, Auxiliary Turbines and Line Shafts, Naval Shipboard,

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MILITARY (cont'd.)

MIL-S-23284 - Steel Forgings, Carbon and Alloy, For Shafts, Sleeves, Couplings, and Stocks (Rudders and Diving Planes).
MIL-P-23917 - Precipitator (Electrostatic) Vent Fog, Shipboard Use.
MIL-S-24093 - Steel Forgings, Carbon and Alloy Heat Treated.
MIL-S-24113 - Steel Plates, Carbon Manganese-Heat Treated by Normalizing Or Quenching and Tempering.
MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) (For Shipboard Application).
MIL-N-25027 - Nuts, Self Locking, 250°F, 450°F and 800°F, 125 KSI FTU, 60 KSI FTU and 30 KSI FTU.

STANDARDS

MILITARY

MIL-STD-271 - Nondestructive Tes.ting Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Pipings Components for Submarine Services.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.
MSl5003 - Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS21208 - Insert, Screw Thread, Coarse and Fine, Free Running, Helical Coil, CllliS.

DRAWINGS

MILI'l'ARY

810-1385850 - Piping, Gage, For All Service.
810-2145504 - Installation of Electrostatic Vent Fog Precipitator and Surge Blow-Off Assembly for Propulsion Reduction Gears.

PUBLICATIONS

NAVSIIIPS

0283-228-1000 - Bearing Babbitting Procedures.
0900-001-7000 - Fabrication and Inspection of Brazed Piping Systems.
0900-003-8000 - surface Inspection Standards for Metals.
0900-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent"'specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A434 - Quenched and Tempered Alloy steel Bars, Hot-Rolled or Cold-Finished.
El86 Reference Radiographs for Heavy-Walled Steel Castings.
E208 Conducting Drop-Weight Test to Determine Nil-Ductility Transition Temperature of Ferritic Steels.
E280 - Reference Radiographs for Heavy-Walled Steel Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code.
Section VIII - Unfired Pressure Vessel.

(Application for copies should be addressed to the American Society of Mechanical E?gineers, 345 East 47th Street, New York, N.Y. 10017.)

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AMERICAN WELDING SOCIETY (AWS)
A2.2 – Nondestructive Testing Symbols.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N.Y. 10017.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
Bl6.ll - Forged Steel Fittings, Socket-Welding and Threaded.
Y32.3 - Welding Symbols.

(Application for copies should be addressed to the American National standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)
000.XX - Standards and Technical Publications Index.
390.XX - Gear Classification Manual.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, N.w., Washington, D.c. 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Definitions. Unless otherwise specified hereinafter, terms and·expressions shall be interpreted in accordance with the definitions contained in the following:

(a) Webster's New International Unabridged Dictionary.
(b) Mechanical Engineers' Handbook by Lionel S. Marks.

3.l.1 speed.Rated speed is the speed requi.red to drive the generator at its rated speed. For alternating current (a.c.) generator sets, rated speed applies to the speed which produces rated frequency. For direct current (d.c.) generator sets rated speed applies to the identification plate design value of speed for rated power output.

3.1.2 Gear nomenclature. Nomenclature for gearing shall be.in accordance with the publications-rI'sted in AGMA 000.XX.

3.1.3 Main]inion, gear, ah or shaft (element). A main pinion, main gear, or main shaft is a respective·e ement wic existsfn the assembly for the purpose of transmitting or transferring torque from an input to the output flange or fitting which drives the generator.

3.1.4 Accessory drive (train). An accessory drive train is any power transmission path driving an attached accessory s ch as a lubricating oil pump, governor, tachometer, etc.

3.2 Materials. Specified materials are listed in table I, however, the manufacturer's List of Preferred Materials (see 3.2.B.1.1), when approved, shall replace table I and is not subject to revision under each contract to reflect later issues of Government specifications unless specified (see 6.2.2(c)). Materials for other parts shall be at the manufacturer's option, however, corrosion-resistant materials shall be used for internal turbine applications of shims, lugs, keys and other removable parts eXposed to the steam atmosphere. Nozzle block bolting which is countersunk and shielded is not included.

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Table I - Materials.

Part A - Materials of principal parts.					
Part or service		Applicable documents	Material and properties	Temperature limit (OF.) (maximum) (see 3.2.2)	Remarks
(a) Antifriction metal		QQ-T-390, grade 2 or 3	Babbitt (tin base)	280°F. hot-spot metal temperature	See note 1
\t>, Bearing t>a.Ll (for accessocy shafts)		FF-B-.L71	Steel	See note 2	See 3.4.JB.5•l
		MIL-B-17931	Steel	see note 2	See **3.4.38.S.l**
(c) Bearing pedestals and caps		QQ-S-691 all classes	Carbon steel and alloy boiler plate	--	See note 3
		MIL-5-.L!>UIS.:S grade B, cw or 70-36	Carbon stee.L (casting)		See note J
		MIL-S-..:.:0 8	Carl;>on steel plate		See note J
		MIL-S-24J.J.3	Carbon manganese steel plate		
(d) Bear-ings, rolle:i:	(l) Radial and loca-ting thrust	FF-B-185	Steel	See note 2 200	See **3.4.38.5.l** ----------
	(2) Combined radial and thrust	FF-B-187	Steel	See note 2 200	see 3.4.38.5.l ----------
(e) Bear-ings, thrust	(1) Shoes	QQ-S-631 or **QQ~S-634,** QQ~S-630	Cl015-Cl025 carbon steel (bar)	--	see note 4
		QQ~S-635 (SAE 1020)	Carbon steel. (plate)	--	----------
		QQ-S-o:,.L, class A, B, C	carnon stee.L (boiler plate)	--	**see** note 4
		MIL-s-.:-.U9.:S, type v, class H	carnon stee.L (forging)		See note 4
		MIL-S-15083, grade B or CW	Carbon stee.L (casting)		-----------
	(2) Level-ing ph,tes	QQ-5-b35 (SAE 1020)	Car.oon steel plate		----------
		QQ-5-b.!4	AJ.J.oy steeJ. (bar)		see note 4
		QQ-S-630	carbor1 steel bar		See note 4
		MIL S-.!4U .:S, type I,II or III, class c, D or E	AJ.J.oy s teeJ. fo ging		Magnetic particle inspection required See note 4

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Table I - Materials (cont'd).

Part A - Materials of principal parts.					
Part or service		Applicable documents	Material and properties	Temperature limit (OF.) (maximum) (see 3.2.2)	Remarks
(e) Bearings, thrust (cont'd).	Collar	MIL-S-, Mv:;,J, class A, types I and II	Alloy steel (forging)	--	Heat treat to 350 ± 50 BHN See note 4
		QQ-S-624	Alloy steel (bar)	--	Heat treat to 350 ± 50 BHN See note 4
(f) Blades, blade holders for stationary blades (intermediate segments), and locking pieces		MIL-S-861 class 403 or 410 condition HT (machined) (forged or rolle.d)	12 Cr corrosion-resisting steel (bar and forging)	9UU	-------
		MIL-S-861 class 422, condition HT (machined, forged or rolled)	12 Cr corrosion-resisting steel (bar and forging)	lOOU	
			Titaniurn and titanium alloys (bars, forgings and forging.. stock)	As approved	When approved by NAVSEC on a case basis
(gJ Bolts, studs and nuts (and other screw-thread fasteners). For casing steam joints, valve chest cover, hold down bolts, nozzle blocks and other highly stressed (2/3 yield) structural applications		MIL-S-1222, type I, symbol B7 and type II, symbol H	Cr-Mo alloy steel (bar)	775	Magnetic particle test is not required
			Carbon steel		
		MIL-S-1222, type I, syrnbol Bl6 and type II, syrnbol 4	Cr-Mo v-alloy steel (bar) Alloy steel (forged or bar)	1000	Continuous threads not required, Also see -3. 4. 28
		MIL-S-isol, class 422, condition HT	12 Cr corrosion-resisting steel (bar)	lOUU	
(nJ casing (nous1ng, covers) (reduction gear)		MIL-S-2409J, all classes	Carbon steel (forgings)		see note ,j
		MIL-S-16113	Steel. pl.ate		see note J
		MJ.L-1:>-15U83, grade B, cw or 70-36	Carbon steel castings		see note J
		QQ-S-69l.-1 class A, B or C	carbon steel. (boiler plate)		see note J
		QQ-S-635, SAE 1020	Carbon steel. (plate)		See note 3
		M.LL-:::;-2O1·66	Carbon steel. (shapes)		see no"t.e ,j
		MIL-S""-'698, type I, class A, B or C	carbon steel. (plate)		See note 3
		QQ-S-b99 1 class 3	CarJJon steel (sheets)	---	See note .J

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Table I - Materials (cont'd).

Part A - Materials of principal parts.				
Part or service	Applicable documents	Material and properties	Temperature limit (°F,) (maximum) (see 3.2.2)	Remarks
Cas ngs an steam chests (turbine)	QQ-S- , class A, B or C	Car on stee (boiler plate)		See c and 3,4.43
	MIL-S-15)83, grade B or CW	Carbon steel (casting)		
	MIL-S-?2698	Carbon steel plate structural		
	MIL-S-2	Carbon-manganese (plate)		
	MIL-S-8	Carbon-Mo alloy steel (casting)	5	
	QQ-S-691 class D	Carbon-Mo alloy steel (plate)		
	MIL-S-15464, class l	1-1/4 cr-1/2 Mo alloy steel (casting)	950	
	MIL-S- class 3	Cr- MoV alloy steel (casting)	0	
	MIL-S-15464, class 2	2-1/4 Cr-1 Mo alloy steel (casting)		
	MIL-S-16993, class 2	12 Cr alloy steel (casting)		
	QQ-S-766, class 410	12 Cr alloy steel (plate)	9	
	MIL-S-16993 class l	12 Cr a oy steel (casting)		
D ap ragms and nozzle blocks (Excluding partitions and shroud rings)	QQ-S-63 condition A	Carbon stee (bar)(0.35 max. carbon)	0	See c and 3.4.42
	M L-S-15083, class B or CW	Car on stee (casting)		
	QQ-S- classes A, B, C	Car on stee (boiler plate)		
	QQ- -6 classes D and E	Car on mo y and manganese moly	875	
	MIL-S-	car on - Mo alloy steel (casting)		
	MIL-S-15464, class l	1-1/4 Cr - Mo alloy steel (casting)		
	QQ-S-763, class 405 or 410	12 Cr corros on resisting steel (bars and fo ging)		
	MIL-S-16993, class 2	12 Cr alloy steel (casting)		

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Table I - Materials (cont'd).

Part A - Materials of principal parts.				
Part or service	Applicable documents	Material and properties	Temperature limit (°F) (maximum) (see 3.2.2)	Remarks
(j) Diaphragms and nozzle blocks (Excluding partitions and shroud rings)(cont'd)	MIL-S-18410, class a	l-l/4 Cr-l/2 Mo alloy steel (bar, billet, forging)	1050	
	l\UL-S-1841.0, class b	2-1/4 cr-1 Mo alloy steel (bar, billet, forqing)	1050	
	MIL-S-l54b4$_1$ class 2	2-1/4 cr-1 Mo alloy steel (casting)	1050	
	MIL-S-16993 class 1	12 Cr alloy steel (casting)	uoo	
(k) Diaphragm and nozzle block partitions and shroud bands	QQ-S-763, class 403 or 410, condi-tion for class 405	12 Cr corrosion-resisting steel (bar and forging)	950	-----
	QQ-S-76b, class 410	12 Cr corrosion-resisting steel (plate, sheet and strip}	00	
	MIL-S-861, class 403 or 410" condi-tion HT, or class 405	12 Cr corrosion-resisting steel (bar and forging)	950	
	MIL-S-861, class 422, condition HT,	12 Cr corrosion-resisting steel (bar and forqing)	l.000	
	QQ-S-76J, class 403 or 410	1.i Cr corrosion-resisting steel (bar and forging)	950	Reamed nozzle blocks
	QQ-S-164, class 416	l.2 Cr, corrosion-resisting steel (bar)	950	Reamed nozzle blocks
(l) Glasses, oil. sight flow indicator	MIL-G-2800	Borosiiicate gl.ass	400	50 psi max.
(m) Packing, labyrinth (gland or diaphragm) and stationary seal strips	QQ-S-763, class 410, (maximum hardness 200 BHN) condi-tion A	12 Cr corrosion-resisting steel (bar and forging)	Use only for 850-950	Seal strips only See note l
	QQ-S-764, class 416 (maximum (bar) hardness 200 BHN), condi-tion A	Cr corrosion-resisting steel	Use only for 850-950	Seal. strips only, See note 1
	MIL-C-l.5345, alloy no. 7	brass, 6 Pb 13 Ni-65 Cu (casting) Leaded-nickel.-	tt5U	Sana castingf are per-mitted,· See note l
	MIL-I-:l4l...l$_1$ class C	percent Ni ductile iron	1000	Centrirug-ally cast **only** Seenote 1

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Table I - Materials (cont'd).

Part A - Materials of principal parts.					
Part or service		Applicable documents	Material and properties	Temperature limit (OF.) (maximum) (see 3•2.2)	Remarks
lm)	Packing, labyrinth (gland or diaphragm) and stationary seal strips (cont'd):	QQ-C-390, alloy numbers os, El, E3, ES, E6₁ E7 and E8	Copper al.l.Oy, (castings)	!>!>O	See note l.
(n)	Packing springs (flat and coil)	ASTM A434, Ni-Cr-Fe alloy	0.08 carbon (max.) 1.00-1.os Mo 1.00-2.00 Mn 1.75-2.25 Ti 0.40-1.00 Silicon :0.10-0.s Vanadium jo.030 (max.) Sulfu1 1·0.35 (max.) Al 0.040 (max.) p 0.001-0.01 Boron 13.5-16.0 Cr 24.0-27.0 Ni balance iron	1000	See 3.2.4
		JAN-W-562, class l or 2	74 Ni-lB er-7 Fe, l Cb (wire)	1050	
		MIL-N-7786,	74 Ni-15 Cr-7 Fe, l Cb alloy (sheet and strip)	10::>U	
(o)	Pins (for notch block and blocking blades)	MIL-S-1222, type I, symbol B16	cr-Mo-v alloy steel (bar)	1000	-----
(p)	Rotating elements, reduction gear, main				
	ll) Pinion	MIL-S-l.9434, classes 2 through 6	Al.l.oy steel. (forgings)	---	With or without nitridina
		MIL-S-006₁ classes 4615 and 8615	Steel bars and billets (for carburizina)	---	carburizea only
		MIL-S-l.6974, classes 414.0 and 4340	All.oy steel. bars, billets, blooms and slabs	---	with or without nitriding
		QQ-S-b:l4₁ classes 4140 and 4340	Al.l.oy steel. bars	---	----
	(2) Gear				
	a. Solid forging (including shaft)	MIL-S-19434, classes 2 through 5	Alloy steel. (forgings)	---	with or without nitriding
		MIL-S-oob, classes 4615 and 8615	Steel. bars and billets for carburizinct	---	carburizect only
	b. Ring (or rim)	UL-S-l.9434₁ classes 2 through 5	Alloy steel (forgings)	---	with or without nitridinq

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Table I - Materials (cont'd).

Part A - Materials of principal parts.				
Part or service	Applicable documents	Material and properties	Temperature limit (op•) (maximum) (see 3,2,2)	Remarks
(p) Rotating elements, reduction gear, main				
(2) Gear(cont.dlc. HUQ	MIL-S-24093	Carbon and alloy steel (forgings)	---	For welded construction
	MIL-S-:lJ:l84	carbon anct al.Loy steel (forgings)		
	MIL-S-15083, grade B	Steel castings		
	QQ-::;-b!I.L, classes A and B	caroon steel (boiler plate)		
ct, wens (side plates)	QQ-S-0:1.L, classes A and B	carbon steeJ. (boiler plate)	---	For weJ.ctea construction
	MIL-S-.l.lb!ll:l1 type I, classes A, B, and C	l:arbon steel (plate)		
e. TUbes (stiffeners, web plates)	QQ-S-o:H, classes A and B	Carbon stee.L (boiler plate)	---	For we.Lctect constructiot
	MIL-S-:l:l698, type I, classes A, B and C	Carbon stee.L (plate)		
	MIL-'.L'-..:u.L!> I	Tune ana pipe, carbon steel		
(J) snart	MIL-S-:l409J	Carbon and al.Loy steel (forgings)	---	-----
	M.LL-S-23..:84	Carbon and alloy steel (forgings)	---	-----
(q) Rotating elements, accessory drive				
(l) Pinion tor w rm) for worm or crossed helical gear drives or pinion and gear for spur or helical gear drives	MIL-S-J.94J4, classes 2 through 6	Carbon and alloy steel (forgings)	---	See 3,4.49 Shall be nitrided
	MIL-S-uob, classes 4615 and 8615	Stee.L bars and billets (for carburizing)	---	Carburized only
	MIL-S-J.b9 ,,.,, classes **4140** and 4340	AJ..Loy steeJ. bars, billets, blooms and slabs	---	snaJ.J. be nitrided
	QQ-S-624, classes 4140 and 4340	Alloy steel bars	---	sna.L.L be nitrided
(2) Gear, driven for worm or crossed helical drives	Cornrnerc1a.L	Bronze	---	-----
(JI Pinion and gear, spiral bevel	**I'U.L-S-800,** classes 4615 and 8615	SteeJ. oars and billets (for carburizing)	---	Carburized only·

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Table I - Materials (cont'd).

Part A - Materials of principal parts.

Part or service	Applicable documents	Material and properties	Temperature limit (oF.) (maximum) (see 3.2.2)	Remarks
(r) Rotors and shafts (turbine)	MIL-S-860, grade B	Ni-Mo-V steel (forgings)	750	See note 5 also see 3.2.7(a) and (b), and 3.4.45
	MIL-S-H60, grade	Ni-Cr-Mo-V alloy steel (forging)	7::,0	
	MIL-S-HtJO, grade G	12 cnrome forgings	·1::,u	
	MIL-S-860, grade F	cr-Mo-V-a.Lloy steel (forging)	1050	
(S) Shrouding for blading	MIL-S-861, class 403, condition HT	12 Cr corrosion-resisting steel (strip and bar)	!:IUU	.UlU aegree bend test required
	MIL-S-861, class 410, condition AN or HT	12 Cr corrosion resisting steel (strip and bar)	900	
	MIL-S-Ho.L, class 422, condition HT	ll Cr corrosion-resisting steel (strip and bar)	.LOOU	
	Conunercia.L	Titanium and titanium alloys (bars, forgings and forging stock)	As approved	Wnen approved(by NAVSEC on a case basis
(t) Supports, flexible (for casing)	QQ-S-691, all classes	Carbon steel and allov grades	To suit service	See note 3
	MIL-S-l4.l.LJ	Carbon manganese steel plate	To suit service	See note 3
	MIL-S-162J.6	HY-80	To suit service	-----
(U) Valves {.LJ control (pop, pets and seats)	Disc, MIL-S-l.H410, class a	l-l./4 Cr-l/2 Mo (bar and forging)	l.OoO	see 3.4.52,ll
	QQ-S-763, class 405 or 410, condition A or T	12 Cr corrosion-resisting steel (bar and forging)	950	
	MIL-S-Hb.L, class 422 condition HT	.Ll Cr corrosion-resisting steel (bar and forging)	.LOOO	
	MIL-S-.LH410, class b	2-1/4 Cr-1 Mo (bar and forging)	1050	
2) Lift· rods} and valv:I stems	QQ-S-763, class 410 with or without nitriding, condition T	12 Cr corrosion-resisting steel (bar and forging)	9::,0	See J,4.!>2,l;
	QQ-S-704, class 416, with or without nitriding condition T	12 Cr corrosion resisting steel (bar)	9..,0	
	MIL-S-86.L, class 422, condition HT	l.l Cr corrosion-resisting steel (bar and forging)	.LOVV	
	MIL-S-869, class A or B	Alloy steel for nitriding	950 and H50 respectively	VDPH HOU (minimum)

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Table I - Materials (cont'd).

Part A - Materials of principal parts.

Part or service		Applicable documents	Materia1 and properties	Temperature limit (°F.) (maximum) (see 3.2.2)	Remarks
(u) Valves control (cont'd)					
	(3) Bush-, ings .	QQ-S-691, class B	Carbon steel (late)	800	See 3•4.52•l
		MIL-S-18410, class a	1- er- 2 Mo (for in)	05	
		MIL-S- , class 410 1 condition AN	2 Cr corrosion-resisting steel (stri and bar)	950	
		MIL-S- , class A or B	A oy steel for nitridin g	950 and 850 respectivel	VDPH (minimum)
		QQ-S- 6 , class 440A, 440B or 440C with or without nitridin g	6- Cr corrosion7 resisting steel , (bar and forging)	1000	
	Seat-ing sur-face	MIL-R-17131, type MIL-RCoCr-A	Rod, welding	10 0	See • •

Part B -- Piping

(v) Piping, (main steam)	MIL-STD-438	As required by referenced standard	As applicable	Submarines
	MIL-STD-117	As required by referenced standard	As applicable	surrace ships

Part C - Oils and greases

(w) Oil	(l) Lubrica-ting	MIL-L-17331, Navy symbol 2190-TEP	Mineral oil	---	Drying agents will be per-rnitted in the oil where the flange tern-perature does not exceed 450°F.
	(2) Flange coating (for making up casing flanges)	TT-L-201, type II, viscosity Z-7, Z-8 or Z-9	Bodied linseed oil		
IX) Grease, lubricating		MIL-L-15719	--------	300	-----

l'lOTES to taole I, parts A, Band C:

1. Fracture and other physical tests are waived; a visual examination is substituted.
2. General purpose use bearings are limited to applications of 200°F.; applications above 200°F. shall utilize bearings heat-stabilized for expected temperature.
3. Nil-ductility transition temperature (NDTT) tests are required only for carbon steels, with less than 1 percent (nominal) of any one alloy, thijt are used for structural members. NDTT shall not exceed plus l0°F. as determined by the method described in ASTM E208. This requirement does not apply to plate thickness less than 5/8 inch or greater than 4 inches. Subject to the Naval Ship Engineering Center (NAVSEC) approval, the use of impact tests may be substituted where there is sufficient statistical data to show correlation between nil-ductility properties and impact values.
4. The 180 degree cold bend test is waived.

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3.2.1 Associated equipment specifications. Documents governing selection of material for associated equipment shall be as follows:

	Equipment or accessory	Applicable document
(a)	controllers {for oil pump motor)	MIL-C-2212
(b)	Coolers (for lubricating oil)	MIL-C-15730
(c)	Couplings (flexible)	MIL-C-23233
(d)	Fittings, lubrication .	MIL-F-3541
(e)	Gages, pressure (for oil or steam)	MIL-I-18997
{f)	Generator sets, steam turbine	MIL-G-3087
(g)	Generator sets, steam turbine, direct drive	MIL-G-19650
(h)	Indicators, pressure (see gages)	
(i)	Indicators, temperature (see thermometers)	
(j)	Motors (for oil pumps)	MIL-M-17060
(k)	Motors (for speed changer)	MIL-M-17059
(1)	Precipitators	MIL-P-23917
(m)	Pumps, rotary (for motor driven oil pump)	MIL-P-19131
(n)	strainers (for oil system)	MIL-S-1784.9
(o)	Strainers (for main steam)	MIL-S-21427
(p)	Tachometers	MIL-T-16049
(q)	Thermometers	IL-I-17244 IL-T-19646
(r)	Valves, relief	IL-V-20065

3.2.2 Maximum temperatures. Table I specified maximum temperature limits for approved materials used in major parts of turbines and gears. The average temperature to which these parts are subjected over any extended operating period shall not exceed the specified limit. The maximum temperature to which parts are subjeceed shall not exceed the temperature limit by more than 15°F., except for short swings. "Sho t swings" are temperature swings during abnormal or unusual operating conditions, during which the temperature shall not exceed:

(a) Design temperature plus 25°F. for more than 5 percent of the total operating life .
(b) Design temperature plus 50°F. for swi!)gS of 15 minutes duration or less, aggregating not more than l percent of the total operating life.

3.2.3 Chemical composition. The chemical composition of the approved materials shown in table I, may he changed to permit use at higher temperatures. In such cases, complete examination and testing shall be performed in.accordance with the basic specification. Approval shall be obtained as required for substitute materials (see 3.2.8).

3.2.4 Packing sprin'ls• Packing springs, in accordance with table I, shall be drawn without the use of metallic lubricant.

3.2.5 Stress criteria and allowable stresses.

3.2.5.1 Stress criteria. Maximum allowable stresses for the specified operating conditions shall be based on whichever of the following is limiting. Centrifugal stresses shall be based on the rated speed.

(a) Yield strength (at maximum temperature with offset of applicab;e material specification).
(b) Stress rupture (minimum of 40,000 hours).
(c) Creep (O.l percent) for life specified (see 3.4.5) wher temperature exceeds 800°F. (Creep is not applicable to specified factors of safety.)

3.2.5.2 Blading. Centrifugal stress (centrifugal load divided by full cross-section area) in the most highly-stressed row shall not exceed stress levels corresponding to·a minimum factor of safety of two, 'except that a factor of safety of 1.5 is allowed for shrouding. Vibratory stress datai (see table VIII) shall be submitted to NAVSEC for approval. The manufacturer's acceptance cri;teria shall.also be submitted.

3.2.5.3 Bolting. Turbine Qa;_sing and steam chest bolting shall be designed to **provide** leak-tight joints for an initial_ eriod (followi g casing assembly) of **at least** 20,000 **hours** and, after retightening, for an a;dditional period of 20,000 to 30,000 hours before again retightening based on relaxation strength of the material at maximum temperature involved.

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3.2.5.4 Turbine rotors. Design primary stresses shall not exceed stress levels corresponding to the following minimum factors of safety:

(a) centrifugal - 2.0
(b) Average tangential - 2.5
(c) Torsional - S.O

3.2.6 Cast iron, Except as noted below and where specifically approved by NAVSEC, the use of cast Irori parts shall not be permitted. Nodular iron, gray iron and close-grain semi-steel are, for the purposes of this specification considered to be cast iron Nodular (ductile) iron conforming to MIL-I-24137, class A, is permitted for governor housings, governor servo housings, valve operator housings and lube oil pump casings, providing they meet the shock requirements (see 3.4.10).

3.2.7 Special material re uirements. Special material requirements for turbines operating on saturated steam are as follows:

(a) solid rotors, where wheels and shaft are forged integral, shall be made of a 12 percent chrome steel alloy material approved by.NAVSEC. The rotor journals or surfaces in way of bearings or babbitted sealing surfaces shall be chrome plated, or surface covered by a process to be approved by NAVSEC. The quality assurance procedures of MIL-S-860 will apply.
(b) On single stage turbines where the shaft and wheel are fabricated separately and assembled,- the wheel shall be made of a 12 percent chrome steel alloy material approved by NAVSEC and the shaft of a material as shown in table I, part A.
(c) Turbine casings including exhaust end casings, steam chest covers, packing boxes, and packing ring holders (inner housing) shall be 12 percent chrome alloy steel in accordance with the material requirements of MIL-S-16993 **or** QQ-S-766 or as approved by NAVSEC. For turbines of 3000 KW or larger, where size may preclude the use of 12 percent chrome, the exhaust casings may be or carbon steel suitably protected from erosion/corrosion by the use of inlays or inserts.
(d) Diaphragms shall be fabricated of 12 percent chrome alloy steel using NAVSEC approved welding procedures.
(e) Nozzle plates or blocks shall be fabricated of 12 percent chrome alloy steel using NAVSEC approved material and welding procedure.
(f) Internal parts of turbine such as diaphragm support lugs, gland and interstage packing antirotation pins, crush pins, internal moisture collectors, drain orifices, steam windage shields, valve lift beam, lift rods and all threaded fasteners such as nozzle block bolts shall be of corrosion/erosion resistant material.
(g) All external drain piping and manifolds from turbine (interstage, gland, etc.) shall be fabricated of corrosion/erosion resistant material.
(h) Valve seat expansion lips or rings used for absorbing expansion between seat and body shall be constructed of nickel-chromium-iron alloy (73-15-7 nominal).

3.2.8 Sulistitute materials. The use of materials (either additional Government specifications or company materials in lieu of specified Government specifications) differing from those specified in table I will be permitted when all of the following conditions are met:

(a) The use of other materials represents no cost increase to the Navy.
(b) Minimum factors of safety can be met with material in accordance with the applicable Government specification.
(c) List of preferred materials has been approved prior to the contract. (The date of the NAVSEC approval letter must be within 2 years of the contract date.)

J.2.a.1 Approval procedure for substitute materials. The manufacturer shall submit a "List of (Company Name) Preferred Materials for Generator Steam Turbines and Gears" (see 3.2.8.l.l) to NAVSEC for approval•. A maximum of 6 months from the date of submittal shall be alloweQ for NAVSEC approval. Once the list is approved, it shall be reviewed at least every 2 years for currency with the referenced Government specifications. NAVSEC will forward a list of the specifications which have been revised for use in conducting the review. The revised list and related comparison sheets shall be submitted to NAVSEC for approval. If necessary a conference may be held for this reapproval. Changes to the manufacturer's materials require approval when they occur if revisions involve physicals, chemistry, processing or relaxation of quality control. Revisions to the List of.Preferred Materials are not retro-active unless mutually agreed upon. As a backup to the List of

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Preferred Materials, material comparison sheets (see 3.2.8.1.2) shall be submitted in order to evaluate the acceptability of the proposed substitute material. Where it is not obvious, the forwarding letter shall indicate the reasons for departures from the quality control requirements of the applicable Government specification. Approval of a comparison sheet constitutes approval to substitute that material only for the applications shown on the List of Preferred Materials.

3.2.8.1. 1 List of preferred material. The List of Preferred Material shall be a drawing similar in formatto the following:

Part or service	Government specification	Company spec.	ASTM spec.	Temperature limit	Remarks	NAVSEC approval letter
(List in same order as table I)			(if applicable)	(Shall not exceed limit of table I)		(Serial nr and date)

Only the Government specifications which are used or substituted for by the company need be shown. Material comparison sheets shall be shown in the remarks column unless material identification number is identical. If the company specification is identical in every respect to the Government specification, such may be noted in remarks column and comparison sheets are not required.

3.2.8.1.2 Material comparison sheets. For each substitute material which is different from the applicable government specification, the manufacturer shall document on a separate sheet of paper a complete comparison in any of the following areas where any difference exists between the substitute material and the specified material:

(a) Chemical composition.
(b) Mechanical or physical properties.

Only significant differences between the substitute material and the specified material require documentation for the following areas1 however, all instances where a requirement of the specified materials is not reflected in the_substitute specification shall be included:

(c) Processing (heat treatment, and so forth).
(d) Examination, tests and quality control provisions.

If the substitute material is identical in all respects with the specified material in any of the above areas, a statement shall be made to that effect in lieu of a comparison; the statement "equal or better[11]·is not acceptable. Each of the above documentations shall constitute one.or more sheets in a multi-sheet drawing, properly indexed and titled. The issue of the Government specification for which comparison is being made shall be shown. If there is no applicable Government specification, the comparison sheet shall provide a comparison with an ASTM specification if applicable; if not, submit information covering only the company material. Each sheet shall be dated when issued and when revised. When a comparison sheet has been approved, it shall be resubmitted for information with NAVSEC approval lette noted thereon.

3.2.8.2 Approval of non-specified materials. Materials for parts not specified in table I shall be self approved by the manufacturer.

3.3 Equipment required.

3.3.l Turbines and gears. Turbines and gears shall be furnished in the quantity, in accordance with the requirements herein and as specified (see 6,2.2(d)).

3.3.l.l Turbine operating steam conditions.

3.3.1.1.1 Normal conditions. Normal (rated) steam inlet pressure and temperature (or percent moisture for class A and B turbines), at the throttle valve inlet and normal (rated) exhaust pressure shall be as specified (see 6.2_2(e))•

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3.3.l.l.2 Maximum and minimum conditions. Maximum and minimum steam inlet pressure and temperature (or percent moisture), at the throttle valve inlet and maximum and minimum exhaust pressure shall be as specified (see 6.2.2(f)).

3.3.2 Additional equipment,. fittings and accessories. Additional equipment, fittings and accessories shall be furnished with the"t::urbines and gears as necessary to meet the requirements herein. The items to be furnished shall include, but not be limited, to those listed in table II, unless otherwise specified (see 6.2.2(g)).

Table II - Additional equipment, fittings and accessories.

Item	Turbine type and class		
y = Turbine manufacturer is required to furnish N = Turbine manufacturer is not required to furnish (N items will be shipbuilder furnished unless otherwise specified).	Type I Classes B, C and D	Type II Classes A	Type II Classes B, C and D
Back oressure trip system (see 3.4.30.1)	y	y	y
Bedolates lsee MIL-G-3087 or MIL-G-19650)	y	y	y
Control systems (see 3.4.52), including:			
Trip throttle valve	y	y	y
Constant speed governing svstem, including:	y	y	y
Speed governor (mechanical/hydraulic) (see note l)	y	y	y
Speed-changer (see note 1)	y	y	y
Speed changer motor (see note 1)	y	y	y
Speed changer stops (see note 1)	y	y	y
Speed control mechanisms	y	y	y
Valve operator	y	y	y
Nozzle control valves	y	y	y
Emergencv control svstem, including:	y	y	y
Overspeed and manual trip system	y	y	y
Orifices	y	y	y
Couplings (see 3.4.39)	y	y	y
Drawings and microfilm (see 3.7 and 3.8)	y	y	y
Gage piping (see 3.4.32.3) and connections	y	y	y
Gears, reduction (where used) (see 3•4•47 and 3.4.48)	y	y	y
Gland seal and leak-off system (see 3.4.53)	y	y	y
Identification plates (see 3.4.34)	y	y	y
Instruments and special fittings (see 3.4.36), including:			
Gages, pressure, steam inlet and exhaust	y	N	y
Gages, pressure, steam, gland sealing	y	y	y
Gages, pressure, steam, first pressure stage of multiple stage turbines	y	y	y
Gages, pressure, oil	y	y	y
Gageboards	y	N	y
Thermometers	y	y	y
Speed indicators	y	y	y
Rotor position indicators	y	y	y
Grease fittings	y	y	y
Clips or fittings for holding insulation and lagging	.Y	y	y
Lubricating oil systems lsee 3.4.51), inciuding:			
Sumps	y	N	y
Attached pumps	y	N	y
Motor driven pumps with controller (750 KW and above)	y	N	y
Hand pumps (under 750 KW)	y	N	y
Strainers	y	N	y
Filters	y	y	y
Precipitators	y	N	y
Coolers	y	N	y
Relief valves	y	See note 2	y
Check valves	y	See note 2	y

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Table II - Additional equipment, fittings and accessories (cont'd).

Item	Turbine type and class		
y = Turbine manufacturer is required to furnish N = Turbine manufacturer is not required to furnish (N items will be shipbuilder furnished unless otherwise specified).	Type I Classes B, C and D	Tvoe II Classes A	Tvoe II Classes B, C and D
Luoricat1ng oi.1 systems (see 3.4.51), 1ncluding: (contd),			
Cooler bv-oass valves	y	N	y
Gages	y	See note 2	y
Alarm switches	y	See note 2	y
Sight :t.LOW :t1 tt1ngs	y	See note 2	y
Thermometers	y	See note	y
Orifices	y	see n0te 2	y
Relief valves, exhaust casing (under .50U0 KW)(see 3.4.JU. J	y	N	y
Repair parts and tool.a (see 3,5 and 3.6)	y	y	y
Rotor grounding device (see 3,4.36.17)	See note 3	See note 3	See note 3
RTE installation, complete except monitors (see 3.4.38.4)	y	y	y
RTE bearing monitors (see 3.4,38.4)	N	N	N
Rupture discs (3000 KW and larger)(see 3.4.30.3)	y	N	y
Steam strainers (see 3.4,31)	y	y	y
Technical data books (see 3,10)	y	y	y
Technical manuals (see 3,9)	y	y	y
Turning devices, manual (see 3,4.35,1)	y	y	y
Turning devices, power operated (for sets of 3000 KW and more)	N	y	N
Turning devices, power operated (for sets under 3000 KW)	N	y	N
Valves, casing relief (see 3.4.30.2)	y	N	y

Notes:

1. Speed governor, changer, motor and stops not required if electric_governors are specified (see MIL-G-3087),
2. Items upstream of shipbuilders connections will be shipbuilder furnished. Items downstream of shipbuilder connection shall be furnished by turbine manufacturer.
3. Rotor grounding devices are to be furnished only for class A and B turbines. The turbine manufacturer shall furnish the device.

3,4 Design concepts and criteria.

3.4.1 Reliability. The principle of reliability is paramount and no compromise of this principle shall be made with any other requirements. It is intended that all parts, except those parts classified as onboard repair parts, shall not be replaced or repaired during the specified life except for damage incurred due to external causes unrelated to design. Planned maintenance actions to be performed shall be in accordance with MIL-G-3087 with replacement or renewal of parts as found necessary. NAVSEC shall be notified if the proposed design requires planned maintenance actions different from MIL-G-3087. Turbine and gear casings will normally not be lifted for routine examination of internal parts unless there are indications of damage or deterioration of performance.

3.4.2 Standardization. Interchangeability of components and parts with units previously furnished is desired with particular reference to repair parts. The manufacturer shall notify NAVSEC when design changes are made which result in non-interchangeability with spare parts previously furnished.

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3.4.3 Accessibility. Design shall, within space limitations, provide the maximum acces--6ibility to turbine and gear parts which require routine examination, maintenance and repairs. It is intended that design provide for minimum effort required to accomplish planned maintenance actions and to effect repairs.

3.4.4 Cost. Cost of turbines and gears is to be considered of such importance that, where there Isa choice in design with each choice providing full reliability and meeting other criteria herein, the choice representing the least cost to the Navy for initial procurement and maintenance in service shall be exercised. It is imperative that design eliminate features which are unnecessary and are of marginal value.

3.4.5 Life. Turbine and gear design shall meet the life requirements of MIL-G-3087.

3.4.6 Margin. Horsepower rating of the turbines and gears shall be as specified in the contract or order. Extra capacity beyond the manufacturer's normal margin is not required.

3.4.7 Weight and (E)ce. When specific weight and space limitations exist, they shall be specified (see 6Y.2 and MIL-G-3087); if not specified, weight and space shall be kept to a minimum consistent with other requirements herein.

3.4.B Extraction, induction and reheat. Except for the introduction of steam into the blade.path of turbine casings frornexcess gland steam and high pressure valve stern leakoffs, no provision shall be made for extraction, induction or reheat.

3.4.9 Exceptions to specifications. Where the manufacturer believes that he can supply turbines and gears of acceptable quality by proposing a substitution for any given design requirement in this specification, he may do so, providing all of the following are satisfied;

(a) The substitution does not compromise reliability or other criteria herein.
(b) The substitution does not increase the contract cost or result in the subsequent higher cost of maintaining the equipment in service after acceptance by the Government.
(c) The substitution represents sound engineering approach to the original specification requirements based upon any of the following:
(1) Previous experience in service (Naval or comparable service).
(2) Test results acceptable to NAVSEC.

Request(s) for substitution may also be submitted after award of contract and are encouraged where either design improvements or cost reduc ions are involved.

3.4.10 Shock resistance. Turbines, gears and accessories shall be designed to meet the shock requiremeiits specified in MIL-G-3087 regardless of ship motion (see 3.4.13), generator load or speed up to and including rated load and speed.

3.4.10.1 Static design method. The static shock factor values for surface ships are as shown in MIL-G-3087. The shock factor values for submarines are as follows:

Vertical - 75 }
Athwartship - 75 at the mounting feet.
Fore and aft - 20

3.4.11 Vibration. Turbines and gears shall be design_ed to meet the vibration limits specified in MIL-G-3087 (see 3.4.27).

3.4.12 Stearn econorn . Stearn economy is a basic design criteria. Guaranteed rates and method of evaliiatI'on sfiai1 be as specified (see 6.2.2(i)).

3.4.13 Trim list roll and pithh. Turbines and gears shall be designed to operate satisfactorily wh n sJui>is"Tn ot er than a normal condition, as shown in table III (trim and list or roll and pitch can occur simultaneously).

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Table III - Trimr list, roll and pitch.

Condition	Class B, C, or D (surface ships)	Class A (submarines)
	Degrees	Degrees
ermanent trim (down, bow or stern)	5	30
ermanent list (port or starboard) l/	15	15
Roll (port or starboard from normal)-	45	60
!?itch (up or down from normal)Y	10	10

l/ Pitch and roll time cycle, if required shall be as specified (see 6.2.2(])).

3.4.14 design.

3.4.14.l Turbines. All generator-turbines shall be of the axial flow impulse type with "straight through" steam flow, (that is, there shall be only one inlet and one exhaust and the steam shall flow from the inlet to the exhaust without any complete reversals).

3.4.14.2 Reduction gears. Reduction gears shall be parallel-offset, either single or double reduction with either single helical or double helical (herringbone) teeth.

3.4.14.3 Rotation and gear offset. All turbines and gears furnished under the same contract or order shall have the same rotation and the same gear offset.

3.4.15 Short circuit torques. All turbines, gears and associated equipment for a.c. generators shall be capable of withstanding without damage a three-phase fully assymetrical short circuit (see MIL-G-3087).

3.4.16 Use of mercury. Classes A and B turbines and gears and associated equipment shall not containmercury in any form and devices employing mercury shall not be used in the manufacture and testing, where the possibility of contamination of the equipment by mercury exists.

3.4.17 Fire hazard. The designs shall be such as to minimize the possibility of fire hazard. The following are areas to be carefully considered.

(a) Oil leaks onto lagging or hot surfaces.
(b) Lube oil gage and piping failures under shock or vibration resulting in spraying oil onto hot surfaces.
(c) Location of lubricating oil strainers to be as remote as possible from surfaces normally at high temperature; and from the open end of the generator where oil fumes could be drawn into the windings.

3.4.18 Arrangement. Sets will be installed with shaft axes in a fore and aft direction and with the shaft axis essentially level when the ship is on an even keel.

3.4.19 Mounting and bracing. Turbines and gears shall not be braced, restrained or otherwise restricted, except by bolts, keys or dowels, securing them to their bedplates or supports.

3.4.20 Expansion and flexibility. Turbines shall include those supports which are necessary to permit free longitudinal and radial casing expansion to allow for rated power tempe ature.

3.4.21 Flexible connection between turbine and condenser. Where flexible connections are used between turbine and condenser, the turbi manufacturer shall furnish the flexible connections for type I turbines and the shipbuilder shall furnish flexible connections for type II turbines.

3.4.22 Bolts. The responsibility for drilling flanges and providing bolting shall be as indicated hereinafter. Whoever provides bolting is responsible for liaison with all others concerned to determine thickness of mating parts (and gaskets if used).

3.4.22.1 Bolts for securing to external systems. Bolts for securing external system connections sha'I'l'"be furnished andinstalled by the shipbuilder. Holes for these bolts shall be drilled by the turbine manufacturer, except that holes for fitted bolts will·be finish-reamed by shipbuilder at installation. A 1/16 inch stock allowance on the bolt hole diameter shall be made by the manufacturer for reaming by the shipbuilder.

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3.4.22,2 Bolts for flexible connection between turbine and condenser. Where turbine and condenser areconnected by a flexible connection, the bolting requirements shall be as specified in 3,4,43.9.3 through 3.4.43.9,3.2.

3.4.23 Alignment, The design shall be such that alignment will not be disturbed or undue stresses set up in any part by normal vibration or contraction and expansion of turbine, gear and piping attached thereto in service. Dowels, keys, rabbets or other positive locating means shall be used as necessary to assure that alignment will be maintained despite slight loosening of hold-down bolts under shock load conditions. The outline drawing shall show the limits of forces and moments that may be imposed by shipbuilder's piping.

3.4.24 Lifting, Means shall be provided for lifting all items weighing 35 pounds or more.

3.4.25 Marking of parts. The use of low stress die stamps for marking on low stressed surfaces is permitted; however, depth of penetration shall not exceed 0.020 inch except for radiographic location markings on castings which may be 1/32 inch in accordance with MIL-STD-271,

3.4.26 Clearance requirements. Design clearances shall be provided to prevent rubbing between stationary and moving parts during all specified operating conditions except that minor contact of packing rings with rotor and minor rubs of oil deflectors, bands and spill strips are acceptable.

3.4.26.1 Packing (dia hragm, gland and dummy) clearances. Radial clearances between packing rings and shaft sha1I (at installation) be not less than either 0,005 inch (tolerance included) or the clearance corresponding to 80 percent of "normal clearance" (clearance expected to exist after preliminary acceptance trials) whichever is larger.

3,4.26.2 Clearance measurement and adjustment. Provision shall be made to measure and adjust the axial position of each rotor; however, such adjustment shall not be possible when rotor is rotating.

3,4.27 Mechanical vibration and balancing.

3.4.27.l Allowable vibration. Unless special structureborne noise levels are specified (see 6.2.2(k)), the maximum allowable levels of forced vibration snall be as shown on figure l.

3,4.27,2 h p balance, Turbine rotors shall be dynamically balanced to meet the vibration limits of ,27,l with all rotating parts installed. Balancing shall be conducted at the maximum safe operating speed of the balancing machine or rated speed of the turbine whichever occurs first. For requirements for balancing gear rotating elements-see 3.4.48.12.

3.4.27,3 Special balance. When contract requires special efforts to obtain noise reduction, maximum efforts shall be made in the production phase to minimize the amount of corrections required during either spin testing or load balancing.

3.4.27,4 In-field balance, Provision shall be made for in-field balancing each turbine rotor without lifting the casing.

3.4.27.4.l Correction planes. A minimum of two correction planes shall be provided for each turbine rotor. Use of weights in in-field balance planes is to be limited, during factory balance, to 15 percent of the total correction available in each plane.

3,4,27,4.2 Access to correction planes. Access to correction planes shall be through either existing inspection openings, outer removable gland covers, bearing caps, coupling guards or special flanged access holes. Minimum effort to install weights is desired,

3.4.27 ,4,3 Balance weiahts. Balance weights shall be either threaded weights for tapped holes or weights secured in ovetail grooves. Provision shall be made to mechanically lock in the weights.

3,4.27,4.4 Phase angfe measurement. Provision shall be made for taking 'phase angle measurements during balancing. Special parts for so doing are not required to be furnished, Shafts shall be marked for use of a stroboscope, photocell or magnetic pick-up. The technical manual shall indicate where the shaft is marked.

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3.4.28 Threaded fasteners. Threaded fasteners shall conform to the specifications listed in table I for the applications indicated and as specified in 3.4.28.l through 3,4.28.1.4. The various types of screw threaded fasteners shall conform to ANSI standards. Threaded fasteners for all steam joints shall be installed using an anti-seize compound which is compatible with the base materials involved.

3.4.28.l Screw threads.

3.4.28,l.l Unified thread series. Screw threads, except as specified in 3.4,28.2.3, shall be of the um.fied thread sen.es in accordance with Handbook H28,

3.4.28.1.2 Coarse versus fine thread series. The coarse thread series shall be used unless the component design indicates a necessity for the use of the fine thread series•

. 3,4,28,1,3 Eight-thread series, For fasteners 1-'inch diameter and larger, the eight-thread series shall be used wherever practicable.

3,4.28,1.4 Preferred thread-series. In tne selection of special threads, preference shall be given to the use of the 20, 28, 36, 44 and 56 thread series.

3.4.28,2 Class of fit,

3.4.28,2.1 Class 2A-2B. Class 2A-2B fit shall be used for the major portion of inter-changeable screw thread fasteners.

3,4,28,2,2 Class 3A-3B, Class 3A-3B fit shall be limited to applications where the necessity for accuracy of lead and angle of thread can be justified,

3.4,28,2,3 Class - Class 5 interference fits shall not be used,

3.4.28,3 General rules for applications.

3.4.28,3,1 Regular versus series. Where necessary for purposes of bearing loads,-pressure tight flanges and other special applications, heavy series shall be used for 1 inch size and larger.

3,4.28,3,2 Thread engagement. Thread engagement for studs and cap screws shall be not less than the nominal diameter. Threads of nuts shall be fully engaged. The maximum pro-trusion of the fastener from the face of the nut shall not exceed approximately 1/2 of the full nominal thickness of the nut for sizes up to 1-1/2 inch diameter or 1/2 of the nut thickness for larger sizes.

3.4.28,3.3 Tapped holes. Tapped holes for stud-bolts and cap screws shall be bottom tapped if the thickness o17:ne remaining material is equal to or less than major diameter of the thread and in such instances they shall have full threads for the specified depth.

3.4,28,3,4 Fitted (body-bound) fastening. Bolts stressed in shear shall be fitted. Holes for body-bound fastenings shall be reamed with coupled parts in position, to dimensions that will assure a tight fit. Tolerances in table IV shall be used for body-bound bolts.

Table IV - Tolerances for body-bound fastenings.

Nominal bolt size (inches)	Tolerances in inches		
	Maximum clearance-body of bolt and hole (Plus)	Diameter of hole (Plus)	Body of bolt (Minus)
1/4 to 3/8, incl.	0,0015	0.0009	0,0006
7/16 to 11/16	.0017	.0010	.0007
3/4 to 1- 1/8	.0020	.0012	,0008
1-1/4 to 1- 7/8	.0026	.0016	.0010
2 to 3	.0030	.0018	.0012
3-1/4 to 4- 3/4	.0036	.0022	.0014
5 to 7, incl.	.0041	.002!>	,0016

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3,4.28,3,5 Locking devices, Nuts on moving parts, internal parts directly exposed to blade path, control mechanisms, and support structures shall be securely locked by means as approved by NAVSEC, Self-locking nuts conforming to MIL-N-25027 may be used. Locking wire shall not be used for securing nuts. Self-locking threaded fasteners shall not have 'been threaded on to an engaging part more than 5 times.

3.4.28,3.6 Internal bolting. The use of internal bolts and studs *in* way of steam path shall be avoided to the maximum extent practicable without unduly complicating the design. Where internal bolting must be removed to lift casing, a warning plate so stating shall be permanently attached to the casing and shall protrude beyond lagging.

3.4.28.3.7 _Acceptable of fasteners..Threaded fastener shall be thr ugh bolts or through studs (two-nut type), Where impracticable to comply with the foregoing, bottoming studs (one-nut type) or tap bolts may be used; however, tap bolts shall not be used for the following;

(a) Steam chest cover,
(b) Horizontal joints.

Bottoming studs shall be either class 2A or 3A fit. Details of bottoming studs shall be approved by NAVSEC. Class 5 interference fits shall not be used. Studs which cannot enter the casing if they come loose .may be treated with "Loctite" or equal for temperatures below 300°F,

3,4.28,3,8 Bolt heating. Highly stressed studs 2 inches in diameter and larger in non-gasketed steam joints tsuch as casing joints and steam chest cover-joint) shall be drilled to permit the use of bolt heaters to obtain required stud extension without undue slugging and overtorquing. The corresponding cover nuts shall also be drilled to permit the use of extensiometers to determi e stud extension.

3.4.28 ,3.9 Bolt heaters. Electrical bolt heaters and extensiometers shall be furnished for submarine machinery. Cable assembly shall be grounded. For surface ships, gas heaters may be used in lieu of electric heaters. Bolt heaters shall be designed with thermal capacity (based upon 15 minutes heating time per bolt) to produce required bolt extensions.

3.4.28.3.10 Stud removal. An internal socket-head machined in end of stud or square end (or hex end) shall be provided to facilitate removal of one-nut studs l inch in diameter or larger where used in non-gasketed steam joints. At the manufacturer's option, stud removers (collet-type) may be furnished in lieu of the internal socket-head recess or external wrenching flats on end of one-nut studs.

3.4.28.4 Torque. For critical applications such as steam joints and support structures, the threaded fasteners shall be tightened to specified tension by either elongation measurement or measuring flats. The turbine manufacturer shall establish these requirements and shall control adherence thereto during manufacture,

3.4,28.5 Pins and dowels. Tapered pins and dowels shall be secured from backing out by staking or other locking devices welding is permitted where the dowel or pin becomes a permanent part of the removable piece. Dowels and pins shall be provided with means of removal such as tapped holes, external wrenching or pulling heads, or shouldered sha ks.

3.4.29 Welding and allied processes. Welding and allied processes shall be in accordance with MIL-STD-278 and MIL-G-3087, Spray metallizing shall not be acceptable.

3,4.30 Protection from high back ressure. Each turbine exhaust casing (and condenser) shall be protected from overpressurizat1on in accordance with 3.4.30.l through 3.4.30,3.l.

3.4.30.1 Back pressure trip system. Each turbine shall be provided with a back pressure trip system.which.shall close the trip throttle valve in the event of high back pressure. The system shall be such that no damage will occur to the turbine exhaust casing or the condenser with the trip set at 5 pounds per square inch gage (p.s.i.g.). Means shall be provided for checking the back pressure trip.

3. .30,2 Relief valves, exhaust casing. Exhaust casing relief valves shall be supplied for classes B, C and D turbines under 3000 KW, The valve shall be sized to limit pressure inside the turbine exhaust casing to 30 pounds gage with the steam inlet fully opened and the exhaust blanked. Design shall conform to type I, composition D of MIL-V-20065, except that the performance requirements and relief valve characteristics of MIL-V-20065 do not apply. The valve shall be a gradual opening type, in lieu of the pop open type, and shall be set to lift at 10 p.s.i.g. and give full relief at 30 p.s.i.g. Calculations shall be submitted

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to show that the valve is sized to meet the above requirements. Sight glasses in exhaust casing relief valves shall be designed for the maximum condition of shock and temperature to which they may be subjected due to design or specification requirements, Flanges shall be in accordance with 3,4,32,

3,4.30,3 Rupture discs, Rupture discs shall be supplied for classes B, C and D turbines of 3000 KW and larger, They shall be set to rupture at 10 p.s.i,g, and shall be sized to limit the pressure in the turbine exhaust casing to 30 p.s.Lg. with the steam inlet fully opened and the exhaust blanked,

3,4.30,3,l Alternate to rupture disc, For classes B, C and D turbines of 3000 KW or larger, consideration will O given to""'\ise of a second back pressure trip system in lieu of the rupture disc, The trip system shall be set to trip at 10 p.s.i,g. and shall close the governing valves in the event the primary back pressure trip system (see 3,4.30.1) does not function. The system shall be of a type that can be tested and proven safe.

3,4,31 Steam strainers. A steam strainer shall be provided upstream of the turbine trip throttle valve, The strainer (all pipe sizes) shall be in accordance with MIL-S-21427 and shall be readily accessible for replacement, examination, and cleaning, and arranged so that all steam to the turbine shall pass through it, Wye type strainers.are preferred, but other types will be permitted where space is limited, Strainers shall have only one inlet and one outlet connection, The inlet connection shall be flanged. The outlet connection shall be either flanged or welded to the trip throttle valve- inlet connection. Strainers shall be composition A, B or C in accordance with MIL-S-21427 as required for the design steam conditions. The design shall be such that all steam will enter the inside of the strainer basket-, so that any foreign material intercepted by the strainer may be withdrawn with the basket, Flanges shall be in accordance with 3.4.32. Blowdown connections are required,

3,4.31.l Stearn strainer gaskets, Gaskets for steam strainer flanges shall be in accordance with MI°"'I:=G=2l032 or MIL-G-16265, as applicable.

3,4.32 Piping, valves and associated equipment.

3,4.32.l General requirements. The main steam inlet flange on the trip throttle valve and the main steam piping, if any, upstream of the trip throttle valve, furnished with the turbines shall be in accordance with MIL-STD-777 (surface ships) or MIL-STD-438 (submarines) as applicable, Spiral wound gaskets may be of the inner-and outer ring type. Other piping furnished with the turbines and gears shall be to ASTM standards with allowable stresses, flange rating and geometry as outlined in the applicable ANSI standards Piping shall be seamless except where such is not commercially available for large sizes, Gaskets to c mercial standards are acceptable except that joint design shall be such that replacement with gaskets to Government specifications can be made. Tapered pipe threads are not permitted,

3,4.32.2 Connections ,'!:2 shipbuilding systems.

3.4.32 ,2.1 Allowable reactions. Stearn and oil connections at turbine shall be such that unacceptable strains are not imposed on turbine by shipbuilder's piping. Data on allowable force and moment reactions which turbine can safely withstand shall be forwarded to the shipbuilder.

3.4.32.2.2 Flanged connections.· All steam pip ng connections to which shipbuilder connects shall be flanged and shall terminate at a point outside of turbine lagging line, All oil piping connections to which shipbuilder connects shall have male-female flanges conforming to ANSI standards, Mating flange, of a material comparable to the turbine flange, shall be furnished with the turbine. Bolting and gaskets will be shipbuilder furnished.

3.4.32.3. Gagi hiping and connections. Piping and connections for all gages shall be in accordance wt Drawing 8Io-T385850, MIL-F-21467 and the following:

(a} Pressure gage connections shall be in accordance with the standard seal requirements of MIL-F-21467.
(b) Only 1/4 inch outside diameter (o.d,) tUbe size shall be used.
(c) The pressure rating of fittings will be upgraded for 1/4 inch o.d, tUbe size and below from 3000 to 6000.pounds per square inch (p.s.i.) working pressure with a minimum test pressure of 10,000 p.s.i.
(d) On gage testers where the connections are 1/4 inch NPT, an adapter shall be furnished with **a 1/4** inch NPT connection on one end and a swivel type connection (similar to the swivel 90 degree elbow end, shown in MIL-F-21467) on the other end.

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(e) Gages shall have connections externally threaded for fittings.
(f) Pressure gages shall not be stem mounted.
(g) Mechanical bite type fittings may be used for flareless tubing.

3.t.32.3.1 Root valve. A root valve of light weight construction, shall be installed between each gage"""'aiid the piping system or component to which the gage is attached,

3.4.32.4 Other component piping and connections, Other component piping and connections may be welded, ITtake-down joint is not:required for maintenance and overhaul, Where take-down joint is required, it shall be flanged using male-female flanges or it shall have straight threads with O-rings. Mechanical bite-type flareless fluid fittings conforming to MIL-F-21467 may be used except that take-down joints shall have straight threads with o-rings. The wall thickness of tubing used with bite-type fittings shall be not less than 0,065 inch in sizes above 1/2 inch. Where flanges are used in oil piping, male-female flanges conforming to ANSI standards shall be used, Tapered pipe threads are not permitted, Steel socket welding fittings in accordance with ANSI Bl6.ll, schedule 40, are suitable for 50 p.s.i. self-contained lube oil systems. Face-feed silver-braze fittings in sizes 1/2 inch and above are not permitted for any components furnished. Only preinserted ring fittings shall be used in sizes 1/2 inch and above. Face-feed fittings may be used in sizes below 1/2 inch. Brazing shall be accomplished in accordance with NAVSHIPS 0900-001-7000.

3.4.33 &ipe . The use of tapered pipe plugs shall be limited to only essential applications ue totlieir susceptibility to coming adrift under shock, Where use is unavoidable, they shall be secured by tack or seal welding.

3.4,34 Identification plates, Metallic identification plates shall be attached to each turbine and gear, Identification plates shall be in accordance with MIL-G-3087 except that the information required is as shown on figures 2 and 3 for the turbines and gears respectively.

3,4,35 Turning device (acking gear). A turning device shall be furnished in accordance with 3,4.35.l or 3.4.35, , to prevent bowing of rotors when turbine is being secured or is at standby, on an operating condenser, with sealing steam on the turbine glands,

3.4,35.l Manual turning device. All turbines and gears, shall have provision for turning rotor, by hand, using either a ratchet turning tool, a bronze headed bar to fit between coupling bolts or a bar to fit in special drilled holes in. the shaft or other means as approved by NAVSEC,

3.4,35,2 Power-o erated turning devices, All class A turbines and all other turbines of 3000 KW or more sha11 be equipped with a power-operated turning device, A three-way valve or an interlock shall be furnished and installed such that the trip throttle valve cannot be opened when the turning gear is engaged and so the turning gear cannot be engaged unless the trip throttle valve is closed, The design shall be such that damage to the device or a personnel hazard will not result from momentary engagement of the device with the turbine rotor rotating or by driving the turbine rotor with steam while switching from one mode of operation to the other. For class A turbines, the shipbuilder will supply steam and control oil to the device as required by the design,

3.4.36 Instruments and special fittings.

3.4,36,l aghs (indicators). All required pressure gages shall be furnished as required by table II, an s all be in accordance with MIL-I-18997,

3,4.36,2 locations.Provision shall be made for pressure taps and piping leading to gages from tneroiiowing locations:

(a) Pressure gage - on the steam chest.
(b) Pressure gage - on first pressure stage in multiple stage turbines.
(c) Pressure or compound gage - on exhaust casing.
(d) Pressure gages as necessary for control of gland sealing steam. Gages shall be of the compound-retard type.
(e) Pressure gages as required by 3,4,51,19 for the lubricating oil system,

3.4.36,3 Gage • All pressure gages shall be 4-1/2 inches in diameter,

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3.4.36,4 Ga!eboard. The turbine manufacturer shall furnish a gageboard, for classes B, C and D turbines or mounting the gages for steam chest pressure, first stage pressure and exhaust pressure, Unless otherwise specified (see 6.2.2(e)), the gageboard shall be mounted on the turbine or gear so that gages are visible to the man operating the throttle valve. If the shipbuilder desire to mount the gageboard elsewhere, the shipbuilder will furnish the required connecting hardware to the gageboard. All other gages shall be mounted on individual or common gageboards or brackets, as desired. These gageboards or brackets shall be attached to the bedplate and shall also support the piping to the gages to prevent relative movement between the gage and piping due to shock or vibration, Gages shall not be stem mounted,

3.4.36,5 9age connections. The connections on the pressure gages shall be in accordance with the figure shown in MIL-F-21467 for standard seat. The hardness and the finish of the connections shall be in accordance with the requirements for hardness and finish specified in MIL-F-21467. Only the 1/4 inch o.d. tube size shall be used.

3.4.36,6 Thermometers. All required thermometers shall be furnished by the turbine manufacturer.

3.4.36.7 Thermometer locations. Thermometers shall be furnished at the oil. cooler inlet and outlet and in each Journal and thrust bearing drain line. All thermometers shall be located so that they are accessible and readable after installation onboard ship. Thermometers may be combined with sight flow fittings.

3.4.36,8 Thermometer range. Thermometers shall have a range of 30° to 240°F,·

3.4.36,9 Thermometers for classes.A·and B turbines and gea . Thermometers for classes A and B turbines and gears sliiITl be in accordance with MIL-I-172 44- or MIL-T-19646.

3.4.36,10 Thermometers for classes C and D turbines and , s. Thermometers for classes C and D turbines and gears shall be in accordance with MIL-I-1 244 orMIL-T-19646.

3,4.36.11 Speed indicators. Permanently installed speed indicators.shall be pl'.ovided on all turbines for driving d.c. generators, so located that they may be readily observed1 in case it is impracticable to mount tbese indicators on the turbines, mounting on th driven auxiliary will be permitted. For a.c. generators a speed indicating device conforming to type IC/EFC or IC/EFD of MIL-T-16049 shall be provided and a 4 to 4-1/2 inch diameter indicating instrument shall be furnished. The meter shall cover a range of speed from zero rpm to 20 percent above rated speed. System inaccuracy shall not exceed 1-1/4 percent in the range of 5 percent below rated speed to 15 percent above rated speed. At all other speeds the system inaccuracy shall not exceed 5 percent. For class A turbines the instrumenb shall be mounted on the shipbuilder furnished gageboard. For class B, C and D turbines the instrument shall be mounted on the gageboard furnished by the turbine manufacturer.

3,4.36,12 Hand tachometers. Hand tachometers are not required but provision shall be made for the use a portable hand tachometer to set overspeed trips.

3,4.36,13 Rotor position indicator. A rotor position indicator shall be permanently installed to permit determination o·f axial blade clearances while the turbine is in·operation. It shall also be capable of determining total rotor movement (float) forward and aft with thrust shoes removed.

3,4.36.14 Bearing RTE, An RTE system shall be furnished (see MIL-G-3087, MIL-G-19650 and 3.4.38,4).

3.4.36.15 Nozzle bowl pressure connection. Nozzle bowl pressure connections are not required1 however, 1f tneturbine manufacturer does provide such connections, they shall either be blank flanged or plugged with a pipe plug which is seal welded.

3.4.36.16 Grease fittin?S• Grease fittings conforming to type III of MIL-F-3S41 shall be provided where grease lubrication is required.

3.4.36 ,17 Grounding device. Provision shall be made on class A and B turbines for grounding turbine rotors to casings to eliminate any electrical potential difference between journals and bearings. Grounding devi·ces shall be furnished by the turbine manufacturer.

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3.4.37 Thermal insulation and la!ging. The shipbuilder will be responsible for furnishing and installing therrnarTnsu ation and lagging. The turbine manufacturer shall dvise the lead (design) shipyard of maximum casing temperatures. The lead shipyard will inform the turbine manufacturer of the requirements for attachments on casing. Such attachments shall be furnished by the turbine manufacturer and shall not involve welding after final heat treatment of the casings, except for carbon steel casings or where the manufacturer elects to weld large nickel-chromium-iron (73-15-7 nominal) "buttons" on alloy casings before heat treatment. The wires or hooks can then be welded on the "buttons" at any convenient time. The turbine manufacturer shall also advise the lead shipyard of any areas of insulation that should be shielded to prevent oil soaking with consequent risk of fire. The type, size and location of attachment devices shall be such as to permit removal of insulation covering examination openings and flanged joints,

3.4.38 Bearings, journal and thrust.

3.4.38.1 Materials. Materials of all bearing parts shall be in accordance with table I.

3.4.38,2 Journal bearing, Bearings for turbine and gear main rotating elements shall be of the plain cylindrical, elliptical, multi-lobed, tilting shoe (or pad) or pivoted shoe (or pad) type, Axial or circumferential grooving is permitted,

3,4,38,2,l Loading, Bearing load based on projected area shall be not greater than 250 p.s.i. under static load, Projected area is defined as the product of the nominal diameter and length.of the journal bearing.

3,4.38,2,2 Wear pattern. Load shall be distributed so as to produce an essentially rectangular wear pattern extending throughout the length of the bearing, Scraping following initial operation to obtain uniform patterns is permissible but shall not increase the original diametral clearance by more than 15 percent of the nominal clearance for the turbine bearings and 0,002 inch deviation for pinion and gear bearings. All contact shall be only in the middle 120 degrees of arc in the lower half of turbine bearings or in the loaded half of pinion and gear bearings.

3.4.38 2,3 Clearance. Nominal design diametral clearance shall be not less than 1.5 mils per inch of journal diameter for.turbine and pinion bearings or not less than l mil per inch of journal diameter for gear bearings.

3,4,38,2,4 Interchangeability, Maximum interchangeability (consistent with design requirements) is required among bearings. Doweling shall be provided when necessary to prevent inadvertent assembly of a bearing in a location or position for which it was not intended,

3.4.38.2.5 stiffness. Bearing shells shall be stiff to prevent warping in service.

3.4,38,2,6 Rolling out bearing. It shall be possible to remove each lower-half bearing without removing the rotor.

3,4,38.2,7 Lifting and handling bearings. Provision shall be made for removal, lifting and handling of bearings 'a's°follows:

(a) Cfp. Each bearing cap shall either have integral lifting lugs or be tapped r lifting eyes which shall be so located as to prevent cocking when lifting the cap,

(b) Bearina shells, Each upper half-shell weighing more than 35 pounds shall be tappe for one of more lifting-eyes, In addition, regardless of weight, each lower half-shell shall be tapped to facilitate removal, when rolling out around journal,

3.4.38.2,8 Reborin* directions. Each rebabbittable spherically-seated bearing and each sleeve bearing whic is not bored concentric with the outer surface of the shell shall be provided with a concentric reference shoulder at each end for reboring. Unless otherwise marked on the bearing, the outside diameter of the shell will be used as the reference shoulder for boring.

3.4.38,2.9 sup ort. Each bearing shall be supported by and contained in a housing consisting of a bearing pedestal and cover. Journal and thrust bearing shall be contained in a common housing. Bearings may be mounted in a fixed position or mounted to provide for self alignment by using a spherical seat or flexible supports. Spherical seats may be either loose or tight according to manufacturer's standard practice, If a loose spherical seat is employed, adequacy of design with respect to fretting damage and tolerance to dirt shall be established.

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3.4.38.2.10 TM for vibration pickup. Each turbine bearing cap shall be provided with a boss for mounting a vibration pickup (used in conjunction with at-sea vibration analysis and shop tests). The boss shall be approximately a 2 inch square located at or near the vertical centerline of the cap with a 1/4 inch 28 thread hole 1/2 inch deep in the center of the boss. Mounting surfaces shall be machined flat and smooth to 125 RHR. If a loose spherical seat is used, the 1/4 - 28 tapped hole shall be located in the body of the ball with a clearance hole 1/2 inch in diameter through the bearing cap above it.

3.4.38.2.11 Bearing surfaces,

3.4.38.2.11.l Babbitt lining. All friction or rubbing surfaces shall be lined with anti-friction metal in accordance with table I. Babbitt thickness shall be not more than 3/16 inch.

3.4.38.2.11.2 Babbitting rocedures. Procedure for babbitting bearings shall be in accordance with NAVSHIPS 0283-2 8-1000 except that bond tests are not required.

3.4.38,2.11.3 Surface finish. Babbitt surfaces shall be finished to 32 RHR or less prior to scraping or initial operation,

3.4.38.2.11.4 Edg 7relief and chamfer. The babbitt shall be relieved along the joint at bore to prevent babbitt carryover and scoring of journals.

3.4.38,2.11.5 End flow control. End leakage of oil from multi-lobe fix d-pad, tilting pad and pivoted-pad bearings shall be controlled by a circumferential dam or seal- strip at each end of bearing. Circumferential dra_in groove(s) shall be provided at one or both ends of other type bearings, with drain areas equal to or greater than end leakage areas.

3.4.38.2.12 Depth micrometer measurements. Provision shall be made to permit the use of micrometer depth gages for measuring wear of all turbine rotor journal bearin_gs without removal of bearing cap. Holes for th!=! depth gage shall be located such that the.i1"'._are not subjected to lube oil pressure during operation. Collar-type plugs shall be used to close the holes and shall be attached to the bearing cover by a short length of flexible chain to prevent loss when removed. The depth micrometer graduations shall increase as spindle._is screwed down. Depth gage constants shall be stamped on small plates attached to each bear- ing cap by the manufacturer.

3.4.38.2.13 Rotor radial position setting. Where adjustable seating shims or pads are used for fixed-position mounting of bearings, means for determining the- radial position pf the rotor shall be provided.

3.4.38.2.14 Crown thickness measurement. Provision shall be made for measuring wear by the crown thickiies's""method, as specified in 3.4.38.2.14.1 through 3.4.38.2.14.6.

3.4.38.2.14.1 Scribe lines on cylindrically-bored sleeve bearing. At each end of the bearing, the unloaded half snaTr have a radial scribe line at the vertical center line, and the loaded half shall have, at each end, three radial scribe lines (the central scribe at the vertical center, and one on each side at an angle approximately 45 degrees). Both ends of bearing shall have an outside diameter machined to provide a reference surface for the micrometer.

3.4.38.2.14.2 Scribe lines on multi-lobed bearing types. A radial line shall be scribed on eacl! end of the bearing shelT at the geometric centerciT'each bearing surface arc whose center is displaced from the bearing center,

3.4.38.2.14.3 Scribe lines on fixed-pad bearings. A radial line shall be scribed on each end of the bearing she!'rat tne geometric center of the concentric arc of each pad.

3.4.38.2.14.4 Scribe lines tilting£!:_ pivoted-pad (or shoe) _bearings A radi l line shall be scribed on each end of each tilting or pivoted-pad or shoe in line with the tulcrum or pivot point on the back of the pad or shoe.

3.4.38.2.14.5 Point of measurement. The crown thickness of each sleeve type bearing shell and each pivoted shoeat the scribe points shall be measured from the end of the shell or shoe during assembly at a distance established by the manufacturer and shown in the manual and on applicable drawings.

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3.4.38,2.14.6 Marking of constants,. Crown thickness constants shall be stamped adjacent to scribed lines at both endsof bearing, For spare bearings, constants shall be stamped at time of installation,

3.4,38,3 Thrust bearin s. Thrust bearings shall be in accordance with materials specified in table I and the requirements of 3,4.38,3,1 through 3,4,38,3,9,

3.4,38,3,1 of thrust bearing. Turbine and pinion thrust bearings shall be of the tilting or pivoted segmental-shoe automatic load-equalizing type, designed to take thrust in both directions by locating equal size thrust elements on each side of a single thrust collar. Thrust shoes shall not be attached to, or derive support from, the journal bearing. Sliding contact between leveling plates and base ring is prohibited,,

3,4,38,3,2 Load. The thrust bearing shall be capable of absorbing any thrust loads which may be developed during specified operation. In addition, the thrust bearing shall be sized to include ample capacity for absorbing loads corresponding to 25 percent (minimum) of the tangential driving force in the flexible coupling connecting turbines and gear. Unit design loading, including the coupling force above, but excluding transients shall not exceed 500 p.s.i. under normal steady state condition. The 500 p.s.i. loading shall be based upon net pad area of the bearing with no area subtracted for chamfers and radii on the edges of the shoes.

3,4.38,3.3 Housin . The thrust bearing housing shall be integral with the journal bearing pedestal. Housing shall be rigid to maintain axial clearances *in* turbine under load. A 1/8 to 3/16 inch permanently-open drain shall be provided from the lowest practicable part of the thrust cavity to ensure drainage at shutdown.

3,4.38,3.4 Shims, Liners or shims shall be fitted to permit adjustment of thrust clearance and of axial clearances within turbine. Stacking of shims is not permitted.

3,4.38,3•.5 Colla:i;, A removable thrust collar keyed to shaft shall be provided, with a sliding fit (0.002 +0,001 inch loose) on the rotor, with a nut to secure the collar against the rotor shoulder. Room shall be incorporated in the design to permit slugging tight the securing nut, which.shall have a right-hand thread, regardless of direction of rotor rotation. A positive means of locking thrust nut shall be provided; the use of a radial set screw is not acceptable. A thrust collar integral with a removable stub shaft is acceptable, Consideration will be given to a thrust collar made as an integral part of the turbine shaft in arrangements where a collar with a sliding fit cannot be tolerated, such as an arrangement whereby the thrust collar serves a double function of a thrust collar and_-coupling flange for transmitting power or where a tight, non-readily removable fit is considered necessary to maintain thrust surface squareness. With an integral thrust collar the thrust bearing must be of the pivoted shoe type having a rating of at least 50 percent greater than the application requirements, and the thrust surface on the thrust collar must have a stock allowance of at least 1/4 inch on the thrust surface for reconditioning purposes. The rotor configuration having an integral thrust collar must be so designed that a replacement of the integral collar with a keyed, shrunk-on and axially locked thrust collar will be possible.

3,4.38.3,6 Shoes, Thrust shoes (or pads) shall be lined with anti-friction metal in accordance with taii!eI. Dovetail grooves in shoes to anchor the babbitt are permitted but are not required. Each thrust shoe shall pivot about a line or point contact, Each thrust shoe.shall be interchangeable with all other thrust shoes on both sides of the thrust collar except for shoes with RTE's installed, Shoes shall be designed so that they cannot be inadvertently installed in a location or orientation for which they were not intended.

3,4,38,3,7 Contact areas and bearing surfaces. Friction and contact surfaces shall be smoothly and accurately"'!Inished. Surface finish and Brinell hardness requirements shall be as follows:

(a) Collar (or runner) - 32 RHR or better (300 Brinell minimum).
(b) Leveling plate (or link) contact areas - 32 RHR or better on surfaces in contact with shoes or base ring; 63 RHR or better on wing surfaces,
(c) Buttons - 32 RHR or better (500 Brinell minimum),
(d) Hase ring contact areas (or hardened inserts, where used) 63 RHR or better.
(e) Shoe babbitted surface - 32 RHR or better.

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3.4.38.3.8 Wear measurement, The detail drawing of the shoe shall indicate by note the place (thickness between button or other indicated surfaces on back of shoe and babbitted face of shoe) and design value of such a measurement, so that a micrometer check can be made to determine when to rebabbitt shoe, Drilling of holes through the body of a shoe for driving out buttons is not permitted in rebabbitting, The button shall be removed from the rear of the shoe by minor machining operations at the rear only.

3.4,38.3,9 Low speed gear thrust bearing, Low speed gear thrust bearings may be flat land, tapered land or pivoted segmental shoe type, Pivoted segmental-shoe type bearings shall conform to the requirements of 3.4,38.3.l through 3.4.38.3,8.

3.4.38.3.10 Clearance. Design clearance shall be at the manufacturer's option, however, allowable increase in clearance shall be at least 50 percent of the maximum initial clearance.

3,4.38,4 RTE installation, Each journal and thrust bearing shall be fitted with an electric resistance type temperature-sensing element refe red to hereafter as RTE. There shall be one RTE in each journal bearing and in one shoe on each side of the thrust collar. Detailed requirements shall be as specified in MIL-G-3087.

3.4,38,5 Accessory drive bearings, Elements of accessory drive may be supported by sliding surface bearings or rolling contact bearings or combination.thereof provided that flexibility is incorporated in each design using both types of bearings in a line of shafting.

3.4.38,5,l Accessory drive noise tested bearings, Noise tested bearings in accordance with MIL-B-17931 shall be used on"aT!class A (submarine) turbines and gears and where required to meet specified noise levels on classes B and C turbines and gears (see 6,2.2(k))•

3.4.38,6 Special bearing tools.

3.4.38.6.1 Bearing yokes and jack bolts. The lifting gear set shall include yokes, jack bolts, and other special items necessary for removing the weight of rotors from, lower-half bearings to permit removal of same from one generator set,

3,4.38.6,2 jack, A special device shall be furnished to provide for jacking rotor forward and aft to measure thrust clearance.

3.4.38 ,7 Oil deflectors, Aluminum or bronze oil deflectors shall be provided to prevent leakage of<nl from bearing housing.

3.4.38.8 Steam shields, A steam shield shall be provided to prevent gland steam from entering bearin

3.4.39 Couplings. Flexible, solid or torque-limiting couplings may be used between the main rotating elements of the turbine, gear and generator.

3.4.39.l Flexible couplin s. Flexible couplings, if used, shall be in accordance with type I, class 1 or 2 of MIL-C-2 233, except as noted in 3.4.39,2.

3.4.39.2 Couplin removal. Design of flexible or torque-limiting couplings shall be such that they canoe isassembled without removing turbine, gear or generator rotors. Lifting of the pinion to remove the coupling is permitted.

3.4.40 Accessory drives. Where accessory drives take-off from the pinion or gear and the bearing reaction of the driving shaft shifts from lower half of bearing_ to upper half of bearing, at any time during generator load changes, the accessory drive shall be connected to the driving shaft by a commercial flexible coupling.

3,4,41 Turbine blading and seal strips.

3,4.41.1 Materials. Blades, locking pieces and pins shall be of material specified in table I.

3.4.41.2 Production method. Blades shall be machined (milled, cut, or broached) from bar or forged stock.

3,4,41.3 Solid blades. Vane $hall be solid and shall be produced with blade base from same piece of blade stock1 weldingiof vane to base is not permitted.

3.4.41.4 Edge radii. Blades shall have vane edge radii equal to or greater than the minimum values shown cm"figure 4.

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3.4.41.5 Surface finish. All blading shall be free from imperfections. Root or dovetail load carrying surfaces shall be finished to 63 RHR or better•

. 3.4.41.6 Stellite erosion shields. Where the combination of moisture and tip speeds are such that erosion of blades may be a problem, blade inlet convex side shall be hard-faced with stellite brazed with a nickel rich silver-brazing alloy. Decision as to whether or not stellite shields are required will be based on previous operating experience under similar conditions for tip speeds in excess of 800 feet per second (fps) and moisture content of 10 percent or greater. Alternate means of erosion protection will be considered.

3.4.41.7 Contour of contact root surfaces. Bucket dovetail surface which contacts wheel outside diameter snail be cut to the same or slightly larger radius than the wheel; bucket dovetail hook load surfaces shall be cut to the same or slightly smaller radius as the corresponding wheel dovetail surfaces.

3.4.41.8 Blade tenon. Shrouding shall be attached to blades by riveted (peened) tenons; welding is prohibited-:--R'ot peening is acceptable1 however, temperature shall be controlled by positive means such as a surface pyrometer or tempi! sticks.

3.4.41.8.1 Fillets. Sharp corners between tenon and tip of blade and between vane and blade base shall be avoided by filleting.

3.4.41.8.2 Tenon hhape. Tenons shall be of round cross section unless other shapes are required due to eit er geometry or strength reasons.

3.4.41.8.3 Tenon head. The riveted tenon shall form a head on outer surface of·shroud. Where holding power of teiion provides the required safety factors, the head and any excess shroud thickness may be machined off to provide flush tenon heads. Tenon shall be peened or riveted to the extent that the chamfer in the shroud band hole is filled.

3.4.41.9 Shrouding.

3.4.41.9.1 Materials !2E. shrouding. Shrouding shall be of the materials specified in table I.

3.4.41.9.2 Number of blades per shroud str p. Number of blades per shroud strip shall be as recommended by theturbine manufacturer an shall be indicated on the applicable drawings.

3.4.41.9.3 Notch block • The shroud strip segment may span the gap where a notch block is used (anatli'e ETaae omitted). Where quietness of operation is a requirement, (see 6.2.2(m)) every effort shall be made to close the gap with a closing blade which can be shrouded.

3.4.41.9.4 Drilling punching of shroud. Shroud tenon holes may be drilled, illed or punched. When shrouds are milled or punched, each set of blades for stock shall include the necessary milling template or punches. Drilled or milled shrouding requires no annealing. All punched shrouds with thickness of 3/16 inch and larger shall be stress relief . annealed after'punching. All tenon holes shall have a chamfer or radius on both sides of the shroud and, unless otherwise approved, shall be located directly from the bladed wheel, or from templates made or set therefroro1 spare shroud bands shall be furnished without tenon holes, unless specifically appfoved otherwise. A note on detail drawings shall indicate that the method of making holes in the shroud is an acceptable method, known not to cause cracks. The method of installation, showing or stating number of holes in each strip, shall also be shown. If a choice of milling or punching shrouding for replacement blading exists, the drawing note shall so state.

3.4.41.9.5 Contact area with vane - Shrouding shall, at installation, have maximum bearing practicable gainst the endoTbTacie vane. The limits in gap clearance shall be as s,aecified in 4.s.2.2.

3.4.41.9.6 End c:@12. between ad acent shroud strips. Shroud strips shall have end clearance to ensure tliat, under maximum operating speed and temperature, they do not expand so as to to·uch each other and iinpose stress on blade tenons.

3.4.41.9.7 Bevel on sealina e1ge. Shroud strips of more than 1/16 inch thickness used to seal axially, inaIT be bevele a ong the sealing edge to approximately 1/32 inch edge thickness.

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3.4.41.9.B Slanted shrouding. The installation of shrouding at an angle is permitted up to 15 degrees maximum angle with rotor axis.

3.4.41.9.9 Channel shrouding. Channel shrouding is permitted only in conjunction with segmental blading.

3.4.41.10 Blade root fastenings in rotor. Rotor blade root fastenings may be either radial-entry or axial-entry. Blade attachment to rotor shall be by mechanical means and shall use either "T" or "Fir Tree" shaped roots.

3.4.41.10.l Axial blades. For axial entry blades, nonhardened steel pins (of size and shape to pernu.tsnearing when driving individual blades out of grooves) shall be provided to secure blades against axial movement in operation. Blades shall obtain necessary blade fit in blade grooves by filing metal ridge on bottom of blade root or by other means. Where pitch of blades is controlled, blades may be installed with clearance in the root; however, maximum movement as measured at the blade tenon shall not exceed 0.020 inch per inch of overall blade height for double tang roots and 0.010 inch per inch of overall blade height for triple tang roots.

3.4.41.10.2 Radial entry blades. Radial entry blade shall be installed with a machined fit as specified by pertinent dovetail drawing. Required fit may be obtained by either filing or trimming a metal ridge on the blade or by use of calking strips. Calking strips shall be rectangular or half-round pieces equal in length to the app oximate thickness of the blade root. Locking pins for the notch block or closing blade shall be of the material specified in table I. The use of shims for circumferential fitting of blades is permitted subject to NAVSEC approval.of the procedure.

3.4.41.11 Stationary blading. Stationary blades may be installed individually or as groups of blades. Intermediate segments and separate blade rings shall be secured to the casing by either mechanical means, or calked in to a dovetail groove of the casing.

3.4.41.12 strips.

3.4.41.12.1 Materials. Materials for seal strips shall be in accordance with table I and 3.2.7.

3.4.41.12.2 Axial seals. Removable axial seal strips may be used in the first stage. Axial sealing in all other stages shall be obtained as an integral part of the blade and diaphragm geometry.

3.4.41.12.3 Radial seals. Radial seals may be used in the first stage only.

3.4.41.12.4 A proval of design. Axial seal strips shall be capable of.being replaced and shall be secure by a method as approved by NAVSEC.

3.4.42 Nozzle blocks diaphragms.

3.4.42.1 Materials. Materials for nozzle blocks, diaphragms, partitions and bolts shall be in accordance with table I and 3.2.7. Crush pins shall be of corrosion-resistant material and may be attached by either welding or mechanical means.

3.4.42.2 Steam flow surfaces. Steam flow passages shall have a surface finish of 63 RHR or better i'ilalrection of steam flow except for approach to throat section which may be 125 RHR maximum.

3.4.42.3 Construction. Nozzle blocks and diaphragms shall be either fabricated, cast, or reamed construction.

3.4.42.4 Lifting and removal The upper-half of each diaphgram and nozzle block shall be lifted with the upper-half of the turbine casing. Each diaphragm half shall be drilled and tapped for one or more lifting:eyes to facilitate lifting and handling. Each lower-half diaphgram shall be drilled an4 tapped at each joint part surface to provide a means for removing lower-half diaphragm {by pulling with eye-bolt or draw-bolt) when lifting out of casing groove.

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3.4.42.5 Partitions. Partitions shall be rolled, cast or machined. Trailing edges of partition vanes shall conform to radii shown on figure 4. The following additional requirement applies to cast partitions:

Precision-cast 13-chrome partitions may be used, provided each casting includes a partition integral with a thick band or ring segment at each end; such segments, when welded together are to form heavy inner and outer bands or rings.

3.4.42.6 Nozzle block details.

3.4.42.6.l Securing block in casing. Nozzle blocks shall be removable from casing. When bolts are used for th!spurpose, the bolt heads shall be recessed below the surface of the surrounding metal and be secured by either peening in at least 3 places, or by suitable shielding. Shielding or retaining segments shall be made of the same material as the nozzle block and shall be retained either by welding center of the segment to the nozzle block or by peening segment grooves at outer ends of segments. Tack welding is not permitted, Where more than one segment is used they shall be installed with gaps between segments. Each gap shall be greater than the maximum thermal growth the segments can experience, but less than the diameter of a nozzle block bolt head. Calking strips and spacers used to retain nozzle blocks in casing grooves shall be secured to prevent their falling out or rotating in the groove.

3.4.42,6.2 Nozzle bridges. Adjacent arcs or groups of nozzles shall be bridged to minimize leakage.

3.4.42.7 Diaphragm joint. The joint between two diaphragm halves shall be tongued and grooved or keyed to minimize leakage through the joint.

3.4.42 ,8 Spari diaphragms. Diaphragms furnished as repair parts shall be finish-machined and ready for insta lation, except that crushing pins shall be 1/8 inch over-size, or furnished long and short.

3.4.43 Turbine casings and chests;

3,4.43,l Materials. Casings, including-gland housings and packing ring holders, and steam chests, shall be of materials specified in table I and 3.2.7.

3,4,43,2 Steam chest. Steam inlet pipe shall be attached to the side or end of the steam chest and interfere with removal of steam chest cover. The joint between steam chest cover and chest shall be metal-to-metal and shall be made-up in the same manner as the casing horizontal joint except that copaltite will not be approved as a sealing compound.

3.4.43.3 Construction. Casings and steam chest may be cast, fabricated or combinations thereof.

3,4,43.4 Horizontal joint. All casings shall be divided at the horizontal centerline.

3.4.43.4.l Access for wrenching. Clearance shall be provided around bolt heads and nuts to permit use of standard tools; however, where joint integrity would be compromised, the use of special tools (to be furnished with the turbine) to accommodate closer bolt spacing is acceptable.

3.4.43.4.2 Joint surface. Flange faces shall be finished to parallel surfaces with a surface finish of"""6'31u!R or better. The sealing surface of each flange shall be a horizontal plane with no discontinuities or steps along the flange except at junction of vertical •joints.

3.4.43.4.3 Flange chamfer. Joint surfaces in way of flange bolts or studs shall be chamfered or counterbored.

3.4.43.4.4 Making-up joint. Joints shall be made-up metal-to-metal at room temperat,ure; however, flange faces may be coated with either a thin coating of bodied linseed oil, or other commercially available sealing compounds. Procedure and compounds tp be used require NAVSEC approval.

3.4.43 ,4,5 Pumping grooves. Flange pumping grooves shall not be provided.

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3.4.43.5 Bearing housings. If the bearing housings are separate from casing, a centering shoulder, spigot, radial keys or radial dowels shall be provided at the joint between housing and oasing. Space shall be provided between bearing and glands to prevent leakage oil from entering the glands and gland steam blow from entering the bearings. Drain pockets and cavities between bearing and gland shall be avoided, so as to reduce the possibility bf contamination of oil and condensate systems due to flooding of an improperly-drained pocket. When the use of a pocket is unavoidable, the cavity shall be as deep as possible with maximum volume and shall be drained by a one-inch diameter (minimum) drain (or equivalent area with minimum dimension not less than 1/2 inch)i such drains shall not be plugged, or otherwise restricted and outline drawing shall so indicate.

3.4.43.6 Vertical oints. Casing may be in two sections longitudinally with a vertical jointi however, the vertical joint shall not require disassembly to lift the upper-half casing.

3.4.43.7 Four-way casing joint. Where four-way casing joints cannot be avoided, flangks may incorporate saw-cuts from the outside of the flange to the bolt hole adjacent thereto. As an alternate, the joint between the steam chest and steam chest cover may be made up using a spiral wound gasket providing:

(a) The required gasket is round or other regular readily available shape. (No odd or irregular shapes will be permitted.)
(b) The joint is provided with dowels and stops to assure proper alignment.

3.4.43.8 Test connection. Special test connections may be provided at manufacturer's option, however-;--such connections shall be closed by threaded plugs (taper or straight thread) or by blank flanges. Plugs, if used, shall be seal welded.

3.4.43.9 Exhaust flange.

3 4.43.9.l Class A turbines. The exh aust flange for class A (submarine) turbines shall be circular for use of a flexible steam joint.

3.4.43.9.2 !.turbines. For type I (package type) turbines, the generator set prime contractor shall be responsible for coordination, between turbine and condenser manufacturers, of flange design and the furnishing of any required gaskets, bolts and templates.

3.4.43.9.3 II turbines. Flange requirements for type II (non-package) turbines shall be as specified in 3.4.43.9.3.1 through 3.4.43.9.3.2.

3.4.43.9.3.l Flange drilling. Where turbine and condenser are bolted directly together, their flanges shall be drilled by the respective manufacturers, except that holes for body-bound bolts will be finish-reamed by shipbuilder at installation. The condenser flange will be drilled to a template or jig to be furnished by the turbine manufacturer. Where turbine and condenser are connected by a flexible connection, instructions for drilling of flanges shall be determined in accordance with shipyard installation procedures, which will be furnished by the shipbuilder with the invitation to bid (see 3.4.22.2).

3.4.43.9.3.l.l Template for condenser flange. The turbine manufacturer shall furnish a thin metal template (for a single unit order) or a drilling jig with hardened bushings (for a multiple unit order) to the condenser manufacturer for drilling the condenser flange. When the condenser manufacturer has drilled the condenser flange, the template or jig shall at turbine manufacturer's option, be returned to the turbine manufacturer.

3.4.43.9.3.2 Responsibility for bolting. Bolting and gasket (if required) will be furnished by shipbuilder except that cap screws or one-nut studs shall be furnished with the component tapped for same. The shipbuilder shall be responsible for providing bolting and gasket (if required) where components attach to a common flexible connection

3.4.43.10 Access and inspection openings. Access and inspection openings shall be provided in turbine casings as follows:

(a) In high pressure and exhaust ends of all turbine casings, to permit in-place balancing of rotors without lifting casings or performing other major time consuming operations.
(b) In upper half of exhaust casings in all turbines, one on each side of the vertical cente:r;line, to permit visual examination of the last row of blading for damage and for manually checking last row of blading for tightness.

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3.4.43.11 Casing lifting.

3.4.43.11.l Jacking bolts. Flange jacking bolts shall be provided as necessary to permit readily breaking joints.

3.4.43.11.2 Diaphragm removal. When upper-half casing is supported in the raised position and rotor is resting in bearings, it shall be possible to remove upper-half diaphragms; removal of all lower-half diaphragms shall be possible in this position or with casing and rotor supported in the raised position but not combinations of the two choices. Removal of horizontal joint studs to accomplish this will be acceptable. However, this requirement for removal of diaphragms is waived if casing supports in excess of 36 inches in height are required to accomplish diaphragm removal.

3.4.43.11.3 Stud removal. The design shall be such that dismantling or assembly does not require the removal of studs.

3.4.43.12 Lifting gear. Lifting and handling gear (including flange jacking bolts) necessary for lifting the turbine and gear upper-half casings, rotors and gear rotating elements of one generator set constitutes one set. Lifting shall be possible when the ship has a list and trim of 5 degrees.

3.4.43.12 ,l Materials for lifting and handling gear. Lifting and handling gear shall be of steel except that aiumiiium may be used for casing supports and rotor guides.

3.4.43.12.2 Lifting and supporting upper-half casin . The lifting gea set shall include at least 3 corner guide pins or posts, dowelled casing supports, special eye-bolts, and such other items as are necessary to lift and support the upper-half casing. Casing supports shall be 36 inches in height or of a height to permit removal of diaphragms whichever is less.

3,4.43,12,3 Lifting and supeorting rotor. The lifting gear set shall nclude rotor guides and saddle support assemblies, special eye-bolts, and any other special equipment necessary for lifting and supporting turbine rotors and gear rotating elements.

3.4.43.12.4 Shipbuilder-furnished lifting gea 7. Traveler bars and trolleys, pulleys, deck-beam clamps, turnbuckles, shackles, hooks, chains, hoists, wire slings with thimbles, and spreader beams (for rotors) will be shipbuilder-furnished.

3.4.43.13 Moisture separation and drainage.

3.4.43.13.1 Internal moisture separation. Provision shall be made for internal moisture separation in stages which normally operate in wet steam. Special requirement for internal moisture separators are as follows:

(a) Water dams. Water dams, if used, shall be of corrosion-resistant alloy steel and snaIT be secured to casing by welding except for 12 chrome casings, in which case bolting is acceptable.

(b) Drain orifices. Removable drain orifices (minimum of 1/8 inch in diameter) sliaI'l be of corrosion-resistant alloy steel and shall be retained in casing by peening in addition to threaded connection.

(c) Casinl drains. All casing drainage, except that from the first stage drain, shal be collected either internally or externally in a common line or manifold and discharged to the turbine exhaust in a manner to prevent collected moisture from impinging on the condenser tubes. A funnel type drain may be supplied in the exhaust trunk for this purpose, or an external connection may be located on the turbine casing for an external drain line to the condenser.

(d) .chest.and stage drains. Blanked-off flanges shall be provided to permit drain ge of steam chest and first stage of casing,

3.4.43.14 Steam shields. Steam (or windage) shields shall be provided for stages having partial admission:-:-

3,4 43.15 Blade throw-out protection. Each casing shall be designed to contain blades which may be thrown from any stage of the turbine at any speed up to 110 percent of rated &peed. Particular attention is directed to last stages.

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3.4.44 Steam inlet connection. The steam inlet connection for class A (submarine) turbines sha1roe-1ocated on the bedplate or foundation for connection of a flexible steam line. The flexible steam line is not required. The location of the connection shall be as specified (see 6.2.2(n)).

3.4.45 Turbine rotors.

3.4.45.1 Materials. Rotors shall be forged from a single billet conforming to materials specified in table I and 3.2.7. Shrunk-on keyed wheels will be considered for turbines to be installed in class D turbines (auxiliary type ships); however, NAVSEC approval is required.

3.4.45.2 Stresses. Stresses shall be in accordance with 3,2!5.4.

3.4.45.3 Balancing. Vibration limits and balancing requirements shall be as specified in 3.4.27.

3.4.45.4 Coupling flange. The coupling flange on the drive end of each rotor shall be shrunk on or shall be integral either with the rotor or with a stub shaft bolted to the rotor. The turbine manufacturer shall collaborate with the manufacturer of the driven component to obtain mutually acceptable coupling design and loads (see 3.4,39),

3.4.45.5 Surface finish, The surface finish of the overall rotor shall be 125 RHR maximum except as follows {maximum RHR values):

(a) Bearing journals - 32 (axial and circumferential direction)
(b) R_adii of equalizing holes.- Polished to 63
(c) Fillets and radii - 63
(d) Axial blade groove.a - 63
(e) Circumferential blade grooves - 63
(f) Load carrying surfaces of blade dovetail - 63

3.4.45.6 Bearing journals. Special attention shall be paid to roundness and concentricity of journals, Deviation from true roundness shall not exceed 0.000050 inch for class A turbines (submarines) and other classes where quiet operation is specified, (see 6.2.2(k) and (m)) and 0.0003 inch for all other applications. Deviation in true roundness is defined as the difference in radii of two concentric coplanar circles which just contain the measured profile of the journal surface.

3,4.45,7 Packing journals. Packing journals shall be smooth or stepped as necessary to match packing teeth, The use of packing sleeves shrunk on or keyed to rotor is prohibited. Lands shall be rugged to withstand contact both radially and axially with the fins of the packing rings.

3,4.45.8 Rotor bores. Rotors may be solid or bored; if bored, the bore shall be chemically cleaned,purged with an inert gas and sealed with air tight plugs.

3,4,45,9 Match marks. At installation of the rotor, axial and circumferential position of the rotor relative to the casing shall be marked for future reference when reinstalling rotor after repair or overhaul, These marks shall be accessible without lifting the casing but removal of bearing caps is acceptable and shall be in tenths of a mil,

3,4.45,10 Critical steeds. he design of the rotors may be such that either the calculated or actual runningirst and second criticals are within the operating range of turbine, For constant speed turbines, the criticals shall be below 80 percent of rated speed or at least 20 percent above rated speed. For variable speed turbines, the critical speeds shall not fall within the ranges of 13 to 17, 28 to 32, 43 to 47 cycles per second (c.p.s.) or 80 to 120 percent of rated speed.

3.4,46 Turbine packing (gland, dummy and diaphragm).

3.4.46 ,1 Materials. Labyrinth packing and springs shall be of the materials specified in table I and 3,2,7, Carbon packing rings will not be permitted.

3,4,46,2 of packin . All gland, dummy and diaphragm packing shall be of the labyrinth type and slial"lnave at 1east four segments per ring.

3,4,46,3 Fin design. Packing rings shall be machined with integral fins thinned on he inner diameter.to an axial dimension of 0.010 + 0.005 inch.

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3.4.46.4 Flexibility, Each ring shall be spring backed to prevent radial movement. Springs may be either of the flat or coil type.

3.4.46.5 Rotation. Each ring shall be secured against rotation or circumferential displacement in its groove while turbine *is* in operation.

3.4,46.6 Surface finish, Steam seal surfaces shall have a surface finish of 63 RHR or better,

3.4,46.7 Marking of rings. To ensure correct installation, each seal ring shall be marked as follows:

(a) The mating surfaces of adjacent segments shall be match marked,
(b) Each ring with "T" shaped retaining shoulders and which can be improperly installed shall be stamped either with an arrow pointing aft or other marking to clearly indicate proper installation,

3.4,46.8 Removeabilit!, The turbine design shall permit removal of all of the outer labyrinth packing rings (al rings outside of the air-steam leak-off area) at each end of each turbine without lifting the turbine casing. All packing ring casings shall be secured to either the upper or lower half of the casing to prevent their falling out when upper-half turbine casing is lifted, After casing is lifted, it shall be possible to roll upper and lower ring segments out of their grooves without removing diaphragms or lifting the rotor.

3,4.47 Reduction gear casing.

3.4.47.l Materials. Casings shall be of material specified in table I.

3.4.47.2 Construction. Casings may be cast, fabricated or combinations thereof.

3,4.47.3 Bearing housings. Thrust and journal bearings shall be directly mounted on or within the gear casing.

3,4.47.4 Bearing bores, Bearing bores shall be properly machined and scraped, as necessary, to ensure correct alignment,

3,4.47.5 Casing selits, Casings shall be divided as necessary to provide accessibility to pinions, gears, bearings and other parts which will require routine maintenance,

3,4.47.6 Access oeenings. Access openings shall be provided to permit examination of the interior of the ca ing. The openings shall provide for examination of the entire tooth length of each element and observation of the oil spray pattern. Each access opening shall have a raised lip and shall be provided with a closure secured by a hinge pin on one end and a clevis on the other. All bolts, pins and nuts shall be riveted on such that parts cannot fall into the gear case during inspection. Each closure shall be designed so that foreign material cannot enter the access opening and oil cannot leak from the opening when the closure is secured.

3.4.48 rotors.

3,4.48.1 Materials. Materials of gear main rotating elements shall be in accordance with table I.

3,4.48,1.1 Through hardened gears. Where through hardened gears are required {see 3.4.48,3), the pinions and gears shall be in accordance with MIL-S-19434 and the pinion shall be at least one class higher than the gear.

3,4.48.l.2 case hardened gears. Where case hardened gears are required, (see 3.4.48.2 and 3.4.48.3), th'e""'i;iinlons and gears shall be either nitrided or carburized unless otherwise specified (see 6.2.2(0)). Where gear elements are to be nitrided, they shall be in accordance with classes 2 through 4 of MIL-S-19434 or AISI class 4140 or 4340 steels. Where gear elements are to be carburized, they shall be either class 4615 or 8615 in accordance with MIL-S-866.

3.4.48.1.3 Nitrided giir properties. Gears and pinions, case hardened by nitriding, shall have properties as fo ows:

(a) The case hardness shall be 50 Rockwell C minimum.
(b) The case thickness shall be 0,015 inch minimum.
(c) The core hardness shall be 27 Rockwell C minimum and 33 Rockwell C maximum.

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3.4.48.l.4 Carburized gear properties. Gears and pinions, case hardened by carburizing shall have properties as follows:

(a) The case hardness shall be 58 Rockwell C minimum.
(b) The case thickness shall be a minimum of 1/6 of the tooth thickness at the pitch circle.
(c) The core hardness shall be 30 Rockwell C minimum and 44 Rockwell C maximum.

3. 4.48.2 Single helical gears. All single helical gears shall be case hardened by nitriding or carburizing, unless otherwise specified (see 6.2.2(p)).

3.4.48.3 Double helical g ars. Double helical gears shall be either case hardened or through hardened, unless otherwi$e specified (see 6.2.2(p)).

3.4.48.4 Shot-peening. Shot-peening of main pinion and gear teeth root fillets is required, if the tooth beam stress (Sk), as determined by the method of MIL-G-17859, exceeds 15 percent of the tensile strength of the material used in fabricating the elements. Shot-peening shall be done prior to finishing teeth by shaving, lapping, grinding or honing.

3.4.48.5 Chordal thickness. Drawings of pinions and gears shall indicate, by note, the minimum acceptable chordal thickness after rehabbing.

3.4.48.6 Matched sets. inions and gears shall be furnished in matched sets. Elements shall be serialized. P nwnbers, housing center distance and actual backlash shall be etched on elements.

3.4.48.7 "K" factor. For through hardened gears the "K" factor for main rotating elements shall not e eed 175 where:

$$K = \frac{Wt}{Fe.a}\left(\frac{Mg + !}{Mg}\right)$$

Wt= tangential tooth load in pounds.

Fe= effective tooth face width.

d = pinion pitch diameter.

Mg= reduction ratio(> l),

3.4,48.8 Helix angle. Single helical gears shall have a maximum helical. angle of 25 degrees. Double1ieI'ical gears shall have a minimum helical angle of 25 degrees. Modified involute profiles, helix angle modifications, and crowned teeth or combinations, thereof may be incorporated to compensate for distortions under load.

3.4.48.9 Contact pattern. Each set of meshing elements shall exhibit the design contact pattern or relationship when meshed together on true centers and when assembled in the gear case. The contact pattern when assembled in the gear case shall be obtained with each bearing supporting each pinion and each bearing supporting each gear having equal wall thickness (in each pair) at the load line within $\frac{0.0015 + 0.00lLS}{F}$ inch, where LS is the span of the midpoints of supporting bearing in inches and F is. the face width in inches.

3.4.48.9.1 Contact pattern area. Mating teeth of gears shall have a contact pattern at the rated load condition of at least 75 percent of the working depth (excepting areas of tip relief) distributed across at least 85 percent of the active face width (excepting areas of end relief) as shown by a wear pattern on a marking compound. A band of at least 5 teeth to represent each mating pair shall be provided to demonstrate this pattern.

3.4.48.10 Tooth finish. Meshing surfaces and root areas of teeth shall blend to form smooth surfaces free of sharp edges. Rounded tooth fillets shall be used. Tooth.profile surf ce finish (all directions) shall be 32 microinches RHR or better.

3,4.48.11 Gear a·ccuracy. Unless otherwise specified (see 6.2.2(q)), pinion and gear accuracies shalr-E'e as follows:

(a) Accumulated tooth spacing error shall be in accordance with figures.
(b) Adjacent tooth spacing (pitch) error shall be in accordance with quality 13 of AGMA 390.XX.
(c) Runout shall be in accordance with quality 14 of AGMA 390.XX.
_(d) Profile shall be in accordance with quality 14 of AGMA 390.XX.

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.. 3.4.48.12 Gear balancing. All pinions, gears and shafts shall be balanced dynamically. Preliminary balancecan be performed after finish machining, before teeth are cut, All balancing shall be by removal of material. Addition of weights will not be permitted due to possibility of weights coming adrift and dropping into the gear mesh. Drawing shall state permissible amount of residual unbalance in inch - ounces. For additional requirements see 3.4.27.

3.4.49 Accessory drive gdars. Gear elements and other parts for accessory drives shall be designed and rated i'n"'acc'or ance with the applicable AGMA publications listed in the latest.AGMA publication index except that materials shall be in accordance with table I. Items (q) (1) and (q)(3) of table I shall have case and core hardness and case thickness as required by 3.4.48.1,3 or 3.4.48.l,4 as applicable,

3.4.50 Lubrication, Turbines, gears and flexible couplings shall be designed for pressure lubrication from the lubricating oil system, except that external parts may use individual grease fittings, Passages, fittings and sprays shall be provided for lubrication and cooling of all parts under all specified operating conditions. Take-down components shall be used to the maximum extent practicable to facilitate cleaning.

3,4,51 Lubricating oil system.

3.4.51.1 Surface ship. For·surface ships a complete, self-contained, forced-feed, lubricating oil system shall be provided, by the manufacturer, to supply all oil required by all bearings (turbine, gear, generator and accessories), couplings and gear sprays at a pressure of not less than 10 p,s.i pressure when the set is operating at rated speed. The requirements of 3.4.51.3 through 3,4.51,26 apply only to the items furnished by the turbine-generator set manufacturer (see 3,3.2 and table II),

3,4,51.2 Submarines, For submarines the requirements of 3.4,51.l apply, except that the shipbuilder will furnish system components not mounted on the set, including the oil sump, pumps, strainer, cooler, cooler by-pass, heater, pressure regulating valve, piping and a flange for connection to the set, The set manufacturer shall furnish all required components and piping mounted on the set and a flange for connecting to shipbuilder's system. Any required gaskets for connecting to shipbuilders connection, will be furnished by the shipbuilder.

3.4.51.3 Control system oil. The lubricating oil system may be used to supply high pressure oil to the control system provided that a relief valve, designed so that it cannot be closed positively, is furnished to reduce oil pressure for lubrication.

3.4.51.4 lubricating oil. oil systems shall be designed for use with Navy symbol 2190 TEP---ruE'r1cating 011 conforming to MIL-L-17331.

3,4.51.5 Oil inlet temeerature. Turbines and gears shall be capable of operation up to and includingfuITToad with inlet oil temperature of 90°F, Normally temperature will be approximately 120°F, and shall not exceed l30°F.

3,4,51.6 ,Oil discharge tem erature. The maximum temperature rise of the oil discharged from any bearing under any speci ied operating condition shall not exceed 55°F, nor shall the temperature exceed 185°F, when measured by thermometers.

3,4,51.7 RTE tem erature limits. Maximum allowable temperature as read by RTE shall not exceed 250°°F.""""for Journal bearings and 270°F. for thrust bearing shoes. Alarm settings should initially be set at maximum limits; however, final alarm setting shall be 20°F. higher than the maximum value observed during trials.

3,4,51.8 Bearing oil suppla and return, Each journal bearing and thrust bearing shall have separate oil connections an separate oil discharge connections; however, inlet and discharge for thrust bearing may be manifolded with those of adjacent journal bearing to permit a single oil supply connection and a single oil discharge connection to serve both bearings,

3.4,51,9 Oil sumps. Oil sumps shall be provided with class B, C and D turbines and gears and shall""'nave a capacity in gallons equal to twice the capacity of the lubricating oil.,pump in gallons per minute (q,p,m,) when operating at rated speed, Oil sumps shall be oil tight (zero leakage),

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3.4.51,9.l Oil level indicators, Oil level indicators having low, normal and high levels clearly indicated shall be furnished, Bayonet type oil gages shall be secured against coming adrift under shock by a quick disconnect means other than a threaded connection.

3.4,51.9.2 Heating cooling - Internal heating or cooling coils are not permitted.

3.4.51,9,3 Paint. Paint shall not be used on surfaces in contact with oil.

3.4.51.9.4 Access holes. Access holes shall be provided as necessary to permit manual cleaning. All areas or pockets in the sump shall be accessible for cleaning.

3,4.51.9,5 Oil sampling valve. A valve shall be furnished to permit taking of oil samples from bottom of sump.

3,4.51,9,6 Purification. Two blanked-off flanges shall be provided, one to permit complete drainage of sump for batch purification of oil or connection to ship's purification system and one for return via separate line.

3.4.51,9,7 Vents. All gear cases and oil sumps shall be vented to atmosphere, The total venting area shall be not less than the following:

Tank capacity (gallons)	Venting area (square inches)	Nominal i.p.s. vent which provides required area Inches
1 to 20	0.53	3/4
21 to 50	•86	1
51 to 100	1.50	1--1/4
101 to 200	2.04	1-1/2

Where oil sump is separate from gear case, vents are required for both gear case and oil sump. Gear case vents shall be in the top of the gear case cover and shall be 3/4 inch i,p.s. minimum.

3.4.51 ,9,8 Precipitators, An electrostatic precipitator, in accordance with MIL-P-23917, shall be furnished and shall be equipped with a blow-off valve to relieve pressure in event of stoppage of the precipitator, The precipitator shall be connected to the highest point of a common header connecting all gear case and oil sump vents. Drawing 810-2145504 shall be used for guidance, except that the precipitator drain shall be arranged for connection to the shipbuilder furnished oily water drainage systems.

3.4.51,9,9 Location. Oil sumps shall be located so that oil drains by gravity from the bearings under all operating conditions, including trim, list, roll and pitch (see 3.4.13).

3.4,51,10 Attached oil pdps, One self-priming attached oil pump shall be provided with each class B, c or D turbTne an gear to provide forced feed lubrication to bearings and to the control oil system.

3.4.51.10.l Pumt drive, The pump shall be driven either directly or by gearing from the turbine or gears a

-3.4,51.10.2 Lubrication while shutting down. Design and size of attached pumps shall be such that lubrication is provided when generators are being shut-down without supplying oil by any other means.

3.4.51,10.3 Pump location. Pumps shall be self-priming under all operating conditions, including trim, list, roll and pitch (see 3.4.13).

3.4.51.10.4 sizing for variable speed generators. When turbine-generator sets are designed to operate over awide frequency range, a motor driven oil pump may be used to supplement the attached pump during operation at the lower frequencies,

3.4.51.10.5 Lubrication whiie idling. Design and size of attached pumps shall be such that lubrication is proviaeaw'hen generators are idling at no greater than one third of rated speed.

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3.4.51.ll Motor driven oil - One positive displacement, rotary type, motor driven pump shall be furnished in accordance with MIL-P-19131 with each class B, C or D turbine and gear of 750 KW or above. Pump flanges shall be in accordance with 3.4.32. Pump shall have a capacity to enable starting of set and, in the event of failure of the attached oil pump, to supply oil to the bearings during coast-down.

3.4.51.ll.l Motor. The motor shall be in accordance with MIL-M-17060 and shall have the following characteristics:

Service ---------- A.
Ambient temperature --- so°C.
Enclosure----------Totally enclosed.
Insulation----------Class B.
Number of phases ------- 3.
Frequency ----------60 c.p.s.
Voltage----------440 a,c,
Duty ---------- continuous.

3.4.51.ll.2 controller. A controller shall be furnished for each motor driven lube-oil pump and shall be in accordance with MIL-C-2212. The controller shall be of the magnetic type with low voltage release (LVR) and thermal overload protection and a selector switch with three positions, "off", "manual", and "automatic". The controller shall function automatically to start the pump in the event of low oil pressure, but shall require manual operation to stop the motor driven oil pump. The type of enclosure shall be dripproof. This controller shall not be used for any other function.

3,4.51.12 Hand oil pum d• For sets under 3000 KW, a positive displacement, rotary type hand pump snaIT m;-provi ed for starting the turbines, The hand pump shall develop the required pressure to open the nozzle control valves and the throttle valve for starting, in the event the motor driven pump is inoperable. It shall also develop the required oil flow at lower pressure for the required lubrication of the turbine-generator set during start-up, The hand oil pump shall be capable of supplying the required capacity without imposing unreasonable physical demands on the operator. To reduce the cranking effort required, the pump may be a duplex type, that is two pump elements on one crank (one element to supply oil to the governing system and the other to the bearings through the strainer and cooler),

3.4.51.13 Oil strainers. The lubricating oil shall pass through an external duplex basket type, magnetic, 011 strainer in accordance with type I, class 4 of MIL-S-17849, modified as specified in 3.4,51.13.l through 3.4.51.13.4.

3.4.51.13.l Strainer baskets. Strainer baskets (elements) shall conform to the following:

(a) Rated flow - As required by system design,
(b) Rated inlet pressure - As required by system design.
(c) Normal operating pressure - As required by system design,
(d) Preferred basket material - Corrosion-resisting steel wire mesh, pleated.
(e) Absolute particle removal - 25 microns.
(f) Oil temperature range - 90°F. to 160°F,
(g) Pressure drop, clean (max.) - 8 p.s.i. at 90°F.
(h) Pressure drop, dirty (max.) - 12 p.s,i. at 90°F.
(i) Collapse differential pressure - 75 p.s.i.
(j) Dirt holdfng capacity (min.) with maximum pressure drop of 12 p.s.i.l/ - 150 grams per 100 gallons per minute (g.p.m.) of rated flow.
(k) Type - Cleanable,
(1) Cleaning method - Ultrasonic, low pressure air and solvent or backflush.
(m) Basket condition after cleaning - Original clean basket pressure drop.
(n) Elapsed time for cleaning (max.) - 45 minutes.

!/ Based upon a contaminant of AC fine test dust or equal,

3,4.51.13,2 Strainer basket covers. Covers shall have an integral lip or deflector to direct oil away from the operator, and hot surfaces, in case of a flange leak. In addition, covers shall be fitted with a l/8 inch iron pipe size (i.p,s,) vent valve and copper tube to permit the operator to direct the vented air and oil to a safe location.

3.4.51.13.2,l Basket cover joints. Basket cover joint design shall be based on the use of carbon steel bolting or Joint sealing materials shall be a type which seals with pressure. The use of materials that require a definite compression and yielding for joint tightness is not acceptable, since these materials rapidly lose their effectiveness,

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3.4.51.13 ,.2.2 Gasket surface. The gasket surface, or o-ring groove, shall be recessed into the body flange so that the cover will make-up metal-to-metal with the body where 0-rings are used and metal-to-metal witl'l the gasket retainer where other types of gaskets are used.

3,4.51.13.3 Body drains. Each strainer body shall be fitted with a drain valve and a drain line, •not exceeding 1/4 inch in diameter, for draining oil from the body back to the lubricating oil sump prior to removing the basket for cleaning.

3.4.51.13.4 Strainer drip pans. Drip pans shall be provided under lubricating oil strainers to catch any oil that might drip or be spilled inadvertently when cleaning baskets. Drip pans shall be fastened securely to prevent their being dislodged due to shock.

3.4.51.13.5 Strainer location. To reduce fire hazards, every effort shall be made to locate oil strainers at an area of the generator set that is most remote from high temperature surfaces.

3.4.51.14 Filters. Filters shall be installed in the control oil system such that all control oil passes through the filters. Where speed governors have an independent, self-contained oil supply, the foregoing requirements shall apply to all parts of the control oil system, except the speed governor proper.

3.4.51.14.1 Filter and size. Filters shall be duplex or dual element, full flow throw-away cartridge type capable of removing particles of 10 microns or larger. Filters shall be arranged and valved such that switching from either element to the other can be accomplished without shutting off the oil supply to the governing system or securing the set. Filters shall have a dirt holding capacity of 100 grams for each 100 g,p.m. flow through the system. The maximum pressure drop through each filter shall be 8 p.s.i. when clean and 12 p.s.i. when dirty, both at 90°F. temperature.

3.4.51.14.2 Filter accessibility and maintainability. Filters shall be located so that they are readily accessible for cleaning maintenance.

3.4.51.15 Coolers. An oil cooler, in accordance with type A, class 2 of MIL-C-15730 shall be furnished with each class B, C or D turbine and gear. The cooler shall have a minimum capacity equal to 50 percent-of the attached oil pump capacity. Cooler size shall be based upon inlet water temperature of 95°F. and oil outlet temperature of 130°F.

3.4.51,16 Relief valves. Relief valves shall be provided at the discharge from each attached and motor driven oil pump and shall be arranged to relieve back to the oil sump.

3,4.51.17 Check valves. Check valves shall be provided in the discharge line from each positive displacement pump (ahead of system relief valve) and other locations as required by the system design.

3.4.51.18 Cooler by-pass valve. A four-way cross cock shall be installed in the oil line ahead of the cooler and designed so that the cooler can be by-passed without interruption of the oil flow. The valve shall be designed so that in no position can the oil to the bearings be completely cut-off.

3.4.51.19 Gag7s and differential ressure indicators. Pressure gages shall be furnished to meet system requirements, and shall include gages to indicate control oil pressure, oil filter and oil strainer pressure drops and pressure at the bearing with the lowest oil pressure. Oil gages shall be in accordance with the requirements of 3.4.36. In lieu of gages to indicate pressure drop across oil filters and strainers, differential pressure indicators may be used provided they meet the shock requirements herein. Gages shall be located near the item being checked,

3.4.51.20 Alarm switches. Alarm switches shall be furnished and installed as follows:

(a) Low oil pressure alarm switch in the piping at the point of lowest oil pressure. The switch shall be single pole, double th ow (SPOT).
(b) Low oil level alarm switch installed such that the alarm will sound in the event of low lube oil level in the oil sump.

3.4.51.21 sivht flows. Each journal bearing cap and each thrust bearing cover or housing shall be fitted with a sight flow integral therewith or bolted thereto. Sight-flows shall provide visual indication of oil discharge from bearing pressure pad or loaded area of the bearing during operation. Bolted-on fittings shall be bronze or brass and windows shall be shock-resistant glass in accordance with MIL-G-2860 (l/4 inch minimum thickness).

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3.4.51.22 Thermometers. Thermometers shall be in accordance with the requirements specified in 3.4.36.6 to 3.4.36.10.

3,4.51.23 Flow-limiting orifices, Orifices shall be provided in the oil supply to each journal bearing and in the inlet to or discharge from each thrust bearing, Orifices may be of a design which permits removal but they shall be installeq in such a manner as to prevent external adjustment of oil flow.

3,4.51,24 Test connections. Test connections {for use of a pressure gage) are not required before eaciibearing, If such connections are provided for use during shop tests, they shall be plugged and seal welded prior to shipment.

3.4.51 ,25 Cleaning of systems All parts of.the lubricating oil system shall be thoroughly cleaned prior to conducting the operating tests.

3,4.51,26 Grease lubrication, Where grease lubrication is required, design shall be such that required lubrication is obtained with grease conforming to MIL-L-15719. Commercially available grease of a type compatible with that complying with the Government specifications may be used during shop tests, Grease fittings conforming to geometry of MS15003 shall be provided with the turbine,

3,4,52 Control systems. A control system shall be furnished that will provide for manual control of speed during start-up, automatic control of speed after start-up and·automatic shut-down in the event of overspeeding, high back pressure or low oil pressure, Means to manually trip the throttle valve closed shall also be furnished,

3,4.52,1 Constant-speed overning system, The governing system shall be as specified in MIL-G-3087, Where a hydrau1ic relay type governor is specified, the system shall be in accordance with 3,4,52,2 through 3,4.52.8, Where an electric type governor is specified, the system shall be *in* accordance with MIL-G-21410 and 3.4,52.6 through 3,4.52.8, Either type shall be capable of controlling and regulating the speed of the turbine in accordance with the requirements of MIL-G-3087 or MIL-G-19650, as applicable, under the range of steam inlet and exhaust conditions as specified (see 6.2,2(e) and (f)).

3.4.52 ,l,l Fail safe feature. The governing system shall be such that.the nozzle control valves wiII"closeupon loss of *oil* pressure,

3,4.52,1.2 Start valve. A start valve shall be furnished, if required, to facilitate starting of the turbine from shutdown or re-starting the turbine after trip-out, at a speed of not less than 50 percent of rated·speed, upon resetting the trip throttle valve.

3,4,52,2 Speid governors (mechanical). The speed governors shall be of the flyweight type. Governor we ghts shall be positively secured to prevent dislodging under shock,· Knife edges are not permitted,

3,4,52,3 Ipeed chanver, A speed changer that can be manually operated by means of a handwheel and a so electrically controlled from a switchboard shall be furnished. The speed changer shall be capable of a range of speed adjust ent as specified in MIL-G-3087. When operated remot ly the speed changer shall be capable of changing the speed an amount equal to the no load speed minus the full load speed, with the specified droop, in less than 60 seconds.

3,4.52.4 Speed changer motor. The speed changer motor shall be a single phase, readily reversible commutator-less a.c.motor in accordance with MIL-M-17059 and MIL-G-3087.

3.4.52.5 1,}ed changer stfipi. Stops shall be provided to limit the range of the speed changer, The hig speed stops al be capable of being retracted manually for testing the overspeed trip, using the speed changer. The high speed stop shall be limiting after test.

3.4.52.6 Spied centrol mechanisms. Speed control mechanisms shall be of the hydraulic elay type using ubricating oil as the medium for the activating force, Loss of oil pressure shall cause the nozzle control valve to close.

3,4.52.J Valve operator, A valve operator shall be furnished, if required by system design, to amplI°fythe governor output to actuate the nozzle control valves.

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3.4.52.8 Nozzle control valves. Nozzle control valves for class A turbines of 2000 KW or larger shall have a minimum total lift of 0.500 inches to minimize governor instability due to the affect of sensitivity and deflections *in* governing mechanisms. Class A turbines of 2000 KW or larger shall have a minimum of 5 single poppet valves or 4 double poppet valves. Other turbines shall have the number of valves required to assure meeting the governing requirements of MIL-G-3087. Valve materials and construction details shall be *in* accordance with 3.4.52.10 to 3.4.52.15.

3.4.52.9 Trip throttle valve. A trip throttle valve shall be furnished and installed between the steam strainer and the steam chest. Throttle valves shall be supported to withstand high-impact (H.I.) shock. Valve type shall be in accordance with 3.4.52.9.1. Valve material and construction details shall be in accordance with 3.4.52.10 to 3.4.52.15.

3.4.52.9.l Trip throttle valve. - Trip throttle valves shall be of the poppet type. Butterfly or gate valves are not permltted. The valves shall be of the hydraulic type, being opened and held open by oil pressure, opposing a spring, which will close the valve upon loss of *oil* pressure. The oil pressure shall be furnished from the units lubricating oil pump. The valves shall be equipped with a handwheel for opening the valve slowly, during start-up, and for controlling steam to the turbine until the governing valves take control. The valves shall be designed so that they will be tripped shut by either the overspeed trip or the manual trip. For condensing turbines, the valves shall also be tripped shut by the high back-pressure trip. In the case of d.c. turbine generators, tripping of the valve shall also open the generator circuit breakers. When the valve has functioned to stop steam to the turbine, it shall be necessary to manually turn the handwheel to the completely closed position, to reset the valve, before reopening to again admit steam to the turbine. The automatic operation of the valve shall be entirely independent of the degree of opening of the valve.

3.4.52.10 Valve materials. Valve materials shall be in accordance with table I. The body for the trip throttle valve shall be of material as specified for the steam chest or casing for comparaole operating conditions.

3.4.52.11 Valve seats. Valve seats shall be removable from the valve body or steam chest. Threaded connections are not permitted. Provision shall be made for expansion between seat and body or chest. Annulus between valve seat and body or chest shall be seal welded to prevent leakage. For saturated steam applications, expansion lips or rings used for absorbing expansion between seat a_nd body shall be constructed of nickel-chromium-iron alloy (73-15-7 nominal).

3.4.52.12 Valve contact area (disc and se t). The valve contact area of disc and seat shall be hard-faced with chrome-cobalt material, MIL-R-17131, type MIL-RCoCr-A, to a minimum thickness of 3/32 inch an approximate width of 1/4 to 1/2 inch.

3.4.52.13 Valve stems and bushings. Soft packing shall not be used. Stems and bushings shall be nitride otherwise""-hardened.

3.4,52.14 Poppet valves.

3.4.52.14.1 Lift-bar. The lift-bar may be either solid or split. If split, the halves shall be aligned by two straight dowel pins or fitted bolts, and countersunk bolts shall be installed between each stem hole. All dowels and bolts shall be staked.

3.4.52.14.2 Button heads. Valve button heads may be either integral with stem or attached to stem *by* screw threads and shall have a spherical seating surface. (Exception to thi.!3will be considered when there is operating experience to indicate other shapes are acceptable,) Removable button heads shall be locked to th ir stems in a manner approved by NAVSEC. Welding to stem will not be permitted.

3.4.52.14.3 Contact area. Lift-bar seat (or seat insert if used) and button head contact surface shall be eithernard-faced with chrome-<;mbalt, MIL-R-17131, type MIL-RCocr-A, or be hardened to a minimum hardness of 200 BHN and stress-relieved.

3.4.52.15 Valve sprin s. Valve gear springs shall be cadmium-plated or otherwise corrosion protected ds sha1l be closed and ground flat and square. Nesting is permitted if necessary to obtain the desired characteristics; however, no compression forces shall be present in springs prior to removal of last retaining bolt or securing device.

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3.4.52.16 Overspeed t ip system. An overspeed trip system, independent of the constant-speed governing system, sha1 be furnished.

3.4.52.17 Definition of overspeed trip system. The overspeed trip system i defined as a system that will shut off all steam to the turbine nozzle control or governing valves, to fully stop the set in the event a predetermined speed is exceeded.

3.4.52.18 Manual trip. Means shall be provided for manual-tripping at any speed, The manual trip shall be located so that it is readily accessible to the man at the trip throttle valve, but cannot be tripped accidentally. If necessary to avoid accidental tripping, a guard shall be installed around the trip button, lever or handle.

3.4.52.19 Trie system basic design. The overspeed trip system shall be reliable, positive and quick acting, of simple and rugged design and construction.

3.4.52.20 Shockproofness. The trip system shall be designed to withstand the shock requirements of MIL-G-3087. In addition, the system shall be designed so that, under shock conditions specified in MIL-G-3087, the system will not be actuated unintentionally to shut down the set, but will function properly if overspeeding occurs.

3.4.52.21 Operation during sudden speed or load change. The overspeed trip system shall not be actuated unintentionally by sudden changes"""'i'ilspeed or sudden removal of any load within rated operating conditions.

3.4.52.22 ipeed setting. The overspeed trip system shall function automatically at 108 to 112 percent o generator rated speed.

3.4.52.23 Maximum speed after tripping. The overspeed trip system shall be designed so that tripping by overspeeding or manually, at any possible load or speed or during start-up or shut-down, shall safely limit the speed of the set to 120 percent of rated speed. The maximum speed that the turbine could attain after tripping and the conditions under which this speed would occur shall be determined by the turbine manufacturer and furnished to NAVSEC for information.

3.4.52.24 Provision for adjusting and testing onboard ship. The overspeed trip system shall be capable of being adJusted and tested onboard ship by shipboard personnel at a range of 108 to 112 percent of rated speed, using the speed-changer motor and by-passing the high speed stop. The high speed stop shall be reset and be limiting after the test. The test procedure shall be described in the technical manual.

3.4.52.25 Speed-sensing devices. Speed-sensing devices shall be of the following type:

(a) Centrifugal (bolt, plunger, belleville spring or two interconnected eccentric rings).

3.4.52.26 Relaying mediums. Relaying mediums may be any one, or any combination, of the following types:

(a) Mechanical.
(b) Hydraulic.

3.4.52.27 Shut-down device. The shut-down device shall be a trip throttle valve in accordance with 3.4.52.9. The trip throttle valve shall be independent of any other control valve and shall control all steam to the turbine nozzle control valves or turbine governing valves.

3.4.52.28 Resettin9. The system shall be designed to reset as follows:

(a) Speed-sensing devices shall reset automatically.
(b) Relaying medium (mechanical or hydraulic) shall reset automatically.
(c) Trip throttle valve shall require manual resetting before staFting up again (see 3.4.52.9).

3.4.52.29 Resetting spe d. The resetting speed for the system shall be not less than 50 percent of rated speed, Th technical manuals shall indicate the maximum resetting speed.

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3.4.53 Gland seal and leak-off system. A steam seal and leak-off system shall be provided to prevent theleakage of steam and water from, or ingress of air, oil or oil vapor into the turbines under any condition of operation. The system shall maintain 1/2 to 2 p.s.i.g. sealing steam pressure at the turbine gland and shall contain provisions for collecting leakage past the packings. The seal steam pressure shall be maintained automatically. For type I turbines (package type), gland leak-off shall be piped to the gland or after condenser furnished as part of the complete turbine-generator. For type II turbines (conventional type), means shall be provided to permit connecting gland leak-off to a gland exhaust system provided by the shipbuilder. The manufacturer shall include in "Bid Data" (see table IX), in technical manuals and on drawings the estimated quantities of steam and air flowa and required vacuum at leak-off connections, for normal conditions and when rings are worn to the maximum allowable clearance as designated by the manufacturer. Make-up steam for use during start-up will be provided by the shipbuilder at a pressure specified (see 6.2.2(r)). When operating at or near rated power sealing steam for the high pressure (HP) end gland shall be supplied from the first stage of the turbine itself, the LP end being sealed by the leak-off from the HP end gland. Excess steam from the HP end gland, not required for sealing shall be piped to the turbine exhaust. Compound pressure indicators, retard type, in accordance with MIL-I•l8997, shall be provided for use in controlling gland seal steam to both glands.

3.5 Onboard tools and maintenance pirts. Items listed in table V shall be furnished as applicable and snalT E'e"shown on NAVSH PS 4786 form. See MIL-G-3087 for tool delivery requirements.

Table V - Onboard_tools and maintenance parts.

Item	Quantity per ship	Reference
1. Special tools (as applicable)!/ For:		
(a) Measuring crown thickness	1 set	**3.4.38.2.14**
(b) Removing and setting overspeed devices	l set	**3.4.52.24**
(c) In-field balancing (weights not included)	l set	**3.4.27**
(d) Alignment of spherical seating bearings	**l** set	**3.4.38.2.9**
(e) Removing thrust collar	1 set	**3.4.38.3.S**
(f) Taking thrust clearance	1 set	**3.4.38.6.2**
(g) Unbolting and remaking casing flanges	1 set	3.4.28.3.10 and **3.4.43.4.1**
(h) Replacing stub shafts	**l** set	**3.4.38.3.5**
(i) charging apparatus and gage for accumulator, if used	1 set	**---**
2. Bearing yokes and jack	1 set	3.4.38.6.l
3. Bolt heaters and extensiometers	2 each size	**3.4.28.3.9**
4. Depth micrometer gage	l each **size**	**3.4.38.2.12**
S. Lifting gear	l set/generator set up to maximum of 4 sets	**3.4.43.12**
6. Turning tool	2 sets	**3.4.35**

-y Special tools are defined as those items not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the DCAS).

3.6 Repair parts.

3.6.l Onboard and stock repair sar sh Onboard repair parts shall be in accordance with table VI and snall beeither furnishe wit the turbine and gear or procured separately by the inventory control point. Stock repair parts (shore-based) shall be handled in accordance with procedures of MIL-P-15137 (see 6.4). Stock (insurance) repair parts to be recommended are shown in table VII.

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Table VI - Onboard repair parts for generator-turbines and reduction gears.

Item	Part, as applicable ito the turbine and gear furnished (A "set" means quantity required fo+ 1 turbine and gear)	Number of turbines and gears of same design per ship							Factor code (For Govern-ment use only)
		1	2	3	4	5-8	9-20	21-50	
1	Accumulator bladder and wearing parts including springs	1	1.	1	1	2	3	4	G
2	Basket, oil strainer, each size and type	1	1	1	1	1	1	1	K
3	Basket, steam strainer, each size and type	1	1	1	1	1	1	1	K
4	Bearings, complete all types, except thrust bearings, sets!/	1	2	3	3	4	6	7	D
5	Bearing parts, thrusts, including collars, complete sets	l	1	1	1	2	3	4	G
6	Bellows assemblies complete *setsY*	1	1	1	1	2	3	4	G
7	Cartridge, oil filter, each size and type	1	1	1	1	l	l	1	K
l;j	Deflector or seal rings, complete sets	l	2	3	3	4	6	7	D
9	Elements, sensing bearings, RTE. and connection-blocks, complete sets	1	1	2	2	3	4	5	E
10	Fasteners, self locking, threaded, complete sets	1	2	3	3	4	6	7	D
11	Gaskets which cannot be fabricated from standard Navy gasket material, complete sets	1	2	3	3	4	6	7	D
1.2	Gl.asses for 011. sight flow fittings and gages, complete sets	l	1	l	l	2	3	4	G
13	Governor parts, overspeed, complete sets	1	1	1	1	2	3	4	G
14	Governor assemol.1.es, mechanical/hydraulic, self-contained type similar to Woodward	1	1	1	l	2	3	4	G
15	Governor system parts, speed regulating, including operating cylinder, pilot valve, bushings, servo-mechanism, linkages, drive gears, and so forth, complete sets	1	l	l	1	2	3	4	G
16	o-:-rings, complete sets	1	2	3	3	4	6	7	D
17	orifices, removable, complete.::rets	1	1	l	1	2	3	4	G
18	Packing, gland, including springs, except when lifting casing is required, complete sets	1	l	2	2	3	4	5	E

See footnote at end of table.

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Table VI - Onboard repair parts for generator-turbines and reduction gears (cont'd).

Item	Part, as applicaole to the turbine and gear furnished (A "set" means quantity required for 1 turbine and gear)	Number of tu nes and gears _of same design per ship							Factor code (For Govern-ment use only)
		1	2	3	4	5-8	9-20	21-50	
19	Packing, valve, and so forth, (other than standard Navy packing carried onboard), complete sets	1	2	3	3	4	6	7	D
20	Pump parts, lube oil, including drive gears, if not included under item 15, gaskets, bearing bushings, or body when bushings are not used, complete sets	1	1	1	1	2	3	4	G
2J.	Springs, not including gland and diaphragm springs, complete sets!/	1	1	1	1	2	3	**4**	G
22	Switches, pressure alarm, each type, range and design	1	1	l	1	l	r	1	K
23	Trip parts, back p_res_sure and lube oil, complete sets	l	l	1	l	l	l	l	K
24	Trip parts, overspeed governor, complete sets	l	1	l	l	2	3	4	G
25	Turning devi ce parts, complete sets	l	l	l	1	l	l	l	K
26	Valve parts, nozzle complete, including lift rods, bush-ings, valves and seats, linkages not included in item 15, complete sets	l	l	1	1	1	1	1	K
27	Valve parts, relief, exhaust casing, complete sets	l	1	1	l	1	l	l	K
28	Valve, complete relief, oil and steam, each .size and type	l	1	1	l	l	l	l	K
29	Valve parts, trip tnrottle, including pilot valve, main valve and seats, snap rings, valve stem with bush-ings, and screw spindle with nut, complete sets	1	l	l	1	2	3	4	G

Y Quantities of bearing, item 4; bellows, item 6; and springs, item 21, are in addition to any such parts being furnished as component parts of other items.

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Table VII - Stock (insurance) repair parts

Item	Part(s)	Number of turbines and gears using part(s)	Stock recommended_/
(1)	(2)	(3)	(4)
1	Trip throttle valve	1 to 50 (incl.) 51 and up	1 2
2	Turbine rotors Y and gear main rotating elements Y	1 2 to 10 (incl.) 11 to 30 (incl.) 31 to 50 (incl.) 51 to 70 (incl.) 71 and up	1 2 3 4 5 6
3	Turbine nozzle blocks Y nozzle diaphragms and intermediate segments y	1 2 to 10 (incl.) 11 to 25 (incl.) 26 to 40 (incl.) 41 to 55 (incl.) 56 to 70 (incl.) 71 to 85 (incl.) 86 and up	1 2 3 4 5 6 7 8
4	Turbine blading J./	1 2 to 10 (incl.) 11 to 20 (incl.) 21 to 30 (incl.) 31 to 40 (incl.) 41 to 50 (incl.) 51 to 65 (incl.) 66 to 80 (incl.) 81 to 95 (incl.) 96 and up	1 2 3 4 5 6 7 8 9 10
5	Labyrinth packing Y and seal strips2./	1 2 3 4 5 to 10 (incl.) 11 to 20 (incl.) 21 to 35 (incl.) 36 to 50 (incl.) 51 to 65 (incl.) 66 to 80 (incl.) 81 to 95 (incl.) 96 and up	1 2 3 4 ... 5 6 7 8 9 10 11 12

y Where the identical part is used more than once in one turbine and gear, multiply the quantity shown *in* column 4 by the quantity required per turbine and gear.

Spare turbine rotors shall be completely bladed and be ready for installation without further machining, Thrust collar, coupling and overspeed sensing devices shall be included. Balancing is required and, unless specifically waived in the contract, the shop operating test of 4.5.9 shall be performed either in a casing for which rotor is intended or by use of a special testing facility.

Spare gear main rotating elements shall be furnished as matched sets. Thrust collars, couplings and all shafts, including quill or· flexible shafts, where used, shall be included. Balancing is required.

y Each stock-numbered nozzle block consists of all material necessary to effect replacement. Included as applicable, are the nozzle block (all segments for admission arc), bolts, calking, and seal strips (if inserted in nozzle block).

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V Bach stock-numbered nozzle diaphragm consists of all material necessary to effect replacement. Included, as applicable, are the nozzle diaphragm, seal strips (if inserted in diaphragm), keys, crushing pins, and any other parts other than diaphragm packing.

Y Each stock-numbered intermediate segment consists of all material necessary to effect replacement. Included, as applicable, are the segment, bolts, seal strips and calking. Blading shall also be included.

7./ Each stock-numbered item of blading consists of 105 percent of all materials necessary to replace one row of rotating or stationary blading. Included, as applicable, are blades, shroud bands, calking, seal strips (if inserted in blade or shroud) locking pieces, screws, pins and any other necessary parts.

Each stock-numbered item of packing consists of one complete replacement metallic labyrinth packing ring for use in gland, dummy or diaphragm. Included as applicab_le, are a ring, spring(s), retaining ring(s), screws, pins and retaining plates.

V A stock-numbered set of seal strips consists of 105 percent of all renewable calked-in seal strips and lock strips in way of rotating and stationary blading, diaphragms, and nozzle blocks for a complete turbine.

3,6.1,1 Inventory control point. Ships Parts Control Center (SPCC), Mechanicsburg, Pa, is the Inventory Control Point (ICP)for all repair parts.

3.6.2 Interchangeability. Repair parts shall be identical and interchangeable with their respective standard-installed parts on all turbines and gears of the same design, and shall be capable of being installed without any special fitting, machining, running-in, or other work, except as allowed below.

(a) Exceptions£!! approved drawings. Exceptions to above interchangeability requirements are permitted, if approved by NAVSEC and so noted on applicable drawings.

(b) Blading exception. All blading material, except shrouding, shall be furnished completely finished and ready for use, except for design allowance if any on blade roots for fitting. Blade material which is to be installed in segmental form shall be furnished assembled in segments.

(c) Shrouding exception. Shrouding shall be furnished cut to length. Surplus stock for axial trimming to obtain specified clearances is acceptable. Unless otherwise approved by NAVSEC (see 3.4.41.9,4), shrouding shall be undrilled and unpunched for blade tenons.

(d) Diaphragm crush pins. Crush pins shall be fitted at installation.

(e) Shims. Shimsfor bearings, oil pump drive, turning gear, etc. may be fitted ei:tTnstallation.

(f) Double seated valves. Double seated valves shall be lapped to fit at installation.

(g) Nozzle control valve excektion. Nozzle control val.v.es, of the button head type, furnished for stoc shall have 1/16 inch extra material on the head to permit fitting at installation.

(h) Nozzle blocks. Calking strips and spacers shall be fitted at installation.

(i) Dowels. Components utilizing dowels will require locating and redowelling with the next larger size dowel.

(j) Gears. In order to obtain.the contact pattern area designated in 3.4.48.9.l it may be necessary to realign the turbine-generator (T-G) set, rebore or rescrape the bearings as determined by the installing activity.

(k) Governors and overspeed trips. Settings of governors and overspeed trips must be rechecked after installing replacement parts.

(1) Bearings, casings, oil seals, packing. Upper and lower halves of bearings, casings, oil seals and packing rings (gland and diaphragm) are manufactured together as complete circles and must be furnished and installed as complete assemblies rather than in halves or segments.

3.6.3 Provisioning list. NAVSHIPS 4786 and 4786A forms are required for all repair parts and shall be prepared in accordance with MIL-P-15137 (see 6.4) except that the turbine manufacturer shall include in the lower right hand margin a drawing title block and revision column. The provisioning list shall be used as a drawing of onboard parts and may be used as packing and shipping list. Applicability of parts shall be indicated in remarks column.

3.7 Drawings. Drawings shall be in accordance with the requirements of MIL-G-3087 and 3.7.1 through 3.7.5.2.

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3.7.1 Marking. Drawings shall sp cify marking detail (where and how) when purchase specifications contain marking requirements and when marking to be done by the manufacturer requires control to. prevent stamping on highly stressed parts.

3.7.2 "Manufacturer's use only" notes. Information intended for manufacturer's use only shall be so designated.-- ---

3.7.3 Material identification, Material for all parts shall be identified on detail drawings or parts list as applicable, Material may be defined by manufacturer's specifications for non-specified parts and for principal parts when approved in accordance with 3,2,8; however, double identification is required. This may be on the same drawing by a reference to a government or national society specification which identifies an equivalent material for use in conducting repairs or in manufacturing the part. An alternate method that will be accepted is to provide a separate drawing (A size), to be called a "Material List", which contains a complete list of the manufacturer's specifications in numerical order with equivalent government or national society specification indicated for each manufacturer's material. If the alternate method is used, reference to the "Material List" shall be shown on all drawings which require material identification. Existing drawings used for new contracts need not be revised solely to reflect changes in Government material specifications referenced thereon, however when drawings require revision for other reasons, the referenced specifications shall be updated at the same time.

3,7.4 Special drawings. The drawings specified in 3.7.4.l through 3.7.4.14 are required in addition to all drawings required by MIL-G-3087 and all drawings required for manufacture, assembly, installation and maintenance of the turbines and gears including the following:

(a) Outline and arrangement (see MIL-G-3087 and 3.7.4.1).
(b) Turbine and gear assembly (see 3.7.4.2).
(c) Subassembly (see 3.7.4,3).
(d) Detail (see 3.7.4.4).
(e) Clearance diagram (see 3.7. .5).
(f) Control system diagram (see 3.7.4.6),
(g) Oil system diagram (see 3.7.4,7).
(h) Lifting and handling-arrangement (see 3,7.4.8).
(i) Lagging and insulation (see 3.7,4.9).
(j) Gland seal and leak-off diagram (see 3.7.4.10).
(k) Instrumentation drawing (see 3.7.4,11).
(1) Drawing list (see 3.7.4.12).
(m) Machinery variation summation drawing (see 3.7.4.13).
(n) Welding and non-destructive testing (see 3,7.4,14).

3.7.4,1 Outline and arrangement (or installation) drawing, In addition to the requirements of MIL-G-3087 theoutline and arrangement drawing shall contain, as applicable, the following, except that no information with a security classification shall be included:

(a) Critical speed,
(b) Low oil pressure alarm switch setting.
(c) Casing relief valve setting.
(d) Sight flow fittings.
(e) Alignment requirements.
(f) Indicate which drains are open only at certain times and therefore can have shut-off valves, and which drains are continuous and will require traps. (Valves and traps will be shipbuilder furnished).
(g) Add to data required by MIL-G-3087 for the gland seal system, the following:
    (1) Gland leak-off flows when gland packing is worn, to maximum clearance designated by the manufacturer.

3,7.4.2 Turbine and gear assembly drawings. Turbine assembly drawings and gear assembl drawings are requi ed. They may be furnished either separately or as a combined drawing and shall include the following:

(a) Side elevation of upper and lower half turbine and gear sectioned longitudinally, with enlarged view of casing strength stay rods, tie webs, ribs and and other parts provided for positioning and strength,
(b) End elevation (upper and lower-half) with transverse section (facing nozzle block) through valve control gear, nozzle control valves, and steam chest. Also gear end elevations.
(c) Principal parts shall be identified by name and material in a list of material which may be integral with, or referenced on, the assembly drawing.

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(d) Special views of valve control parts are required to indicate details of cam lift bar, poppet stem button head1 discs, seats, and bushings or seals (and how retained).
(e) Rotor bore, if applicable.
(f) All planes for shop and in-field balancing shall be shown and identified, and method of attaching balance weights shall be indicated.
(g) Gland, dummy, and diaphragm packing rings shall be numbered consecutively (preferable beginning at the thrust end)i geometry, construction and installation clearance of a representative ring of each category shall be indicated by special enlarged views and sections.
(h)
(i) Blade clearance (representative radial and axial values) shall be indicated.
(j) Journal and thrust bearing clearances (and thrust collar position) shall be indicated.
(k) Detail of thrust nut locking feature.
(1) Detail of overspeed protective devices.
(m) Nozzle block and diaphragm support and retention features, along with diaphragm crush pins, shall be shown by enlarged view or section.
(n) Identification of gland seal and leak-off connections. (Unless identified in outline).
(o) Identification of oil supply and return connections. (Unless identified in outline).
(p) Identification of oil deflector, waste oil-and-water pockets (if any) and drainage of same.
(q) Water-removal features and casing drains shall be indicated, along with representative dimensions involved.
(r) Representative views of threaded fasteners shall be included, along with means of securing and retaining same.
(s) Dowels·1ocating major subassemblies shall be shown by enlarged view or section.
(t) Flange stud and bolting schedule, giving size, projection, and whether or not heater hole is provided.
(u) Means of providing for longitudinal and radial expansion (flex-leg, keys, clearance bolts, etc.) of casing shall be shown or described in a note.
(v) Attached instrumentation and fittings shall be clearly shown or enlarged.
Miscellaneous small or loose parts not covered or specifically referenced on other drawings.

3,7.4.3 Subassembly drawings. These drawings shall be provided as necessary to show details of subassemblies not included in the cross-sectional assembly. Included in these drawings shall be one drawing or drawing list to describe the completely bladed and assembled turbine rotor with all necessary rotating parts such as thrust cqllar oil pump drive gears and overspeed trip bolt or plunger (if installed). This drawing is required for ordering stock rotors.

3.7.4.4 Detail drawin s. Details of all parts including identification plates, must be delineated, except thatetails of standard commercial items such as nuts, bolts, antifriction bearings, and other similar items shall not be furnished when they can be described by catalog number, size, and so forth, on the cross-sectional assembly drawing or a subassembly drawing.

3.7.4,5 Clearance drawing and overhaul report form. A combined clearance and overhaul report drawing shall be preparedfor each turbine anugear designi the drawing shall contain a longitudinal assembly view of the rotor with detail views indicating "cold" clearances for parts as listed in 3.7.4.5.1 through 3.7.4.5.7 in tabular form with columns entitled "design" (values filled in) and blank spaces for "as found" and "as closed" clearances for use by repair activities. Drawing shall indicate tolerances for installation and maximum allowable clearances for which replacement or reconditioning of parts is recommended.

3.7.4.5.1 Blading clearances. Clearances (radial and axial) shall be shown for rotating blading at its closest approach to casing, diaphragm, nbzzle block, seal strips and other · non-rotating parts.

3.7.4,5,2 Bearing clearances. Diametral clearances shall be shown for journal bearings, The clearance or fit between each bearing shell and its cap shall be indicated. Axial float of the· thrust collar shall be shown with the rotor hard up against either set of shoes.

3.7.4.5 ,3 Diaphragm and glan packing clearance. Axial and radial clearances shall be shown for each labyrinth packing ring. Provision for recording axial clearances is not required.

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3.7.4.5.4 Q!! seal clearance. Radial clearances shall be shown for each oil ring.

3.7,4.5.5 Rotor radial position setting, Where reference measurements are used for checking radial position of rotors utilizing bearings with adjustable seating pads, the radial reference measurements to be used as constants and method of measurement shall be shown.

3.7.4.5.6 osition. Provision for determining axial position of rotor (see 3.4.45.9) shall be indicated.

3.7.4.5.7 Notes. There shall be notes on the drawing similar to the following:

Note / - All radial and diametral clearances shown are with the rotor journals--- (manufacturer shall state here whether journals are concentric with casing bore or resting in the bottoms of bearings). Two or more radial clearances shall be shown for parts having their axes eccentric with rotor axis,
Note 2 - All clearances shown are "installation clearances", which should be used as a guide when blading, bearings, thrust shoes, packing rings or oil seals are reconditioned or replaced.
Note 3 - If applicable, include caution note regarding taking total float measurements which can damage packing teeth. If total float can be taken, show expected value.

3.7.4.6 Control system diagram (drawing). A schematic drawing of the control system shall be furnished. Drawing shall indicate valves and linkages and include information for setting and adjusting control mechanisms. The control system diagram shall contain a description of how the control system operates and procedures for start-up, operation, and shut-down.

3.7.4.7 Oil system diagram. This drawing may be combined with the control system diagram for those turbines having hydraulic governors.

3.7.4.7.l Schematic. The drawing shall indicate schematically bearing inlet and drain connections and, as applicable, oil connections to overspeed mechanisms, control mechanisms and gland seal regulator. Piping shown, but not furnished by the turbine manufacturer, shall be dotted or phantomed, while solid lines represent items and connections furnished with turbines and gears. Size of each connection shall be shown

3.7.4,7,2 Flow data. Values of oil flow (g,p.m.) shall be shown at each connection. Direction of flowsli'aIIoe indicated by an arrow. By note or table, drawing shall also indicate conditions for which oil flows apply (state percent power, and pressure).

3.7.4.7.3 Bearing orifices. Size and location of bearing orifices shall be indicated. Where orifices are integral with bearing shell, a note to this effect (giving a size) is acceptable,

3.7.4.7.4 Heat rejection. The drawing shall show heat rejection data used in sizing the lube oil cooler.

3.7.4.8 Liftin and handling arrangement drawings. Drawings shall illustrate the use of lifting and handling gear furnished for rotor and casing. Weights of parts to be lifted or handled shall be shown.

3.7.4,9 La ging and insulation. The turbine manufacturer shall furnish a drawing showing the application of--rii'sulation and lagging (see 3.4.37).

3.7.4.10 Gland seal and leak-off diagram.

3,7.4.10.1 Schematic. This drawing shall indicate schematically the turbine connections for gland seal, leak-offs from glands and valve sterns, and casing and chest drains. Piping shown, but not furnished by turbine manufacturer, shall be dotted or phantomed, while so1id lines represent items and connections furnished with turbines.

3,7,4,10,2 Direction of flow. The direction of flow in each line shall be indio.ated by an arrow or other means.--

3.7.4,10.3 Connection sizes. The size of each line and connection shall be indicated,

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3,7.4.10,4 Flow data, The values (seal and vent) of steam flow, air flow and temperature at each gland seal shall be tabulated for specified full load and standby conditions with normal clearances. Capacity of regulator and dump valve shall be shown. An ambient air tempeature of l20°F. shall be used for calculations.

3.7.4.10.5 - There shall be notes on the drawing similar to the following:

Note l - The capacity of the gland seal system, gland exhauster system, and leak-off piping shall be based upon "maximum clearance" flows: (corresponding to "maximum clearances" which require replacement of packing due to to excess leakage)i these maximum flows, calculated for 2 p.s.i,g in the seal pocket and 5 inches of water vacuum in the leak-off pocket, are as follows: (manufacturer shall indicate values here),

3,7,4,11 Instrumentation drawing, An instrumentation drawing shall be furnished and shall contain an outline of the turbine and gear (with cut-out views as necessary) indicating all points of instrumentation either furnished with_ the turbine and gear or by shipbuilder. A table showing maximum expected values of temperature and pressure shall be provided, Relief valves (with settings given), *rotor* position indicator, sight flows, and bearing RTE's (include location of external connections) shall be included. In lieu of furnishing a separate instrumentation drawing the information required above may be shown on the outline drawing providing the drawing is not unduly cluttered.

3.7.4.12 Drawing list. A drawing list is required for each turbine and gear and all drawings applicable to eq\iTpment furnished shall be listed thereon. The list shall provide a complete breakdown with an alphabetized index thereunder by major assemblies. It is the intent of this requirement to carry part generation breakdown to the detail drawings which support the subassemblies even though the detail parts may lose their identity in the assembly, The list of preferred materials and comparison sheets are not included in the drawings to be listed,

3,7,4.13 Machinery variation summation drawing. A machinery variation drawing shall be furnished for each class of ships. (Class of ships as used here refers to those units covered by the same technical manual.) This drawing shall contain a post production summation of all variations which involve special repair parts or affect maintenance actions (see 3,12) and shall indicate for each variation a brief description of the deviation from the basic drawing, the serial number of unit affected, the hull number of the ship to which assigned, and the necessary correlation with special parts and repair parts which result from the variation.

3.7.4 ,14 Welding and nondestructive testing (NOT) information, Drawings shall show all essential welding fabrication details and areas requiring nondestructive testing. Welding and NOT symbols shall comply with ANSI Y32,3 and AWS A2,2 respectively. Fabrication drawings shall reflect the kind of weld and shall state that welding is in accordance with MIL-STD-278.

3.7.5 Drawing approval.

3,7,5.1 Drawings requiring approval: (categories MIL-D-1000),

(a) Category A, The following drawings require NAVSEC approval,
(1) Assembly drawing(s) (see 3.7. 4,2)•
(2) Clearance drawing (see 3,7.4.5),
(3) Applicable drawings indicating radiographic inspection (see 4,4,3),
(4) Onboard repair parts list (see 3,6,3).
(5) Instrumentation drawing (see 3.7.4,11).
(6) Welding drawings (only those drawings of welds that require radiography) (see 3,7,4,14).

(b) categories Band G. For shipbuilder-furnished equipment, the following drawings require approval by the Cognizant Supervisor,
(1) Outline and arrangement drawing(s).
(2) Lifting arrangement drawing.
For government-furnished equipment, the above require procuring activity approval.

(c) Categories H and I. Except for-the above, other drawings in these categories shall be self-approved by the turbine manufacturer.

(d) Revisions to ant category drawing. Revisions to drawings which have been approved by t e Government and contractor approved drawings which depict a part or component as furnished require approval of NAVSEC when the revision affects either installation, performance, maintainability or stock repair parts.

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3,7.5,2 Approval rocedure, The detailed procedure for submitting original and revised drawings for approvals all be as specified in MIL-G-3087,

3,8 Microfilm. Microfilm shall be furnished in accordance with MIL-G-3087.

3.9 Technical manuals, Technical manuals shall be furnished in accordance with MIL-G-3087,

3,10 Technical books.

3.10.l Data categories. Technical data shall be categorized as shown in 3.10,l.l through 3.10,1,3.

3.10,l.l Category! data. Category I data covers design details.

3.10.l.2 Category II - Category II data covers performance.

3.10.l.3 Cateaory III data. Category II data shall be submitted to NAVSEC only and shall be safeguarde as proprietary data to protect the manufacturer's commercial interests. Each data sheet in this category shall contain. "CAUTION: FOR NAVSEC USE ONLY - DO NOT MICROFILM OR OTHERWISE REPRODUCE WITHOUT CONSENT OF (NAME OF MANUFACTURER)--".

3.10.2 Content of data - Data books shall contain all.data listed in column 7 of table VIII.

Table VIII - List of technical data requirements.

Data category	Item number	Data description	Ordering data	To be furnished by manufacturer in: Bid or proposal	Technical manual	Tech data book	Remarks
(1)	(2)	(3)	(4)	(5}	(6)	(7}	(8)
I	.L	Nozz.te block					
		(a) Number of groups	-	-	-	X	
		(b) Number of nozzles per group	-	-	-	X	
		(c) Throat area per group		-	-	X	
		(d) Active arc	-	-	-	X	
	2.	Dia12hragms stage)					
		(a) Nozzle height	-		-	X	
		(b) Nozzle throat area		-	-	X	
		(c) Active arc	-		-	X	
		(d) Number of nozzles	-	-	-	X	
	3•	Blading stage}					
		(a) Bla es per row indicating number of gaps	-		-	X	
		(b) Mean diameter	-	-	-	X	
		(c) Blade height (excluding root and tenon)	-	-	-	X	
	4.	Reauction gears					
		(a) Gear efficiency	-	-	-	X	
		(b) BHP	-	-	X	X	
		(c) Pinion r.p.m.	-	-	X	X	
		(d) G ar r.p.m.	-	-	X	X	
		(e) Reduction ratio	-	-	X	X	
		(f) Pitch diameter (inches)	-	-	X	X	
		(g) Number of teeth (pinion)	-	-	X	X	
		(h) Number of teeth (gear)	-	-	X	X	
		(i) Addendum (inches) (pinion)	-	-	-	X	
		(j) Addendum (inches) (gear)	-	-	-	X	
		(k) Dedendum (inches) (pinion)	-	-	-	X	
		(1) Dedendum (inches) (gear)	-	-	-	X	
		(m) Diametral pitch	-	-	-	X	
		(n) Helix angle (degrees)	-	-	X	X	

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Table VIII - List of technical data requirements (cont'd).

Data category	Item number	Data description	Ordering data	To be furnished by manufacturer in			Remarks
				Bid or proposal	Technical manual	Tech data book	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
I	4.	Reduction gears (cont'd.)					
		(o) Normal pressure angle (degrees)	-	-	-	X	
		(p) Normal pitch	-	-	-	X	
		(q) Tooth thickness on pitch diameter	-	-	-	X	
		(r) Face width	-	-	X	X	
		(s) Pitch line speed (ft/min)	-	-	-	X	
		(t) Total load (pounds)	-	-	-	X	
		(u) Tooth pressure per inch of face	-		-	X	
		(v) "K" factor	-	-	X	X	
II	1.	Material tabulation	X	X		X	Principal parts
	2.	Size and space	X	X	X		Outline drawing
	3.	(a) Weight target (total)	X			-	
		(b) Weight estimate (component (specific weight)	-	X	-	X	
		(c) Final weights (total and of principal parts)	-	-	X	-	May be shown on drawings
	4.	Induction pressure and flow data for each point	X	X	X	X	
	5.	Curves for correcting operating (trial) conditions to design conditions	-	-	-	X	See figure 6
	6.	Nomograph, relating expected variations in KW, gear r.p.m. pressure, temperature, vacuum	-	-	X	-	
		Rotor criticals (rigid bearing and running)	-	-	X	X	
	8.	Tabulation of the following design data for specified steady state conditions.	-	-	X	X	
		(a) R.p.m. (ratea,					
		(b) KW output					
		(c) (Throttle :t:low (lb.;hr)					
		(d) Inlet steam conditions pressure and temperature)					
		(e) Steam rate (lb./KW-hr)					
		(f) Calculated maximum speed after actuation of over-speed trip					
		(g) Stage pressures and temperatures at measured points					
		(h) Exhaust vacuum, temperature moisture content and enthalpy					

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Table **VIII** - List of technical data requirements (cont'd.),

Data category	Item number	Data description	Ordering data	To be furnished by manufacturer in			Remarks
				Bid or proposal	Technical manual	Tech data book	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
II	9,	Tabu.Lation at following data for each bearing:	-	-	-	X	
		(a) Size (diameter and active length)					
		(b) Projected area (in^2) (inside drain grooves)					
		(C) Load (p.s.L) on proJected area					
		(d) 011 required (g.p.m, and press) at full load					
		(e) Surface speed (r.p.m.) at rated speed					
		(f) Diametral clearance (minimum and maximum)					
		(g} Location and size at orifices					
	10.	Operational limitations for principle parameters for which instrumented (pressure, temperature, r.p.m. and so forth) include curves of stage pressures (where measured) versus load	-	-	x	-	
	.Ll.	curves, on a base of kW over entire oEeratin ran9e, snowin9:					
		(a) Turbine steam rates (including all losses and all gland sealing steam) and flow indicating, valve points and assumed gear efficiencies at design points	-	-	-	x	Include Willans line
		(DJ Maximum vacuum the turbines can usefully (and safely) employ	-	-	x	-	
		(C) Gland sealing steam and leakoff quantities (design condition)	-	-	X	x	
		(d) Stage pressures wnere measured	-	-	-	X	
		(e) HP output at turbine coupling	-		-	x	Horsepower values to be based on gear output

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Table VIII - List of technical data requirements (cont'd.).

Data category	Item number	Data description	Ordering data	To be furnished by manufacturer in			Remarks
				Bid or proposal	Technical manual	Tech data book	
(1.)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
II	12.	Table of values on a base of r.p.m. at rated conditions, showing the following torques under short circuit conditions					
		car Torque at each turbine coupling	-	-	-	x	
		(bJ Torque at gear coupling of reduction gear	-	-	-	x	
	.Lio	curves, on the base of r.p.m., over entire operating range showing:					
		(a) Rate of luoe 011. circulation required and an estimate of the required pressure at the bearing side inlet connections	-	-	-	x	
		(b) Rate or neat removal required from the lubricating oil	-	-	-	x	
	14.	Torsional vibration analysis, curves and data (including frequencies, highest shaft stress relative to design stress for high and low speed spans, WR^2 and stiffness values used in calculations)	-	-	-	x	
	15.	curves, valve J.ift versus load			x	x	
	16.	Mo.uier charts (full load and 60 percent load)	-	-	-	x	Show actual state lines
III	l.	Curve of turbine e 1c1ency versus turbine horsepower	-	-	-	x	
	. 2.	Curves of OJ.aae ana rotor material rupture strength and creep (O.1 percent) strength versus temperature for 1,000, 10,000, 40,000, 60,000 and 100,000 hours	-	-	-	-	Include with material com-p rison sheets
	3.	Blaae and rotor tabuJ.ation {by sta2esJ includin2:	-	-	-	x	
		{aJ Stage temperature (metal)					
		(b) Material strength at temperature					
		(C) mnimum factor of safety					
		(QJ Maximum or .11miting va.1ue and direction of stress in rotor section (groove, disc, wheelneck, etc) and blading (tenon, vane, tang, hook, neck, shank, etc)					

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Table VIII - List of technical data requirements (cont'd.).

Data category	Item ntlln-ber	Data description	Order-ing data	To be furnished by manufacturer in			Remarks
				BJ.d or pro-posal	Tech-nical man-ual	Tech data book	
(lJ	(2)	(JJ	(4J	(5J	(6J	(7)	(BJ
III	3.	Blade and rotor tabulation IDV stagesJ inc.Lucting: \cont•Ct.J					
		(e) General notes statin2:					
		(1) Stresses are direct centrifugal unless otherwise indicated					
		(;lJ Limitlng or maximum stress and safety factors indicated are for rated.kw and r.p.m., unless other-wise stated					
		(3) Ratio of allowaole sheer stress to tensile stress					
	4.	Tabulation \DY stages) of tne following (0 to 125 percent rated soeed)	-	-	-	X	Numoer of nozzles per diaphragm, number of blades per shrouded group and numbeI of shrouded groups shall be indicated.
		(a) Frequency eva.1uation ot tangential, axial and torsional modes	-	-	-	-	
		(DJ Maximum v10ration stress and location (tenon, vane, etc.)	=	=	=	=	
		(C} Steam bena1ng stress					
		(dJ R.p.m.(SJ at wn1cn resonance occurs	=	-	=	-	
		(e) Centrifugal stress	-	=		-	
	5.	Short circuit torque multiplying factor			-	X	
		(aJ Ca1cu1ations snowing stresses of affected parts under maximwn short circuit conditions	-	-	-	-	

3.10.3 Data format and identification. Data sheets may be typewritten or hand written and reproducecl"'"'Ey photographip or offset methods and shall be contained in paper binders. Cover of binder shall indicate ship application involved. Each sheet shall be dated. A listing of contents (with data listed in order required by table VIII) shall be included.

3.10.4 Submittal of data. Data books shall be submitted for information as soon after award of contract as availaE!i, (but no later than 120 days) and shall be as cdlnplete as practicable. Thereafter, data sheets (or pages) shall be submitted as prepared or revised. Distribution shall be in accordance with table IX.

Table IX - Distribution of data books

Data category	NAVSEC (6145) (direct mai1)	Each building shinvard	Cognizant SUPSHIPS (for info)	NAVSEC Division Phila. Pa.
I	j	0	0	l.
II	3	3	1	l
III	3*	0	0	0

*Denotes use of registered mail.

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3.10 :5 List ff technical requirements. Table VIII is a composite list of technical data requirements for technical manuals and technical data books and shows interrelationship with ordering data and bid data.

3.11 Reports. A schedule f9r submission of the items listed below shall be prepared by the turbine manufacturer and shall be forwarded to the following within 90 days after award of contract:

NAVSEC - 3 copie$
Each shipyard concerned - 3 copies
Each supervisor involved - 3 copies

The schedule shall list each item, indicate expected date for submission, and contain a column to be filled in when the item has been completed. Schedule shall be revised as necessary with final submittal to be used as a check on completion of the items involved. Items to be included, as applicable., are as follows:

(a) Drawings listed by title and drawing number if available (see 3.7)
(b) Blade stress data (see 3,2.5.2)
(c) Technical data books (see 3.10)
(d) Microfilm (see 3.8)
(e) Vibration test report .(see 4.5.10)
(f) Blade vibration test agenda and report (see 4.5.12)
(g) As-shipped clearances (see 4.5.3)

3.12 Equipment variations. Equipment variations shall be handled in accordance with the procedures in MIL-G-3087.

3.13 Quality control records. Quality control records shall be in accordance with MIL-G-3087.

3.14 Workmanship. Workmanship shall be first class in every respect.

3.15 Desi approval, As indicated within this specification, NAVSEC approval is required for a nu er of general design details and for other specific areas applying to a particular design of turbine and gear to be furnished for a given application. Approvals required and procedures for obtaining approval shall be as specified in 3.15.2.

3.15.1 Summary approvals required.

3.15.1.1 General design practices.

(a) Method of protecting 12 percent chrome rotor journals (see 3.2.7).
(b) Substitute materials (see 3.2.8).
(c) Method of locking removable button heads on nozzle control valves (see 3.4.52.14.2)•
(d) Acceptance criteria for blade stresses (see 3.2.5.2).
(e) Details of bottoming studs (see 3.4.28.3.7).
(f) Means of locking nuts (see 3.4.28.3.5).
(g) Use of pre-punched shrouding (see 3.4.41.9.4).
(h) Use of shims in blade installation (see 3.4.41.10.2).
(i) Method of installing seal strips (see 3.4.41.12.4).
(j) Exceptions to spare part interchangeability (see 3.6.2).
(k) Repair procedures (equipment variations) (see MIL-G-3087).
(1) Welding procedures (see MIL-STD-278).
(m) Instrumentation system for blade tests (see 4.5.12).
(n) Procedure for making-up steam joint (see 3.4.43.4 4).

3.15.1.2 NAVSEC approvals required for design.

(a) Exceptions to specifications (see 3.4.9).
(b) Use of ductile iron (see 3.2.6).
(c) Blade vibration stress data (see 3.2.5.2).
(d) Use of shrunk-on wheels (see 3.4.45.1).
(e) Drawings specified by 3.7.5.
(f) Preliminary technical manual (see MIL-G-3087).
(g) Blade vibration test agenda (see 4.5.12).
(h) Equipment variations (see MIL-G-3087).
(i) Requirements for stellite shields on blades (see 3.4.41.6).

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3.15.2 Approval procedures.

3.15,2.1 General desi n practices, Approval of the items listed in 3.15.1.1 are required once and do not require re-approval under each contract. Accordingly, the manufacturer may prepare a booklet of "design practices" and obtain across-the-board approval from NAVSEC for these items.

3.15.2.2 Specific turbine and g ar designs. Design approval conferences shall be held to review the overall design. It is intended that drawing approvals be handled concurrently at the conferences to minimize delays and repeated exchange of letters.

3.15.3 Design release conferences. As soon as practicable after award of contract, the manufacturer shall request a conference at NAVSEC (request to be forwarded via shipbuilder and cognizant supervisor if applicable) for the purpose of obtaining approval of basic design and ordering of material. Areas to be covered at the conference include but are not necessarily limited to the following1 however, partial coverage of these items will be considered if necessary to grant timely approval of long lead time components.

(a) Interchangeability of design or parts thereof to previously furnished designs.
(b) Materials of principal parts (minimum of items listed in table I) including identification of approved substitute materials proposed.
(c) Basic design features.
(d) Predicted performance.
(e) Arrangement.
(f) Weights.
(g) Specification clarifications if applicable.
(h) Control system.
(i) Need for stellite shields on blades,
(j) Lubricating oil system.

3.15,3,1 Data to be furnished prior to conference. At least 2 weeks prior to the conference, themanufacturer shall forwarda proposed agenda with background information regarding items to be discussed. Preliminary drawings (such as the outline and arrangement drawings and assembly drawings), material lists, technical data, and other documentation as necessary to permit review by NAVSEC prior to the conference shall be included.

3.15.4 Design review conferences. Other conferences shall be held as necessary to approve the final design and to discuss progress or problems that have arisen. As in the case of the design release conference, request for conference shall be mad at least 2 weeks prior to the desired date and shall include a proposed agenda with supporting information. It is also intended that NAVSEC comments on drawing subrnittals be discussed at these conferences.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier"'"Is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements,

4.2 Quality program. The supplier shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858 (see 6.2.2(s) and 6.4).

4.2.l Quality provram re uirernents. The program shall be described formally in a quality control manual in dept to assure that technical and quality requirements will be met. The manual shall be s·ubrnitted to the Government for determination of acceptability. Such determination will not pf'eclude the request of the Government for additions, refinements or change when evidence indicates the need for such action. Once the manual has been accepted, any subsequent changes shall be submitted to the Government for review, The,contractor shall implement, as a minimum, the program described in the quality control manual. Conformance **with** these requirements shall not be construed as relieving the supplier of final responsibility to furnish equipment meeting specification requirements. Unless otherwise approved by NAVSEC, the Product Inspection type A requirements which will be imposed are as follows:

(a) Overspeed test - (each unit) (see 4.5.16 and 4.5.17).
(1) Verifytfiat c'oritractor•s test procedures and Instrumentation are adhered to,
(1) Verify vibration data.

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(b) Post-test examination (periodic test unit) (see 4.5.2.4, 4.5.3 and 4.5.7).
(1) Rotor assembly.
a. Verify that blades are properly installed; check confprmance to drawing requirements on alignment, pitch and tightness.
b, Verify tenon peening for conformance to requirements; assure absence of cracks and overworked material.
c. Verify radial and end gaps of shrouds for conformance to drawing requirements.
(2) verify proper assembly of seals and examine.for condition.
(3) Examine each row of blades for shroud lift.
(4) Examine throughout for rubbing and damage to internal parts.
(5) Verify that all balance weights are staked in accordance with drawings.
(c) Post-test examination (each unit) (see 4.5.4).
(1) Examine all journals for scoring.
(2) Examine all bearings for damage.
(d) Radiography (each unit) (see 4.4.3)•
(1) Rev ew each ND'l' (radiography) result and submit those required by 4.4.3 to the contractually-designated Government representative for approval.

4.3 Material examination. Where substitute materials have been approved, examination requirements of the substitute specification shall apply. Examination of materials for parts not specified in table I shall be to the company's specification or standard even though identification by government or technical society specification is required (see 3.7.3). The use of materials other than those called for on drawings is permitted on a case basis subject to review and acceptance by the local Defense Contract Administration Services (DCAS)•. For parts listed in table I, alternate material must have been approved by NAVSEC, and in every case, the material to be used must be such that required factors of safety are met and where applicable, the essential variables of repair welding procedures for the material called for by the drawing remain applicable to the alternate material.

4,4 Nondestructive testing. Nondestructive testing methods shall be in accordance with the methods of MIL-STD-271,

4.4.1 Hydrostatic test. Hydrostatic tests shall be performed for the items shown in table X. Test pressure shall be determined as follows:

$$Pt = 1,5 \times P_0 \times \frac{st}{so}$$

•. Where:

pt = Hydrostatic test pressure (never less than 30 p.s.i.g, nor more than that necessary to produce the minimum yield of the material).
st = Allowable stress under test conditions based on ASME boiler code for material used (see section VIII of ASME Boiler'-and Pressure Vessel Code),
so = Allowable stress at operating temperature based on ASME boiler code for material used (see section VIII of ASME Boiler and Pressure Vessel code).
P_0 = Maximum operating pressure (setting of lowest steam generator safety-valve to be used for steam chests, excluding nozzle bowls).

Table X - Nondestructive testing requirements,

Part or service	MT/PT	RT	HT	UT	Visual
1. Thrust bearing collar	X-2	-	-	-	-
:/, Bl.ades and locking pieces	X-VA				
J. Nozzle partitions					X
4. Packing and packing springs					X
S. Packing casings	X-3				
6, Pins {notch bl.QCk ana l.ocking pieces!	X-1				
7. Fabrication welds in control mecharusms	X VA			X-VA	
8. Shrouding				-	X
and ooerating gear	X-3	-	-	-	-
10. control. va.1.ves					
(a) Poppets and seats, life rods, valve. stems, bushings and. lift bars	-	-	-	-	-
(b) Seal welds and exoansion rings	x-2	-	-		X
11. G.l.ana seal. regulator noav invaraulic tvoel	X-3		X		
12, Pipes (steam)					
(a) Below 300 p.s.i.	-	-	X	-	-
(b) Above 300 p,s.i.	-	-	X	-	-
l.!!S, Lube 01.1. sumps					Xcsee 4,5,13)

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NOTES: 1. Inspections required in table X do not eliminate requirements of applicable material specifications.

MI' - Magnetic particle test
PT - Dye penetrant test
RT - Radiographic test
HT - Hydrostatic test
UT - Ultrasonic test

X - Indicates te t required
X - Followed by number indicates acceptance level
X - Followed by VA indicates vender levels as approved by NAVSEC

2. Inspections required in table X are in addition to the requirements in Appendix A of MIL-STD-278.

4.4.2 Magnetic particle and dye-penetrant tests. There shall be 100 percent magnetic particle or dye penetrant testing of all accessible areas for the items shown in table X. Acceptance standards shall be in accordance with NAVSHIPS 0900-003-8000 except that note 1 of table II does not apply.

4.4.3 Radiography. Radiographic inspection shall be conducted for the items shown in table X and Appendix A of MIL-STD-278. Extent of coverage shall be as follows:

(a) Casings and steam chests.
(1) ExterriaTp'ressure boundary walls above valve seats (nozzle control and by-pass) 2-1/2 inches and larger.
(2) Steam inlet - 2-1/2 inches and larger.
(3) Steam chest flange and cover (castings only).
(4) Front wall and first stage shell area (horizontal flange excluded) where inlet pressure to chest is 900 p.s.i. or greater.
(5) All fabrication welds in the above areas.
(b) Piping - 100 percent coverage is required.

Acceptance standards shall be in accordance with table XI for castings and NAVSHIPS 0900-003-9000 for welds except that incomplete fusion and penetration is not acceptable. Drawings showing extent of coverage and RT symbols shall be submitted for NAVSEC approval (a separate sketch or drawing properly referenced on the engineering drawings is also acceptable).

Table XI - RT inspection standards.

Thick-ness	Shrinkage			Porosity		Inclusions		Unfused chaplets
	ASTM standard	Type	Reference radiograph	ASTM standard	Reference radiograph	ASTM standard	Reference radiograph	
Less than 1 inch	El86	1 2 3	CA 2 CB 2 CC 2	El86	A2	El86	B2	None acceptable
J. 1ncn up to 3 inches	El86	J. 2 3	CA 3 CB 3 CC 3	El86	A3	El86	B3	None acceptable
3 inches and over	E280	1 2 3	ca 2 Cb 2 Cc 2	E280	A3	E280	B3	None acceptable

NOTES:
Where ASTM E186 *is* specified, use the following:
(a) 1 to *2* MEV x-ray films for x-ray sources *in* this range and lower and for iridium 192.
(b) Gamma ray films for cobalt 60 sources (and radium if a y).
(c) 10 to 24 MEV x-ray films for x-ray sources in this range.
Where ASTM E280 is specified, use the following:
(a) 10 *to* 24 MEV x-ray films for x-ray sources in this range.
(b) 9balt 60 films for other sources.

4.4.3.1 Radiographic-standard shootin9 sketches. Shooting sketches in accordance with MIL-STD-271 shall be providea to assist in interpretation of the applicable radiographs; however, approval of the sketches is not required.

4.4.4 Ultrasonic test. Ultrasonic test shall be conducted for the items shown in table X. Specific stanaards ofacceptance shall be established by the turbine manufacturer and shall be submitted for NAVSEC approval.

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4.4.5 Exception for piping. All piping which is furnished in sections which will be fitted and welded to form final assembly by the shipbuilder and which will be hydrostatically tested by the shipbuilder does·not require hydrostatic tests by the turbine manufacturer. Drawings shall clearly indicate which piping is to be tested in the field. Small sections of low pressure piping (such as gland seal vent lines, valve stern leakoffs, and lube oil connections) which are to be welded to nipples on turbine casing do not require hydrostatic tests. Mr/PT to class 2 acceptance standards may be substituted.

4.5 examination and tests.

4.5.1 Heat stability test. Requirements of MIL-S-860 apply for turbine rotors except that the maximum deviation Uo'Ilowing the second cold reading) of turbine rotors for submarine applications shall not exceed 0.0005 inch.

4.5.2 BladinV• Geometry checks of tenons, vane and roots shall be performed to ensure conformi y to drawing requirements.

4.5.2.1 After assembly before shrouding. After assembly of blading in rotor and before shrouding, blades shall be checked to ensure that they are installed to drawing requirements.,

4.5.2.2 After shrouding. End gaps between shrouded groups and radial gaps (clearance between shroud and blade measured parallel to tenon axis) shall be checked for conformance to drawing requirements. The shroud shall bear against some portion of the vane of every blade being shrouded. Approval of the manufacturer's engineering department is required for either acceptance or re-peening of blading where radial gaps are in excess of 0.010 inch for cylindrical shrouds and 0.015 inch for conical (slant) shrouds. If clearances larger than these val.ues are accepted, the maximum value for each stage so affected shall be recorded on the machinery variation summation drawing (see 3.7.4,13).

4.5.2.3 Acceptance test for tenon peening. The turbine manufacturer shall prepare and submit for information a tenon peeniiig'"""procedure. The procedure shall include standards of acceptance including sectioned pieces showing acceptable and non-acceptable peening. Workmen who will do peening shall each be qualified to the procedure by demonstrating that they can produce a minimum of 5 test pieces representative of normal blade/tenon design in accordance with the standards of acceptance.

4.5.2.4 After sh?P operating (see 4.5.9). On the periodic test unit, all blading shall be exami'iiearor indications of deformed tenons, shroud lift (see 4.5.2.2) and other damage. On all turbines, the last row of blading shall be examined, for tightness and damage, through the access openings (see 3.4.43.10).

4.5.3 As shipped clearances. On the periodic test unit, after completion of the shop operating test (see 4.5.9), complete clearance data shall be taken and recorded on the overhaul report form (see 3.7.4.5). Bearing crown thickness measurements and depth gage constants are not required to be included since they are stamped on the bearing shells and bearing cap respectively. one copy each shall be forwarded to the cognizant Supervisor and building yard (for the applicable ship).

4.5.4 Bearings and journals. After completion of the shop operating test (see 4.5.9), bearings (journal and thrust), thrust collar, thrust shoes, oil deflectors and journals shall be examined for damage. Journals shall not be reworked for appearance only, when it is evident that performance is not affected. Minor scraping of bearings is acceptable. During load or overspeed tests, RTE readings shall be recorded for information and for confirmation of proper installation. Temperature limits of 3.4.51.6 do not apply to the overspeed conditi.cm.

4.5.5 Onboard tools. All special tools which involve critical fits for proper usage shall be demonstrated on the first unit of each contract or order, by actual use on the turbine and gear for which they are intended unless acceptability can be verifled by dimensional checks of the applicable parts. Where special tools involve a procedure not previously used, such procedure shall be demonstrated on the first uriit of a given design by using the· tools provided.

4.5.6 Threaded fasteners and dowels. It shall be verified that steam joint bolting has been tightened to specified values In accordance with documented procedures. The use of helical screw thread inserts in accordance MS2l208 to compensate for damaged threads is acceptable. ·

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4.5.7 1 rinth packin . On the periodic test unit, after completion of the shop operating test (see 4.5.9), iaphragm and gland labyrinth packing rings shall be examined for damage. If damaged, they shall be either repointed or replaced as necessary. If examination of the packing rings during the manufacturing process reveals surface porosity or defects that are not detrimental to the service of the packing, such rings may be accepted on a case basis by the local DCAs.·

4.5.B Steam valve tightness test. The turbine manufacturer shall demonstrate by contact dye checks that valves and seats havea continuous line contact.within the seating area,

4.5.9 Shhp operating test. Each turbine and gear shall be subjected to an operating shop test to c eek the operating dynamic balance of rotating parts and the strength and rigidity of all parts of the completed assembly, as well as the adequacy of internal clearances, Lubricating oil used during the test shall have the same viscosity as oil in accordance with MIL-L-17331, Tests may be run either with or without the generator.

4.5,9,l Duration of test. The test shall be conducted for a period of at least 30 minutes continuous operatio'n":'""" The rotors shall run in their own casings.

4.5.9 ,2 Test speedh The test shall.be conducted at rated speed to demonstrate that the speed regulating mec anisms operate properly.

4.5.9.3 Overspeed test. The shop operating test shall also include operation to demonstrate the ability ofthe rotor to be safely overspeeded for 3 minutes to the maximum speed expected after actuation of the overspeed trip and trip throttle valve, if such speed is calculated to be greater than 120 percent of the rated speed,

4.5.10 Vibration test. Vibration amplitudes shall be recorded during the shop operating test (see 4.5.9) at various speeds up to and including rated speed. Amplitude and frequency measurements shall be made with instruments having sensitivity consistent with figure 1. The maximum acceptable reading, as measured on the bearing journal or adjacent part of rotor at each end of each turbine, at rated speed shall not exceed the allowable amplitudes shown on figure 1. If shaft motions are measured directly by probes on journal or other means data shall be included in report of vibration results to be submitted to NAVSEC for information. Vibration levels shall be such that testing of overspeed devices can be safely conducted without damage to the units.

4.5.ll Lifting gear. Lifting gear for the first turbine and gear of each contract or order shall be fitted and demonstrated by the manufacturer before delivery to the ship. Inclining of machinery to obtain list or trim effects is not required.

4.5.12 Blade vibration test. When specified (see 6,2,2(t)), a blade vibration test shall be conducted to determine frequencies and stresses of one selected stage. Stage to be tested shall be as approved by NAVSEC and will be chosen considering calculated stresses, type of vibration mode, size, and similarity to other stages or designs. Test shall be conducted using either the actual turbine or a special test vehicle. Test agenda and instrumentation used shall be as approved by NAVSEC.

4.5.13 S mps, All oil sumps shall be tested by filling with water or clean hot oil and carefully examined for leakage any slight leakage shall be corrected. Any visual leakage shall result in rejection of the sump,

4.5.14 — of . pressure trips. Ea h back pressure trip shall be tested at least twice. Testing may be with either steam or air. Bench testing is acceptable.

4.5.15 Check of exhaust casing relief valves. Each exhaust casing relief valve shall be tested at least twice to determine conformance with 3.4.30.2. Bench testing is acceptable.

4.5.16 Check of overs!eed trke settings. Overspeed trip settings shall be checked at actuating spe'e'd"s.'"'"° Trip sha 1 be c ecked 5 times and tripping speed shall be repeatable within plus or minus 2 percent.

4.5.17 Check of overspeed rit system. The overspeed trip system of the first turbine-generator set undera contracts a be tested, with the throttle valve in the wide-open position, by suddenly opening the nozzle control valves while the unit is operating at no load and at rated speed. The rate the nozzle control valves are opened shall be not less than the rate that would be caused by the speed regulating governor suddenly sensing zero speed while the turbine is operating at rated speed. The steam inlet pressure used during the test shall be the maximum design operating steam pressure. The turbine shall not exceed the design maximum overspeed. This speed shall be listed in the technical manual and the technical data book. During this test the time for the trip throttle valve to close shall be ascertained.

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4.5.18 Check of oil system. During performance tests required by MIL-G-3087, the adequacy of the oiTsystem shall be demonstrated to show that there is no detrimental effect on the bearing lubrication as a result of governor transients. Special attention shall be given to the thorough cleanliness of all parts of the integral lubricating system, and care shall be taken to ensure that full flow can be circulated throughout the entire system.

4.5.19 Check of drains. Careful visual examination and checking of all details in connection withtli"earainage system shall be made to ensure that all turbines and gears are properly and completely drained. This requirement includes drainage of turbine casing, casing drain manifolds, oil and water pockets_of bearings and gland housings, and oil drainage from bearings.

4,5.20 Checks gear main rotating elements.

4.5.20.1 Forgings.

4.5,20.1.1 Forgings for gear shafts and pinions over 3 inches in diameter shall be subjected to ultrasonic test and surface defects examination.

4.5.20.l,2 All teeth of main elements shall be examined for surface defects by magnaflux.

4.5,20.2 Main pinions gears.(accuracy).

4,5.20,2.1 The manufacturer shall demonstrate or provide objective evidence of the fact that the procedure for coating teeth with copper or layout lacquer produces the desired film thickness. The procedure shall then be approved by NAVSEC in accordance with drawing approval procedure (see 3.7 and 3,4.48,9,1),

4.5,20,2,2 Tooth geometry errors (see 3,4,48.11) shall be measured on main elements produced on each contract in accordance with the following sampling plan:

(a) First gear of each design and first pinion of each design produced,
(b) From each succeeding lot of 5 to 10 elements of each design, one such element· randomly selected.

4.5.20,2,2.1 Pitch diameter runout, pitch variation, total accumulated pitch error and profile error (relative to design profile) shall be measured,

4.5.20,2.3 Each set of meshing elements shall show design tooth contact and relative helix angles when meshed on true centers and when assembled in the gear case, For these purposes, a light coating of pigment in oil may be transferred between teeth, and feeler gauges may be used (see 3.4.48,9),

4,5.20,2.4 Hardness for through hardened elements shall be determined after final heat treatment. For pinions, the hardness readings shall be taken at 4 points on the outside diameter of each helix approximately 90 degrees apart, before rough machining. For gear rims, readings shall be taken at 4 points on the end of each helix, approximately 90 degrees apart and slightly below the root diameter of the teeth but before teeth are cut.

4,5,20,2.5 Surface hardness of case-hardened elements shall be determined after final heat treatment. Hardness readings shall be made by the superficial method when feasible on flanks of teeth. Otherwise indentations may be made on ends slightly below the root diameter, Readings shall be taken at 4 points at each end, approximately 90 degrees apart. The case depth and core hardness shall be determined from coupon samples.

4,5.20,2.6 Tooth surface finish shall be checked either by comparator block or by measurement, on at least one tooth in each quadrant of each helix of each pinion of each gear (see 3.4,48,10) as follows:

(a) First pinion and gear of each design.
(b) One of each element, randomly selected, from each succeeding lot of 5 to 10 elements of each design.

4,5,21 Performance tests. Performance tests shall be conducted in accordance with the requirements of MIL-G-J08"T"o"r'MIL-G-19650 as applicable.

4.5.22 Shock, noise and vibration tests. Shock, noise and vibration tests of complete turbine-generators sna!rhe"!'n accordance with MIL-G-3087 or MIL-G-19650 as applicable.

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4,5.23 Inclined tests. Inclined operation tests shall be in accordance with the requirements of MIL-G-3087 or MIL-G-19650 as applicable.

4.6 Inspection of e,;eparation for deli . The packaging, packing and marking shall be inspected for compITance with sectionSof this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed *in* section 2, see 6.5.)

5.1 Preservation, packaging, packing and marking. The preservation, packaging, packing and marking shall be in accordance with MIL=c;::-3087 or MIL-G-19650 as applicable.

5.2 of polystyrene (loose-fill) material,

5.2.1 For domestic shipment and early equipment installation and level C rackaginl and packing.--Unless otherwise approved by the procuring activity (se 6.2.2(u) , use o polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.2.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited-for level A ackaging-and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. This equipment is intended for service in Naval ships where it is expected to withstand continuous use for long periods, without benefit of overhaul. The equipment is vital; failure can result in serious interruption of ship and fleet operating schedules.

6.2 Orderin data.

6.2.1 Specific application. Since this specification is general in scope, the details listed in 6.2.2 should be specified in the contract or order. (It is not necessary to delete inqpplicable portions of the basic specification.)

6.2.2 Detail requirements.

(a) Title, number and date of this specification.
(b) Type and class of generator-turbine required (see 1.2).
(c) Materials, if other than specified in 3.2 and required changes (if any) to List of Preferred Materials.
(d) Quantity (see 3.3.1).
(e) Normal rated steam inlet and exhaust conditions (see 3.3.1.l.l, 3.4.52.l and MIL-G-3087 or MIL-G-19650).
(f) Maximum and minimum steam inlet and exhaust conditions (see 3.3.1.1.2 and 3, 4,. 52.l).
(g) Any other items to be furnished with turbines and gears, in addition to those listed in 3.3.2.
(h) Weight and space limits, if any (see 3,4.7 and MIL-G-3087 or MIL-G-19650).
(i) Required steam ecoriomy in.eluding method of evaluation (see 3.4.12).
(j) Pitch and roll time, if required (see 3.4.13, table III).
(k) Specify airborne and structureborne noise levels over noise frequency range and vibration targets or limits (see 3.4.27.l, 3.4.38.5.1, 3.4.45.6 and MIL-G-3087 or MIL-G-19650) if more stringent than limits in 4.5.10 and figure 1.
(1) If gageboard is to be mounted other than as specified in 3.4.36.4.

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(m) If quietness of operation is a requirement (see 3.4.41.9,3 and 3.4.45.6).
(n) Location of steam inlet connection (see 3.4,44},
(o) If other than nitriding or carburizing is to be used on pinions and gears (see 3,4.48,1.2),
(p) Specify method of gear hardening, if other than as specified in 3,4.38,2 and 3.4.48.3,
(q) Specify gear accuracy, if other than as specified in 3.4.48.11.
(r) For class B, C and D turbines, specify pressure of the make-up steam for the gland seal systems (see 3.4,53).
(s) Quality assurance provisions (see 4.2),
(t) If a blade vibration test is required (see 4.5.12).
(u) When use of polystyrene "loose-fill" is approved (see 5.2).
(v) Information required with bid or proposal (see 6,3.l and 6,3,4).
(w) In the event turbines and gears are purchased separately, not as part of a complete turbine-generator set, the applicable items under ordering data of MIL-G-3087 or MIL-G-19650 shall also be specified.
(x) Specify waivers, revisions and additio al requirements,

6.3 Information required with proposal£=: -

, 6.3.l Quantity of £;:proposals. For AVSHIPS and NAVSEC procurement 5 complete copies of bid shall be sub11U.tted, unless otherwise stated (see 6.2.2(v)).

6.3.2 Drawings The bid or proposal shall include an outline drawing with dimensions and preliminary turbine and gear longitudinal sectional assembly drawings with materials for princip'al parts. Drawings shall be to scale.

6.3.3 Other data and state ents. The bid or proposal shall include curves, tabulations, sketches, diagrams-;-s£atements and such other data as is indicated in column 5 of table IX.

6.3.4 List of items to be furnished. A list of items being furnished is not required if in accordance wI'th the specffications7 however, if such a list is included in the proposal or bid, it is mandatory that each item listed be identified with tpe applicable part of 3,3.2 and 6.2.2(v).

6.4 Mana ement control system documents. The following management control system documents shou1d be included on DD Form 1660:

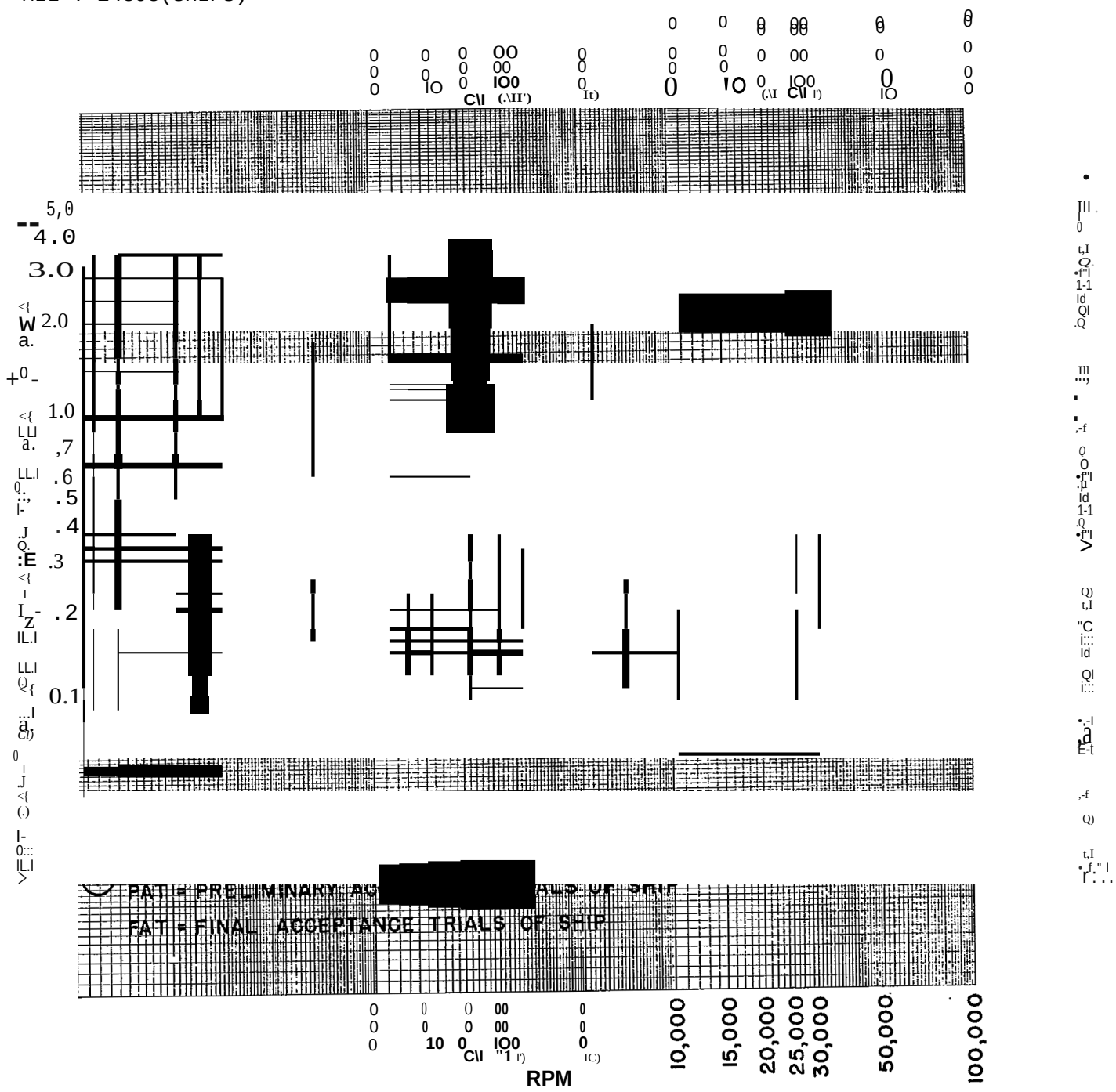
(a) MIL-P-15137 (see 3,6.l and 3,6.3).
(b) MIL-Q-9858 (see 4.2).

6.5 Supersession data. This specification supersedes that part of MIL-T-17523C(SHIPS) for generator turbines gears.

6.6 Sub-contracted material and pirts. The preparation for delivery requirements of referenced documents listed in Section do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy .:.. SH
(Project 2825-N006)

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NOTE: Rotational (filtered) vibration readings shall not exceed the above limits for any operating condition within the following speed ranges:

(a) For constant speed turbines and gears. 80 to 120 percent of rated speed.
(b) For variable speed turbines and gears. 13 to 17, 28 to 32, 43 to 47 cycles per second (c.p.s.) and 80 to 120 percent of rated speed.

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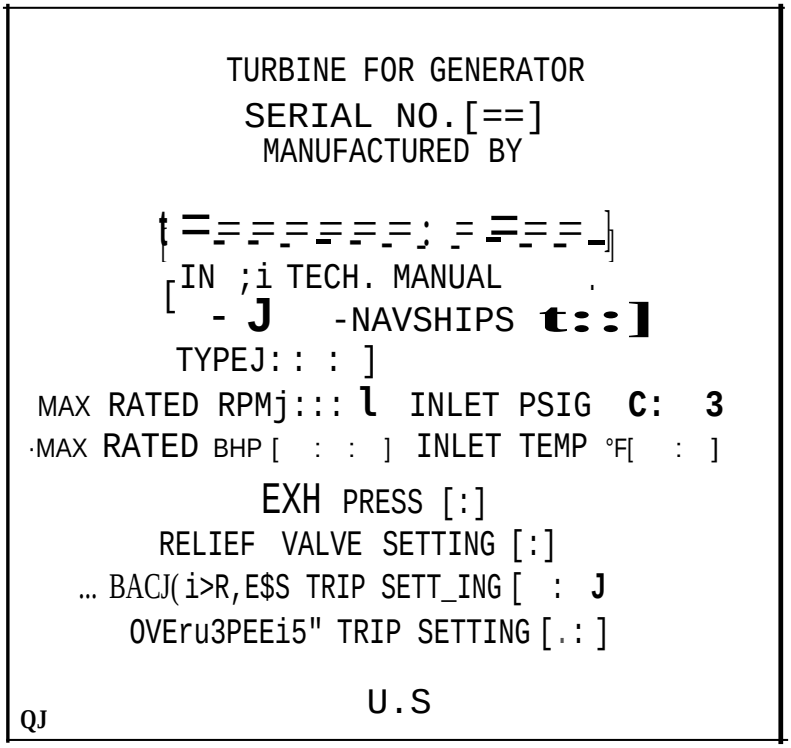


Figure 2 - Typical identification plate for generator-turbines.

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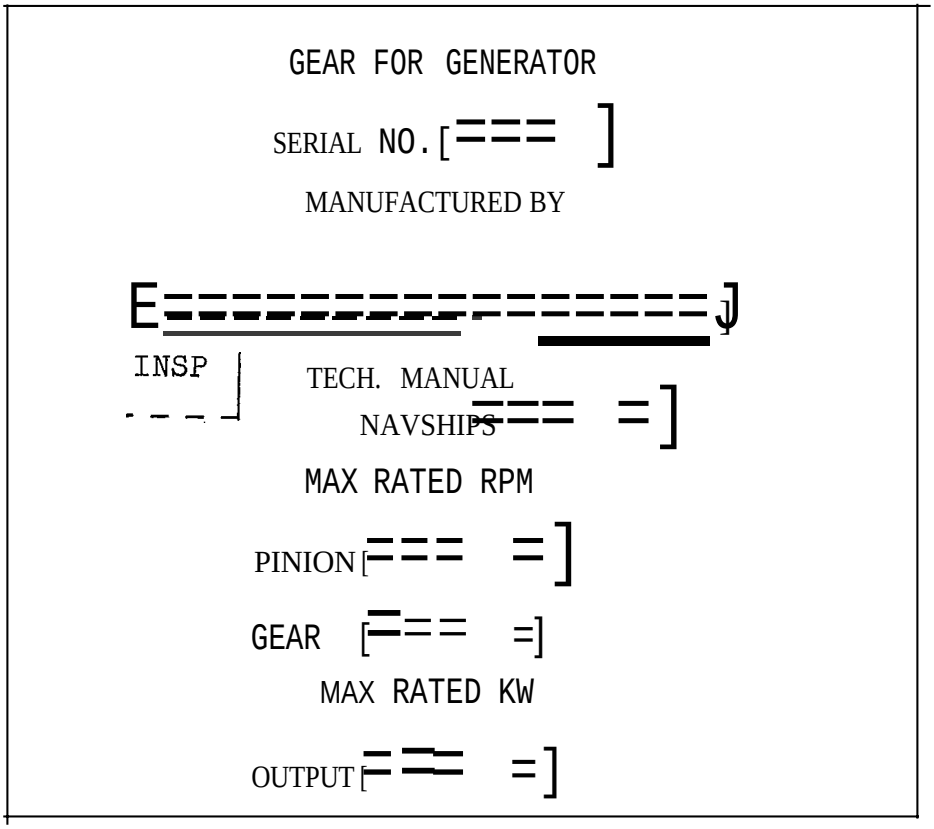
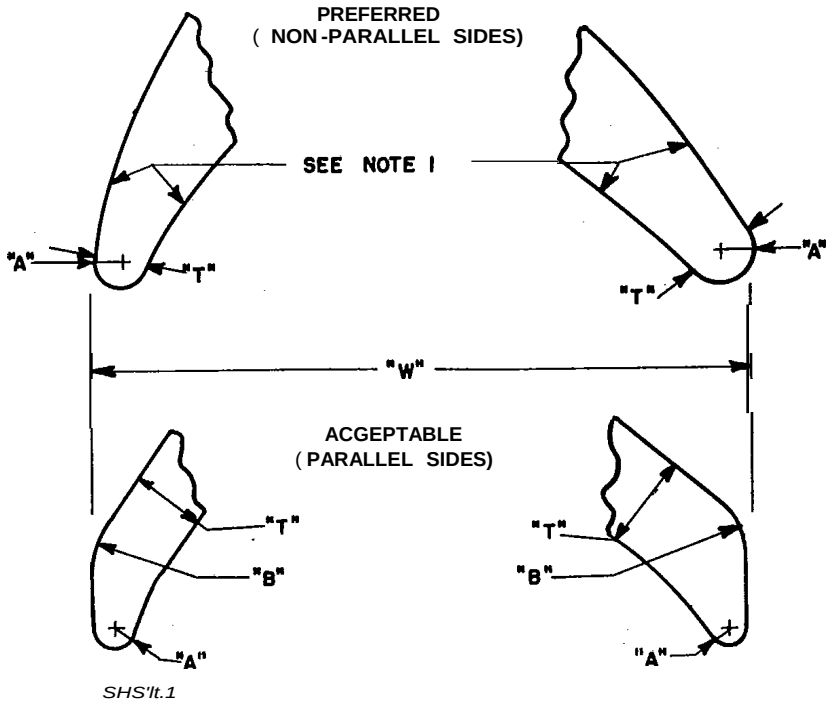


Figure 3 - Typical identification plate for generator gear.

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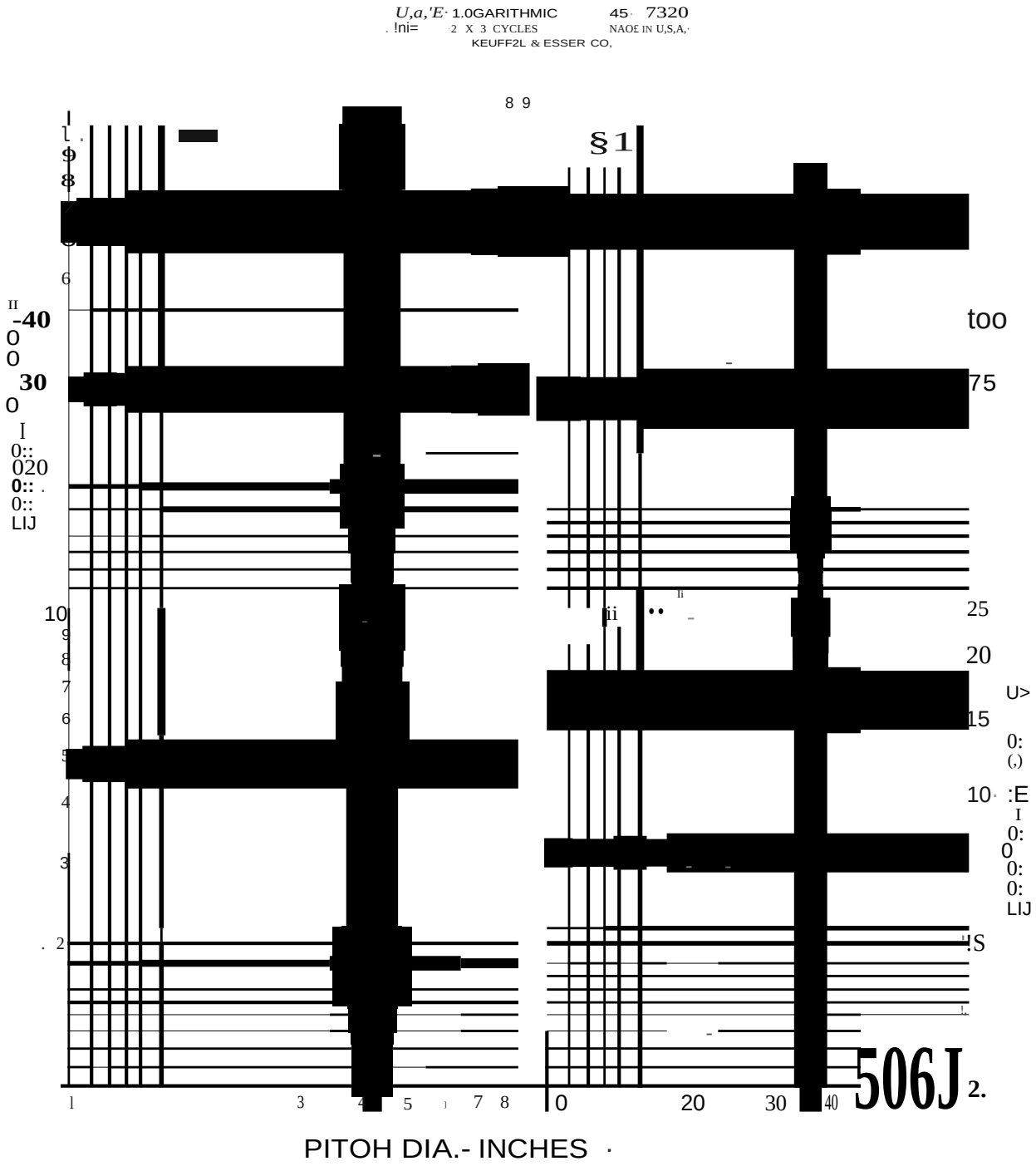
Vane width "W"	Min. val.ue of "T"	Min. value of t'A"	Radius _"Bl'
Less than l.	0.010	0.005	See note·.2
Above 1	•020	.010	

Notes to figure 4:

1. Concave and convex surfaces of diaphragm and nozzle, partition shall not be parallel at any point:.
2. Radius "B" is not specified quantitatively, but this area shall be polished smooth. A definite break between planes will not be acceptable.

Figure 4 - Vane metal section (blading and partitions).

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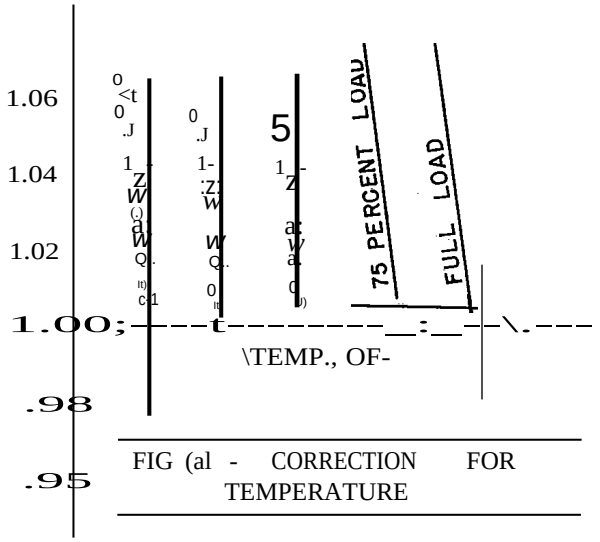


FIG (al - CORRECTION FOR TEMPERATURE

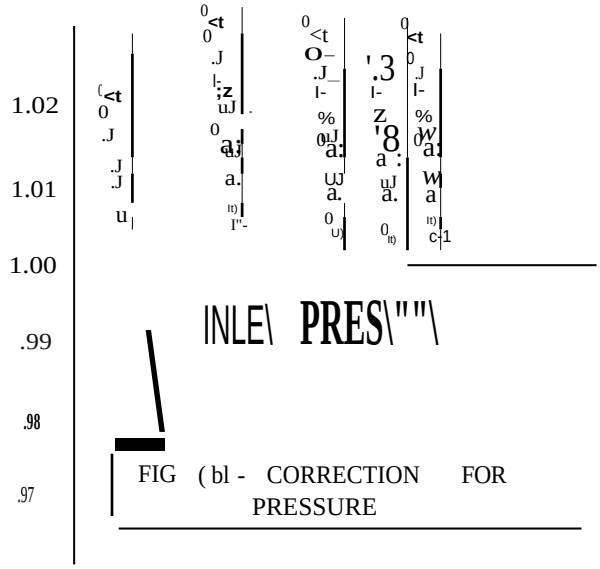


FIG (bl - CORRECTION FOR PRESSURE

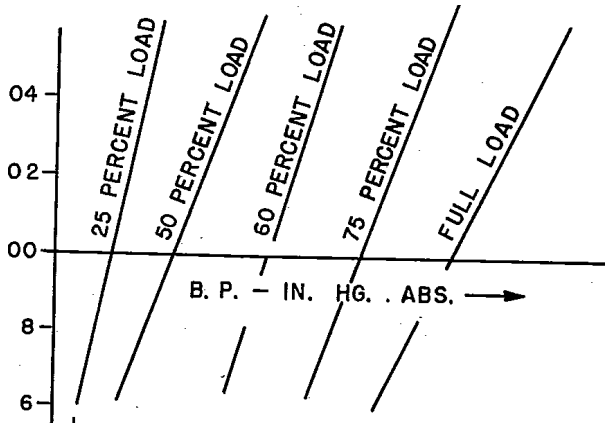


FIG (cl - CORRECTION FOR VACUUM

SH 984-c

Figure 6 Correction factors at constant valve settings.

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